

RTL Design Sherpa

DDR2 / LPDDR2 Family Controller Hardware Architecture Specification 0.9

October 8, 2026

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1 Pumice Has Index

Generated: 2026-10-20

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2 FUB Breakdown

This section inventories the post-Phase-2 pumice block breakdown. Pumice is now a consumer of the common-ip memory-controller stack plus a small set of rock-specific FUBs. The family-wide architecture, ownership rules, and plane stack are documented in common-ip HAS ch05; per-block microarchitecture is in common-ip MAS. The rest of this chapter only records which common blocks pumice uses and where they are wired.

2.1 1. What pumice consumes from common-ip

Pumice imports the family package and instantiates five common planes directly, with no pumice-side wrappers; its choices are parameters on those planes. The shared leaf FUBs live inside the planes; pumice does not instantiate them directly.

Common block	common_mas chapter	Pumice instantiation site
mc_common_pkg	ch01 — package	imported in pumice_core.sv:30 and the common macros that compose it
mc_axi4_plane	ch02 — host-side plane	pumice_core.sv:376 (u_axi_plane)
mc_axi_burst_chopper	ch02 — host-side plane	inside mc_axi4_plane (pumice_core.sv:376)
mc_wr_splitter	ch02 — host-side plane	inside mc_axi4_plane (pumice_core.sv:376)
mc_wr_intake	ch02 — host-side plane	inside mc_axi4_plane (pumice_core.sv:376)
mc_rd_intake	ch02 — host-side plane	inside mc_axi4_plane (pumice_core.sv:376)
mc_rd_return_ring	ch02 — host-side plane	inside mc_axi4_plane (pumice_core.sv:376)
mc_addr_mapper	ch02 — host-side plane	inside mc_wr_intake/mc_rd_intake of mc_axi4_plane (pumice_core.sv:376)
mc_storage_plane	ch03 — entry-pool plane	pumice_core.sv:504 (u_store_plane)
mc_wr_data_cam	ch03 — entry-pool plane	inside mc_storage_plane (pumice_core.sv:504)
mc_rd_cmd_cam	ch03 — entry-pool plane	inside mc_storage_plane (pumice_core.sv:504)
mc_scheduler_plane	ch04 — scheduler plane	pumice_core.sv:601 (u_sched_plane)
mc_bank_timers	ch04 — scheduler plane	inside

Common block	common_mas chapter	Pumice instantiation site
		mc_scheduler_plane
mc_bank_timer	ch04 — scheduler plane	inside mc_bank_timers inside mc_scheduler_plane
mc_global_timers	ch04 — scheduler plane	inside mc_scheduler_plane
mc_cmd_arbiter	ch04 — scheduler plane	inside mc_scheduler_plane
mc_training_plane	ch05 — training plane	pumice_core.sv:756 (u_train_plane)
mc_dfi_2p1_plane	ch07 — DFI 2.1 plane	pumice_core.sv:803 (u_dfi_plane)
mc_dfi_cdc	ch07 — DFI 2.1 plane	inside mc_dfi_2p1_plane (pumice_core.sv:803)
mc_dfi_cmd_path	ch07 — DFI 2.1 plane	inside mc_dfi_2p1_plane (pumice_core.sv:803)
mc_dfi_wr_serializer	ch07 — DFI 2.1 plane	inside mc_dfi_2p1_plane (pumice_core.sv:803)
mc_dfi_rd_aligner	ch07 — DFI 2.1 plane	inside mc_dfi_2p1_plane (pumice_core.sv:803)
mc_dfi_3p1_cmd_path	ch08 — DFI 3.1 plane	not used by pumice (scoria)
mc_dfi_4p0_cmd_path	ch09 — DFI 4.0 plane	not used by pumice (andesite)

2.2 2. What pumice owns

Pumice owns no leaf FUBs any more. The DDR2/LPDDR2-generation initialization, mode-register, refresh, page-policy, training, and DFI-encoding FUBs it used to own now live in common-ip and are composed directly by the common `mc_scheduler_plane`, `mc_training_plane`, and `mc_dfi_2p1_plane` macros in `pumice_core`. Pumice owns the tops (`pumice_top_geared`, `pumice_top`, `pumice_core`), `pumice_pkg`, the `pumice_csr` register description, and the build-time selection of these children:

Block	Role	Location
`mc_xddr2_init_sequencer`	DDR2/LPDDR2 JEDEC MR/init cold-boot engine.	`common-ip/rtl/fub/mc_xddr2_init_sequencer.sv`, composed inside `mc_scheduler_plane` (`pumice_core.sv:601`)
`mc_xddr2_mode_register`	Per-rank MR shadow + live decode to CL/CWL/BL/AL.	`common-ip/rtl/fub/mc_xddr2_mode_register.sv`, composed inside `mc_scheduler_plane`
`mc_xddr2_refresh_ctrl`	tREFI down-counter, postponer, and REFab/REFpb dispatch.	`common-ip/rtl/fub/mc_xddr2_refresh_ctrl.sv`, composed inside `mc_scheduler_plane`
`mc_page_policy`	Runtime page-policy engine and command telemetry.	`common-ip/rtl/fub/mc_page_policy.sv`, composed inside `mc_scheduler_plane`
`mc_xddr2_cmd_formatter`	JEDEC command encoding into DFI v2.1 wires (DDR2 truth table / LPDDR2 CA bus, OP `0x56`/`0xAB` for ZQ).	`common-ip/rtl/fub/mc_xddr2_cmd_formatter.sv`; instantiated inside common `mc_dfi_cmd_path`, which pumice consumes via `mc_dfi_2p1_plane`.
`mc_lpddr2_zq_ctrl` (common-ip)	Periodic LPDDR2 ZQ calibration maintenance (MRW to MR10).	`common-ip rtl/fub/mc_lpddr2_zq_ctrl.sv`, composed inside `mc_training_plane` (`pumice_core.sv:756`)

Block	Role	Location
`mc_lpddr2_dq_cal` (common-ip)	One-shot LPDDR2 DQ calibration interface (MRR MR32/MR40).	`common-ip rtl/fub/mc_lpddr2_dq_cal.sv`, composed inside `mc_training_plane`

2.3 3. Retired and absorbed blocks

The pre-Phase-2 pumice-local plane macros and their leaf FUBs have been replaced by common-ip equivalents. The old names may still appear in retired sentinel tests and filelist comments; they are not part of the current build.

Retired/absorbed block	Status after Phase-2
`pumice_axi4_plane`	Replaced by common `mc_axi4_plane` (`pumice_core.sv:376`).
`pumice_wr_intake` / `pumice_rd_intake`	Replaced by common `mc_wr_intake` / `mc_rd_intake` inside `mc_axi4_plane`.
`pumice_wr_data_cam` / `pumice_rd_cmd_cam`	Replaced by common `mc_wr_data_cam` / `mc_rd_cmd_cam` inside `mc_storage_plane` (`pumice_core.sv:504`).
`pumice_bank_timers` (local)	Replaced by common `mc_bank_timers` inside `mc_scheduler_plane`.
`pumice_cmd_arbiter` (behavioral core)	Replaced by common `mc_cmd_arbiter` inside `mc_scheduler_plane`. The pumice port-adapter wrapper was **removed 2026-10-11** — never instantiated in the default build (it survived only as the issue-rate test DUT); history in git.
`pumice_scheduler_plane` (behavioral macro)	**Removed 2026-10-11** — its init/refresh/page-policy children are now composed directly by common `mc_scheduler_plane` (`pumice_core.sv:601`).
`pumice_training_plane` (behavioral macro)	**Removed 2026-10-11** — its LPDDR2 ZQ/DQ-cal clients are now composed directly by common `mc_training_plane` (`pumice_core.sv:756`).
`pumice_dfi_cdc` / `pumice_dfi_cmd_path` /	Pumice-local DFI wrappers retired; common `mc_dfi_2p1_plane` owns the

Retired/absorbed block	Status after Phase-2
`pumice_dfi_wr_serializer` / `pumice_dfi_rd_aligner`	CDC and datapath (`pumice_core.sv:803`).
`txn_queue`	Replaced by the two common CAMs: `mc_wr_data_cam` + `mc_rd_cmd_cam`.
`bank_machine`	Replaced by FSM-free common `mc_bank_timer`, stamped by `mc_bank_timers`.
`xbank_timers`	Replaced by common `mc_global_timers` (turnaround) + `mc_bank_timers` (per-bank).
`cmd_encoder`	Replaced by common `mc_dfi_cmd_path` (which instantiates common `mc_xddr2_cmd_formatter`).
`odt_ctrl`	ODT handling now lives inside `mc_xddr2_cmd_formatter` / `mc_xddr2_mode_register`; no standalone block.
`page_predictor` (standalone)	Open-page decision is inline in common `mc_cmd_arbiter`; runtime policy is provided by common `mc_page_policy`. The adaptive modes retired 2026-09-27 (`mc_page_policy`: the `adapt_time`/`adapt_access` encodings 4/5 now fall through to the build default).
`wr_cmd_cam`	Replaced by common `mc_wr_data_cam`.
`scheduler`	Replaced by common `mc_scheduler_plane`.
`gear_dfi`	Replaced by common `mc_dfi_2p1_plane`.
`axi_frontend` / `axi_intake` / `*_macro`-as-architecture	Replaced by common `mc_axi4_plane`, generated `pumice_csr`, and `pumice_core`.
`powerdown_ctrl`	**Removed 2026-10-11** — was never

Retired/absorbed block	Status after Phase-2
	wired into the default `pumice_core` build; no replacement block (low-power entry is a scheduler maintenance concern, not a standalone FUB).
`dfi_signal_pack`	**Removed 2026-10-11** — final DFI register stage dropped; common `mc_dfi_cmd_path` drives the bus from the formatter's registered outputs.
`addr_mapper` (local copy)	**Removed 2026-10-11** — the local copy was never instantiated; address mapping is the common `mc_addr_mapper`, instantiated inside `mc_wr_intake` / `mc_rd_intake`.

Verified host-width freedom via `pumice_top_geared` remains {64, 128, 256}.

---

3 Training: LPDDR2 ZQ and DQ Calibration

The training plane is the common `mc_training_plane` (`common-ip/rtl/macro/mc_training_plane.sv`), instantiated directly by `pumice_core` as `u_train_plane` between the scheduler plane and the DFI plane ("Plane 2b" in the RTL, `rtl/top/pumice_core.sv:756`). Pumice builds it with both LPDDR2 clients enabled (`HAS_LP_ZQ = 1`, `HAS_LP_DQ_CAL = 1`), so it holds the two calibration FUBs — `mc_lpddr2_zq_ctrl` (periodic ZQ calibration) and `mc_lpddr2_dq_cal` (one-shot DQ calibration via mode-register reads) — muxes both onto a single maintenance command channel into the scheduler's arbiter, and, through `mc_training_shell`, owns the `mc_clk` ↔ `dfi_clk` clock crossing for the read-aligner calibration sideband. Until 2026-10-11 these FUBs were the pumice-local `pumice_zq_ctrl` / `pumice_lp_cal` behind a `pumice_training_plane` wrapper; both moved into common-ip and the wrapper was removed.

Both FUBs run only when `init_done` is set and `memtype == MEMTYPE_LPDDR2`; DDR2 has no ZQ pin and no MRR mechanism, so the entire plane is inert for DDR2 builds. That is an elaboration-free inertness — same RTL, gates low — so the existing DDR2 test suites run unchanged beside it.

3.1 The family rule: interfaces in hardware, searches in firmware

Scoria settled this for DDR3 write leveling (decision D2), andesite carried it to read leveling and LPDDR4 CA training, and pumice makes the same decision for the same reasons: the controller provides the interface — the MRR engine, the timing windows, the capture sideband, the telemetry — and the search runs in firmware. A DQ-vs-read-delay-tap sweep's step size, tap count and temperature behaviour are PHY properties, not controller properties; a calibration bug should be a script edit, not a respin. Concretely, the DQ-cal sweep (walk the board's FPGA IDELAY taps, capture, compare) lives in board firmware driving the PHY primitives directly. The core exposes `cal_start` and the captured MRR data; it deliberately contains **no** PHY-primitive logic and gained **no** new top-level ports for training.

LPDDR2's JEDEC menu is also smaller than DDR3/LPDDR4's, and the plane does not invent the missing entries. Write leveling, read leveling, DLL calibration and CA training do not exist in LPDDR2 — verified against the JESD209-2F mode-register map (MR41–47 are "Do Not Use", MR48–62 are reserved; CA training is an LPDDR4 MPC mechanism owned by andesite's training plane). What LPDDR2 actually offers is two mechanisms, both implemented here: ZQ calibration through MR10 writes, and DQ calibration through MRR reads of MR32/MR40.

3.2 mc_training_plane: arbitration and CDC

The plane composes the two FUBs around mc_training_shell, which does the slot arbitration and the clock crossing; run gating lives inside the two FUBs.

Table 3.0: mc_training_plane responsibilities

Responsibility	Detail
FUB holder	u_zq (mc_lpddr2_zq_ctrl, slot 0) + u_lp_cal (mc_lpddr2_dq_cal, slot 1); each removable by HAS_LP_ZQ / HAS_LP_DQ_CAL
One-active command mux	mc_training_shell puts both FUBs on one trn_cmd_* channel; fixed priority slot 0 (ZQ) > slot 1 (DQ cal), because ZQ is periodic maintenance while lp_cal is a firmware-initiated one-shot. trn_cmd_mrr_o is 0 for ZQ (MRW) and 1 for lp_cal (MRR)
Calibration sideband CDC	In mc_training_shell: cal_expect crosses mc_clk → dfi_clk as a sync_pulse (one mc_clk assertion produces exactly one dfi_clk pulse); the captured beat crosses back through a cdc_synchronizer (data held stable by the aligner until the next arm) with a sync_pulse for the valid strobe
Status aggregation	cal_busy_o = zq_busy || lp_busy; cal_done / cal_err come from lp_cal; zq_busy / zq_overdue / zqcs_total from ZQ

pumice_core wires the plane exactly as andesite wires its training plane: the trn_cmd_* channel nets into the scheduler plane's generic maintenance slot (maint_*, Ch 3.2), and cal_expect_o / cal_data_i / cal_valid_i net into the DFI plane's read aligner. pumice_top connects the CSR side (CAL_CTRL fields, timing registers) and harvests status back into CAL_STATUS and the CAL_MRR32/40_DATA slices.

3.3 `mc_lpddr2_zq_ctrl`: periodic ZQ calibration as maintenance traffic

LPDDR2 has no dedicated ZQCS/ZQCL command (unlike DDR3); both are **MRW writes to MR10**, OP = `0x56` (short, ZQCS) or `0xAB` (long, ZQCL), per JESD209-2F §5.13.3. ZQ Init (`0xFF`, tZQINIT) already exists in `init_sequencer` state `S_L_ZQ`; this FUB owns everything after init: the periodic calibration that keeps the output impedance inside its ±15% window as temperature and voltage drift.

The FUB **never generates the command itself**. It raises a request and, on grant, the arbiter issues the FUB's payload verbatim — `OP_MRS` with the row packed as `{MR10_index[5:0], OP[7:0]}` in the row field (bank 0; the MR index rides the row field, the same packing init MRW uses). The four-state FSM:

1. **Interval countdown** — `CAL_ZQ_INTERVAL` MC cycles between calibrations; `0` disables the engine entirely. `zq_en` gates the whole FUB.
2. **Request** — on expiry, raise `zq_req` and hold it. While waiting ungranted under demand, the FUB marks itself `zq_overdue`.
3. **Mode-C deferral** (`zq_defer_en`) — at expiry, if the scheduler reports sustained demand, the request can be held in `ZQ_DEFER` instead, up to `zq_overdue_max` MC cycles (13-bit, `0` = no cap), after which the request fires regardless. A deferral that overruns its cap — or any nonzero deferral that exits on demand loss — escalates the request to **ZQCL**: after deferred calibration the long calibration re-establishes the ±15% RON budget that the short one only trims.
4. **Post-grant hold** — on grant the FUB loads `t_zqcs` or `t_zqcl` (`CAL_ZQ_TIMING`) and asserts `cal_busy` for that window: JEDEC forbids other data-bus activity during calibration, so the arbiter must keep demand quiet until the window expires. Only then does the interval reload.

Two integration facts worth recording. First, the FUB's `demand_i` port exists for the Mode-C policy but is **tied to `1'b0` in the committed plane** — the scheduler's demand sideband is not wired through the training plane yet, so in the current integration expiry requests immediately (ZQCS, never deferred), and `zq_overdue` can never assert — both the deferral entry condition and the request-state overdue set read `demand_i`. The deferral policy is implemented and FUB-tested; the hookup is the remaining wiring. Second, the MR10 writes are **transient commands, never shadowed**: the ZQ MRW flows through the arbiter's maintenance path straight to the formatter, and `mode_register`'s shadow is written only by `init_sequencer`, so a periodic ZQ calibration can never disturb the decoded CL/CWL/BL. Multi-rank ZQ (a shared ZQ resistor across ranks) is left as a documented `generate` hook; the board targets are single-rank, and the arbiter's pick is single-rank in v1 anyway.

ZQ bus placement follows the all-banks-idle grant contract rather than the JEDEC minimum: LPDDR2-S4 requires all banks precharged for ZQ calibration while LPDDR2-N permits Idle-or-Active; the conservative contract satisfies both.

3.4 mc_lpddr2_dq_cal: one-shot DQ calibration over MRR

JESD209-2F §5.12.2 defines DQ calibration: MRR reads of MR32 (pattern A, 1010) and MR40 (pattern B, 0011) make the DRAM drive a known, training-only pattern onto DQ, so firmware can observe the read-capture timing without any memory array in the loop. mc_lpddr2_dq_cal is the one-shot sequencer that issues those two reads and captures the answers; what firmware does with the answers — the tap sweep — is deliberately outside the core.

The eight-state FSM, started by a cal_start pulse (a self-clearing CSR write):

1. Wait for the maintenance grant (the arbiter's all-banks-idle contract plus the w_out_safe demand gate leave no read or write data in flight — which also satisfies the JEDEC MRR spacings: RD→MRR ≥ BL/2, WR→MRR ≥ WL+1+BL/2+tWTR).
2. Issue MRR to **MR32** — OP_MRS with the MRR flag set, row {MR32_index[5:0], OP=8'h00} — and pulse cal_expect for one cycle, arming the DFI-plane read aligner.
3. Capture the first dfi_rddata_valid beat (MRR data lands RL·tCK + tDQSCK + tDQSQ after the command; BL=4, only beat 0 carries the MR data on DQ[7:0], with x16 devices mirroring the pattern on DQ[8]). A t_readout timeout without a beat sets sticky cal_err and ends the run.
4. Wait t_mrr (JEDEC §5.12: 2 clocks, the CSR reset default), then repeat for **MR40**.
5. On the MR40 capture set sticky cal_done. Both captured beats are exposed to firmware through CAL_MRR32_DATA / CAL_MRR40_DATA (full DFI read width, four 32-bit slices each at the default 128-bit geometry).

cal_abort (or losing the run condition mid-sequence) returns the FSM to idle; abort additionally drops the captured-data valids. The sticky cal_done / cal_err survive both — they are cleared only by soft reset — so a host can always distinguish "never ran", "ran and converged", "ran and timed out", and "ran, aborted" after the fact.

3.5 The MRR path: one new bit, not a new opcode

`dram_op_e` stays exactly as wide as it was — all 16 opcodes are consumed and MRR never enters the demand path. Instead MRR rides the existing `OP_MRS` as a new 1-bit flag, end to end:

- The arbiter passes the granted maintenance op **verbatim** (no re-encoding) and latches the requester's `trn_cmd_mrr_i` into the command word.
- The scheduler's output cmd FIFO word grows by one bit to `{mrr, ap, col, row, bank, rank, op}` (pack site `pumice_core.sv`, `w_cmd_data`; unpack site `mc_dfi_cmd_path.sv`). Demand commands simply carry `mrr = 0`.
- In `mc_xddr2_cmd_formatter`'s `OP_MRS` branch, `cmd_mrr_i` drives **CA3r**: MRW and MRR share the same MA/OP field packing — MA[7:0] on `{CA1f, CA0f, CA9r..CA4r}` — and MRR is MRW with CA3r inverted, bit-exact to JESD209-2F §5.12 (the OP field is don't-care-zero for MRR; the lp_cal row packs it 0). The `lpddr2_mrr` formatter test encodes the same table.
- No formatter change was needed for ZQ: LPDDR2 has no ZQCS/ZQCL opcodes, so those encodings remain NOP at the formatter's default branch — LPDDR2 ZQ is pure MRW.

On the return side, `mc_dfi_rd_aligner` gains a calibration capture sideband in the `dfi_clk` domain: a single-cycle `cal_expect_i` arms a capture register; the first `dfi_rddata_valid` beat is latched into `cal_data_o` (full DFI read width, held until the next arm) with a one-cycle `cal_valid_o` pulse, and the register disarms. The sideband cannot collide with the normal read FIFO push path because it only fires while armed, and the arbiter's grant contract guarantees no read is outstanding when calibration runs.

3.6 Arbitration policy

The scheduler's arbiter gains exactly one maintenance client (`trn_cmd_*`), in the family's standing shape — the same contract scoria and andesite use for ZQ and training. Priority, descending: **init** > **refresh** > **trn** > **demand** (column, then ACT, then PRE). The training branch fires only under `w_trn_safe`: every bank idle in the registered view, no row-affecting command in flight or inside its 2-cycle guard window, no tRFC recovery running, and no grant in flight. The granted op passes through verbatim; the grant is a single-cycle pulse on the fire. Refresh outranks training, so a pending refresh drains banks and wins the bus first; training outranks every demand class.

The post-grant hold is where BUG-003 discipline matters. Both FUBs drive `cal_busy` during their bus-quiet windows (ZQ hold, lp_cal until done), and the arbiter folds `cal_busy_i` into `w_out_safe` for **every demand class** — ACT, RD/WR and PRE all carry a `!cal_busy_i` term, while the training class itself is unconditional. The gate lives in the fire stage, never the pick cone: `cmd_valid_o = r_pick_valid && w_out_safe` and `w_fire_out = r_pick_valid && cmd_ready_i && w_out_safe`, so a pick the safety gate rejects is never pushed to the cmd FIFO and the fire is the push by construction. That is precisely the invariant BUG-003 restored for the bank timers; the training gate had to land there, not in the pick pipeline, or it would have reopened the same divergence.

3.7 CSR summary

The plane is controlled and observed through seven registers in the 0x0A0 block of `pumice_csr` (field definitions and resets: `rtl/macro/pumice_csr.rdl`):

Table 3.1: Training-plane CSR summary

Address	Register	Access	Contents
0x0A0	`CAL_CTRL`	rw	`zq_en[0]`, `zq_defer_en[1]` (Mode-C), `cal_start[4]` (wo, self-clearing), `cal_abort[5]` (wo), `zq_overdue_max[28:16]` (max deferral, 0 = no cap)
0x0A4	`CAL_ZQ_INTERVAL`	rw	ZQCS interval in MC cycles, 0 = disabled
0x0A8	`CAL_ZQ_TIMING`	rw	`t_zqcs[15:0]`, `t_zqcl[31:16]` post-grant hold windows
0x0AC	`CAL_TRAIN_TIMING`	rw	`t_mrr[15:0]` (reset 2), `t_readout[31:16]` MRR issue→data timeout
0x0B0 +4×	`CAL_MRR32_DATA[0..3]`	ro	Captured MR32 beat 0, four 32-bit slices of the DFI read width
0x0E0 +4×	`CAL_MRR40_DATA[0..3]`	ro	Captured MR40 beat 0, four 32-bit slices

Address	Register	Access	Contents
0x0F0	CAL_STATUS	ro	zq_busy[0], cal_busy[1], cal_done[2] (sticky), cal_err[3] (sticky), zq_overdue[4], zqcs_total[31:16]

The data arrays start at 0x0B0 and 0x0E0 because the retired 0x0C0–0x0DC hole is never reused (pumice doctrine) — 0x0E0 clears it. `cal_done` / `cal_err` are sticky and clear only on soft reset.

3.8 Verification and integration notes

- **FUB / macro**: none in the pumice area. The training FUBs, the formatter and the scheduler plane are common-ip `mc_*` modules, and their directed tests – `test_pumice_zq_ctrl.py` (`mc_lpddr2_zq_ctrl`), `test_pumice_lp_cal.py` (`mc_lpddr2_dq_cal`), `test_dfi_cmd_formatter.py::lpddr2_mrr` (`mc_xddr2_cmd_formatter`, MRR CA encoding against JESD209-2F §5.12) and `test_pumice_scheduler_plane.py` (training priority vs refresh / demand, all-banks-idle gating, demand blocked during `cal_busy`, fire==valid) – were removed from the pumice area 2026-10-13 (pumice tests target only pumice DUTs); no common-ip replacement exists yet. The read-aligner cal capture is `mc_dfi_rd_aligner`'s. The command-stream checker allows MRR and MRW-MR10 in its stream rules.
- **Top**: none yet. No test under `dv/tests/top/` writes `CAL_CTRL.cal_start` or enables ZQ (checked 2026-10-13), so calibration currently has no coverage in the pumice area at any level.
- **TB tie-off**: `pumice_core_tb_top.sv` ties every training input to 0 (`zq_en`, `zq_defer_en`, `zq_interval`, `t_zqcs/t_zqcl`, `zq_overdue_max`, `cal_start`, `cal_abort`, `t_mrr`, `t_readout`) and leaves the status outputs open — that TB scores the datapath and paging sweep, not calibration. Enables at 0 make both FUBs inert by construction.

4 What Moved to Common (Phase-2 extraction)

On 2026-10-09 → 2026-10-10 the shared logic of the three research controllers — pumice (DDR2/LPDDR2), scoria (DDR3/LPDDR3), andesite (DDR4/LPDDR4) — was extracted into `mem-ctrl-ip/common-ip/`, and pumice was re-integrated onto it the same weekend. This chapter records what moved, what stayed, and where each is documented now. The family architecture the extraction implements — the plane stack and the descriptor seam — is owned by the family HAS (common_has); this chapter is pumice's delta against it, and it is the only place in this book that tells the move story.

4.1 What pumice consumes from common-ip

pumice_core instantiates six common planes directly, and every leaf FUB beneath them is common-ip as well:

Common block	Role in pumice	Microarchitecture
`mc_axi4_plane`	AXI4 slave intakes, burst chop/split, coalescing, read-return ring	common_mas ch02
`mc_storage_plane`	rd/wr CAMs, read snooping, bank image	common_mas ch03
`mc_init_plane`	JEDEC init sequencing, mode-register shadow + CL/CWL/BL decode, DFI PHY init handshake, `init_state_o` telemetry (extracted from the scheduler plane 2026-10-15; granite instantiates the identical plane)	common_mas ch10
`mc_scheduler_plane`	pick machinery, timers, generic maintenance slot, in external-init mode (`HAS_INIT_SEQUENCER=0`, `HAS_MODE_REGISTER=0`), plus its DDR2/LPDDR2 children `mc_xddr2_refresh_ctrl` and `mc_page_policy`	common_mas ch04
`mc_training_plane`	LPDDR2 ZQ (`mc_lpddr2_zq_ctrl`) and DQ calibration (`mc_lpddr2_dq_cal`) behind the two-slot arbiter and `mc_training_shell` CDC	common_mas ch05

Common block	Role in pumice	Microarchitecture
`mc_dfi_2p1_plane`	DFI 2.1 command/data pipeline (`mc_xddr2_cmd_formatter` inside `mc_dfi_cmd_path`) + the single aclk↔dfi_clk CDC	common_mas ch07

The package is the family's `mc_common_pkg`; `pumice_pkg` remains and re-exports it (the shipping-CSR widening argument for not migrating the package itself is unchanged — see the family package design).

4.2 What stayed pumice-side

At the 2026-10-10 extraction everything generation-specific stayed, by the family's two-customer rule: the init sequencer, mode register, refresh controller, page-policy engine, power-down controller, the DFI command formatter and signal pack, ZQ calibration (`pumice_zq_ctrl`), DQ calibration (`pumice_lp_cal`), and a local `addr_mapper`, held together by two thin wrappers, `pumice_scheduler_plane` and `pumice_training_plane`.

That state lasted one day. On 2026-10-11 the DDR2/LPDDR2 children moved into common-ip as the `xddr2` / `lpddr2` generation FUBs (`mc_xddr2_init_sequencer`, `mc_xddr2_mode_register`, `mc_xddr2_refresh_ctrl`, `mc_xddr2_cmd_formatter`, `mc_page_policy`, `mc_lpddr2_zq_ctrl`, `mc_lpddr2_dq_cal`), the common planes gained `HAS_*` parameters to compose them, and both wrappers were removed. `powerdown_ctrl`, `dfi_signal_pack`, and the unused local `addr_mapper` were removed outright. What stays pumice-side now is only the tops (`pumice_top_geared`, `pumice_top`, `pumice_core`), `pumice_pkg`, the `pumice_csr` register description, and the parameter choices on the common planes (FUB Breakdown).

4.3 What this book does with the moved blocks

Chapters whose subject moved carry a **“lives in common-ip now”** banner under their header, pointing at the owning common_mas chapter; their pumice provenance and measured history are kept — that history is invariant, and the numbers in it (bandwidth, timing margins, bug regressions) are pumice’s own measurements, not the common block’s. Chapters whose subject stayed are unaffected except where names changed.

4.4 The measured record survives the move

The Phase-2 re-integration was gated on pumice's full suite passing on the common stack (top/FUB/macro/phy, SEED=12345) and on the Nexys A7 timing closures being re-established at 66.67 MHz and 75 MHz after the `rd_crc_check` popcount fix. Those numbers — and every pre-extraction measurement in this book — describe pumice as it runs today; the extraction changed where the logic lives, not what it does.

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5 mem-ctrl-ip Family — Overarching Hardware Architecture Specification

Version: 0.3 **Date:** 2026-10-20 **Status:** v0.1, the vision edition. This book is the family-level HAS: the plane-stack architecture the partner set out during the Phase-2 common-IP campaign, written down as the binding contract the rocks and the products are built against. The per-block microarchitecture lives in common_mas; this book owns the shape of the whole and the rules no single block owns.

> **Read this first.** A controller book (pumice/scoria/andesite HAS/MAS) and a family doc disagree → the family doc wins and the book is the defect. That rule predates this book (the family INDEX states it); this book is one of the docs it protects. Every claim here is either (a) measured in a named rock, (b) a cited JEDEC/DFI clause, or (c) an explicitly marked open question. Nothing else is asserted.
>
> **Warning:** every timing CSR in this family is a count of MC-clock cycles, correct only at the MC clock frequency it was derived for. A frequency change invalidates the whole timing set at once; software must re-derive and reprogram it before traffic resumes. Stale values present as data corruption, not errors. Each rock's HAS carries this warning at the register map; this book makes it family doctrine (Ch 4).

5.1 Document Organization

Note: All chapters linked below for automated document generation.

5.1.1 Front Matter

- Document Information

5.1.2 Chapter 1: Introduction

- Purpose, Scope, and the Two Lines

5.1.3 Chapter 2: The Plane Stack

- The Architecture: axi4 → storage → scheduler → (ras) → training → dfi
- The Descriptor Seam

5.1.4 Chapter 3: The DFI Generations

- Why `dfi_2p1_plane` / `dfi_3p1_plane` / `dfi_4p0_plane` Are Separate Blocks

5.1.5 Chapter 4: Doctrines Added by the Common-IP Campaign

- New Family Doctrines

5.1.6 Chapter 5: The Line-Up

- The common-ip Layout
- Products and Research: granite / gabbro / diorite

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6 mem-ctrl-ip Common IP — Micro-Architecture Specification

Version: 0.3 **Date:** 2026-10-20 **Status:** v0.1. One chapter per common block. The architecture these blocks implement — the plane stack, the descriptor seam, the doctrines — is the family HAS (common_has); this book is the per-block microarchitecture and never restates the architecture. Where the two disagree, the HAS wins and this book is the defect.

> **Read this first.** Every claim in a chapter is one of: (a) read from the RTL it documents (cited `file:line`), (b) inherited from a named rock's measured history (cited to the rock's book or bug record), or (c) an explicitly marked open question. Nothing else is asserted. Silicon-bug history (`BUG-*`, `ISSUE-*`, measured regression numbers) is invariant documentation and is carried verbatim from the RTL comments.

6.1 Document Organization

Note: All chapters linked below for automated document generation.

6.1.1 Front Matter

- Document Information

6.1.2 Chapter 1: The Family Package

- `mc_common_pkg` — types, memtype enum, shared opcodes

6.1.3 Chapter 2: `mc_axi4_plane` and the Host-Side FUBs

- `mc_axi4_plane` — AXI slaves, intakes, coalescing, return ring

6.1.4 Chapter 3: `mc_storage_plane` and the Entry-Pool FUBs

- `mc_storage_plane` — CAMs, read snooping, bank image

6.1.5 Chapter 4: `mc_scheduler_plane` and the Pick Machinery

- `mc_scheduler_plane` — schedulable-entry view, timers, arbiter

6.1.6 Chapter 5: `mc_training_plane`

- `mc_training_plane` — LPDDR2 clients, generic shell, maintenance slot

6.1.7 Chapter 6: `mc_ras_plane`

- `mc_ras_plane` — skeleton, transform pair, reserved CSR surface

6.1.8 Chapter 7: `mc_dfi_2p1_plane` and the Shared DFI Pipeline

- `mc_dfi_2p1_plane` — DFI 2.1 command/data pipeline + CDC

6.1.9 Chapter 8: `mc_dfi_3p1_plane`

- `mc_dfi_3p1_plane` — the 3.1 generation delta

6.1.10 Chapter 9: `mc_dfi_4p0_plane`

- `mc_dfi_4p0_plane` — the 4.0 generation delta

6.1.11 Chapter 10: `mc_init_plane`

- `mc_init_plane` — JEDEC init, MR shadow, PHY handshake

  Command-flow order (per the family HAS plane stack) places the init plane between the storage plane and the scheduler plane; it is filed as

chapter 10 so chapters 2–9 keep stable paths for inbound links. The architecture statement is the HAS stack, not the chapter number.

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7 CSR Register Map

The controller's configuration and observation registers are defined in `rtl/macro/pumice_csr.rdl` and generated with PeakRDL into a passthrough-cpuif register block (`bin/peakrdl_generate.py`). The DV mirror is `dv/tbclasses/pumice_regmap.py`; software addresses every field **by name** through that generated map — hardcoded offsets are forbidden. This chapter is the authoritative narrative for that RDL; every register, offset, and reset value below is reconciled against it.

The map occupies a 4 KB space (12-bit address, `CSR_ADDR_W = 12`). All registers are 32 bits wide, little-endian, 4-byte aligned. The register bus is the PeakRDL **passthrough cpuif** (`s_cpuif_*`), not a hand-written APB slave — see §4.3. Config fields are consumed `hw = r` (`hwif_out.*`) and drive the core by name; status/observation fields are `hw = w` (`hwif_in.*`).

7.1 Control and Status (0x000 – 0x00F)

7.1.1 CTRL (0x000, R/W)

Bit	Field	Reset	Description
0	init_start	0	Write 1 to start init (self-modifying strobe)
1	init_force_restart	0	Write 1 to force re-init even mid-sequence
3:2	RSVD_3_2	0	Reserved
4	pwr_req_low_power	0	Request power-down state
5	pwr_req_dpd	0	Request DPD (LPDDR2 only)
6	pwr_req_active	0	Request return to ACTIVE
7	pwr_req_self_refresh	0	Request self-refresh
30:8	RSVD_30_8	0	Reserved
31	soft_reset	0	Write 1 to assert internal soft reset (self-clearing)

7.1.2 STATUS (0x004, R only)

Bit	Field	Description
0	init_done	1 when init complete
1	init_error	1 on init failure
7:4	power_state	Current power-state FSM state (encoded)
8	pasr_active	LPDDR2: PASR mask is non-zero
23:16	init_step_dbg	Current init step number (for bring-up)
31	version_match	1 when build matches expected version

7.1.3 STATUS_HISTORY (0x008, R only)

Last 8 power-state transitions, 4 bits each. Most recent in `[3:0]`; new transitions push the oldest off. Useful for bring-up debugging of power-state oscillation.

7.2 Timing Parameters (0x010 – 0x01F)

> **Warning — every value in this section is a count of `aclk` cycles, correct only at the MC clock frequency it was derived for.** The reset values below are the design-point derivation (the Nexys A7 build runs `aclk` at 75 MHz; §6.1). Change the MC clock frequency — a retarget to 66.67 MHz for timing margin, a different board — and **every cycle-counted register in this map must be re-derived from the JEDEC ns/nCK value at the new frequency and reprogrammed before traffic resumes**: this Timing Parameters block, the init waits (`INIT_TIMING0` / `INIT_TIMING1`), the ZQ and calibration intervals (`CAL_ZQ_INTERVAL`, `CAL_ZQ_TIMING`, `CAL_TRAIN_TIMING`), and the PHY round-trip latencies (`PHY_TIMING`). A stale value after a frequency change violates a real DRAM timing and presents as data corruption, not as an error. There is no hardware ns→cycle conversion and no frequency-change protocol in this generation — reprogramming is software's job. (LiteDRAM's `BUILD_CLK_HZ` reading 100 MHz on a 75 MHz engine clock is this exact failure in the wild — 2026-10-03 char results.)

Packed timing parameters in MC (`aclk`) cycles. All fields are `sw = rw`, `hw = r` and drive the core scheduler / DFI plane.

7.2.1 TIMINGS_RC_RCD_RP_RAS (0x010, R/W)

Bits	Field	Reset	Description
7:0	tRC	60	tRC in cycles
15:8	tRCD	15	tRCD in cycles
23:16	tRP	15	tRP in cycles
31:24	tRAS	40	tRAS in cycles

7.2.2 TIMINGS_RFC_REFI (0x014, R/W)

Bits	Field	Reset	Description
15:0	tRFC	16	tRFC (or tRFCab) in MC cycles — mission-mode REF recovery, consumed by the arbiter
31:16	tREFI	1950	tREFI in cycles

7.2.3 TIMINGS_RRD_FAW_WTR_CCD (0x018, R/W)

Bits	Field	Reset	Description
7:0	tRRD	6	tRRD, in MC cycles to block (spacing enforced is N+1)
15:8	tFAW	35	tFAW, in MC cycles
23:16	tWTR	4	tWTR, in MC cycles to block (spacing enforced is N+1)
31:24	tCCD	4	tCCD, in MC cycles to block (spacing enforced is N+1)

Every JEDEC timing field in this map is a count of MC cycles to BLOCK, so the command spacing the DRAM sees is **N+1**. Proved and bounded tight in `formal/mc_common/{mc_bank_timer,mc_global_timers}`; see `pumice_mas/ch04_apb_config/03_runtime_overrides.md` for what that means when programming a part from its datasheet, and pumice TASK-034 for the open question of recovering the cycle.

7.2.4 TIMINGS_CL_CWL_WR (0x01C, R/W)

Bits	Field	Reset	Description
7:0	CL	6	CAS latency
15:8	CWL	4	CAS write latency
23:16	tWR	15	Write recovery
31:24	tRFCpb	70	LPDDR2 per-bank tRFC

7.3 Mode Register Values (0x020 – 0x02F)

7.3.1 MR0, MR1, MR2, MR3 (0x020, 0x024, 0x028, 0x02C, R/W)

Low 16 bits (VAL) contain the MR value; upper 16 bits reserved. These hold override MR values; the default init walk uses the JEDEC-standard values built into `init_sequencer.sv` (§6.2). Reset = 0.

7.4 LPDDR2-Specific (0x030 – 0x03F)

7.4.1 PASR_BANK_MASK_RANK0 (0x030, R/W)

Bits	Field	Reset	Description
7:0	pasr_banks	0	LPDDR2 PASR per-bank mask for rank 0 (MR16); bit N = 1 masks bank N

7.4.2 PASR_SEG_MASK_RANK0 (0x034, R/W)

Bits	Field	Reset	Description
7:0	pasr_segs	0	LPDDR2 PASR segment mask for rank 0

7.4.3 TEMP_DERATE_RANK0 (0x038, R only)

Bits	Field	Description
1:0	temp_class	From rank 0's MR4: 00 = nominal, 01 = 2x refresh, 10 = 4x refresh

The current RDL defines the single-rank instances only (NUM_BANKS = 8, rank 0). Multi-rank builds would extend this block; the map as generated today does not allocate per-rank PASR/temperature registers beyond rank 0.

7.5 Scheduler / Refresh / Page / Addr / Init Tuning (0x040 – 0x05F)

7.5.1 SCHED_TUNING (0x040, R/W)

Bits	Field	Reset	Description
3:0	lookahead_active	0	Active lookahead window (0..LOOKAHEAD_DEPTH_MAX). 0 disables lookahead.
4	force_inorder	0	1 = force first-ready FIFO (disable row-hit reordering).
5	happy_enable	1	1 = HAPPY predictor active (only meaningful if synthesized).
15:8	age_max_runtime	0	Runtime AGE_MAX override (0 = use build-time default).
23:16	txn_queue_high_water	0	Backpressure-assertion threshold for the txn queue.
27:24	lookahead_max_obs	—	(R only, hw = w) Echo of build-time LOOKAHEAD_DEPTH_MAX.

7.5.2 PAGE_PRED_TUNING (0x044) — RETIRED

Retired 2026-09-27 with the HAPPY predictor (pumice TASK-011: the history-based modes measured inert or harmful on silicon). The address is a hole in the current RDL, as is 0x120 (OBS_PAGE_PRED_ACCURACY). Earlier revisions documented warmup_cycles / hysteresis here; neither field exists in the generated register block anymore. The page-policy telemetry (PAGE_STATS_*, OBS_ROW_HIT) is unaffected.

Former contents for history:

Bits	Field	Reset	Description
15:0	warmup_cycles	1024	Warmup cycles before predictor is trusted (retired)
23:16	hysteresis	2	Saturating-counter hysteresis (retired)

7.5.3 REFRESH_TUNING (0x048, R/W)

Bits	Field	Reset	Description
1:0	refpb_policy_or	0	00 = build-time; 01 = RR; 10 = OLDEST_FIRST; 11 = DARP.
3:2	page_policy_or	0	00 = build-time; 01 = OPEN; 10 = CLOSE; 11 = HAPPY_HYBRID encoding (**retired — the RTL maps it to OPEN**, pumice TASK-011).
7:4	refresh_defer_active	1	Active refresh deferral count (1..REFRESH_DEFER_MAX). 1 = no batching.
31:16	zqcs_freq_hz	1	Periodic ZQCS interval in Hz. 0 disables periodic ZQCS.

7.5.4 ADDR_MAP (0x04C, R/W)

Replaces the retired ADDR_MAP_TUNING register. The AXI-to-DRAM mapping is driven by a single knob rather than a scheme selector — see §5.4 and §3.1, and common-ip/rtl/fub/mc_addr_mapper.sv.

Bits	Field	Reset	Description
4:0	bank_lsb	0x0A	Bank-field LSB in the word address. Default 10 = COL_WIDTH = ROW_MAJOR placement. RTL clamps to [0, COL_WIDTH].
8	hash_en	0	Enable bank XOR-hash: bank ^= fold(row) ^ hash_seed.
23:16	hash_seed	0	XOR-hash seed (seed[BW-1:0]).

The register reset value is 0x0000000A (bank_lsb = 10, hash off). Setting bank_lsb = COL_WIDTH gives ROW_MAJOR; bank_lsb = log2(DRAM_BL) gives maximum bank interleave with burst locality preserved; hash_en folds an XOR-hash on top of any placement (the old XOR_HASH scheme). There is no separate scheme-selector field. (DRAM_BL, not cols/burst — the mapper indexes device words; pumice BUG-022.)

7.5.5 INIT_TUNING (0x050, R/W)

Bits	Field	Reset	Description
3:0	zq_retries	3	ZQ calibration retry count before raising init_error. Range 1–8.

Bits	Field	Reset	Description
15:8	init_timeout_ms	10	Per-step init timeout in ms. Range 1–255.

7.5.6 INIT_TIMING0 (0x058, R/W)

JEDEC init-sequence waits (MC cycles), consumed by `init_sequencer.sv`. Previously hardcoded.

Bits	Field	Reset	Description
15:0	t_init_wait	512	CKE / tINIT settle
31:16	t_dll_wait	256	DLL lock (tDLLK)

7.5.7 INIT_TIMING1 (0x05C, R/W)

Bits	Field	Reset	Description
7:0	t_mrd_wait	8	tMRD (post mode-reg)
15:8	t_rp_wait	8	tRP (post precharge)
23:16	t_rfc_wait	16	tRFC (post refresh)

7.5.8 TIMINGS_RTP_RTW (0x054, R/W)

Read-to-precharge and read-to-write turn-around. Previously `tRTP` was hardcoded (8'd4) and `tRTW` was tied to it; both are now independent configs.

Bits	Field	Reset	Description
7:0	tRTP	4	Read to precharge
15:8	tRTW	6	Read to write

7.6 DFI / PHY Timing (0x060 – 0x067)

7.6.1 DFI_PHASE (0x060, R/W)

Which DFI sub-phase carries the READ vs WRITE command, to match the PHY's `rdphase` / `wrphase` contract (see §4.2). Fields are sliced to `clog2(DFI_RATE)` bits downstream; upper bits are ignored when `DFI_RATE` is narrower than the field. Defaults 0/0 preserve the legacy all-on-phase-0 behavior.

Bits	Field	Reset	Description
2:0	`rd_phase`	0	READ command DFI sub-phase
6:4	`wr_phase`	0	WRITE command DFI sub-phase

7.6.2 PHY_TIMING (0x064, R/W)

PHY / DFI data timing plus memory-type selection. All fields `hw = r` so they drive the controller core.

Bits	Field	Reset	Description
7:0	`t_phy_wrlat`	0	WRITE command -> `dfi_wrdata_en` (Nexys A7 bring-up tuple programs 1)
15:8	`t_rddata_en`	6	RD command -> `dfi_rddata_en` window
16	`memtype`	0	0 = DDR2, 1 = LPDDR2 (drives the whole core `memtype_e` path)
23:20	`refresh_burst`	1	REFs drained per request (1..8)

`PHY_TIMING.memtype` is the runtime memory-type select. It is decoded into the `memtype_e` enum at the top and fanned out to the addr/scheduler/DFI/init logic; geometry (DFI_RATE, DRAM_BEAT_WIDTH, NUM_BANKS, ROW/COL width, BL) remains build-time — see §5.1.

7.7 Per-Bank Observation (0x080 – 0x09F)

Single-rank (rank 0), `NUM_BANKS = 8`. Each is a `regfile` of 8 word-wide registers striding by 4 bytes.

7.7.1 OBS_ROW_HIT[0..7] (0x080 + N×4, R only)

Per-bank ROW HITS: column ops issued to a bank whose row was **already open**. `VAL[31:0]`, `hw = w`, driven by `mc_page_policy` from the issued command stream. Free-running — they clear only on `aresetn`, so read two and subtract for a window, as with `PAGE_STATS_*`.

These are the sound row-hit count. Do **not** re-derive hits as `PAGE_STATS_HIT - SCHED_STATS_ACT`: that goes negative under a background-close mode (ISSUE-014).

Until 2026-09-28 these were `sw = rw` with `onread = rclr`, which could never have worked — the field is `hw = w` and driven every cycle, so the hardware rewrites whatever a read clears.

7.8 Calibration / Training (0x0A0 – 0x0FF)

LPDDR2 ZQ calibration and MRR DQ calibration, driven and harvested by `mc_training_plane` (Ch 3.9). Both FUBs are inert for DDR2 builds, so every register in this block reads its reset (or holds zero) when `memtype == DDR2`.

7.8.1 CAL_CTRL (0x0A0, R/W)

Bits	Field	Reset	Description
0	zq_en	0	Enable periodic ZQCS/ZQCL
1	zq_defer_en	0	Mode-C deferral: hold ZQ under demand (see Ch 3.9 — the ZQ FUB's `demand_i` is tied off in the current integration, so this is dormant until the sideband lands)
3:2	RSVD_3_2	0	Reserved
4	cal_start	0	Write 1 to start a DQ-cal sequence (self-clearing strobe)
5	cal_abort	0	Write 1 to abort the current DQ-cal sequence
15:6	RSVD_15_6	0	Reserved
28:16	zq_overdue_max	0	Max ZQ deferral cycles before overdue (0 = no cap)
31:29	RSVD_31_29	0	Reserved

7.8.2 CAL_ZQ_INTERVAL (0x0A4, R/W)

`VAL[31:0]`, reset 0. ZQCS interval in MC cycles; 0 disables periodic ZQ.

7.8.3 CAL_ZQ_TIMING (0x0A8, R/W)

Bits	Field	Reset	Description
15:0	t_zqcs	0	Post-grant bus-quiet hold for ZQCS (MC cycles)
31:16	t_zqcl	0	Post-grant bus-quiet hold for ZQCL (MC cycles)

7.8.4 CAL_TRAIN_TIMING (0x0AC, R/W)

Bits	Field	Reset	Description
15:0	t_mrr	0x2	MRR-to-MRR spacing, MC cycles (JEDEC tMRR = 2)
31:16	t_readout	0	MRR issue → captured-data timeout

Bits	Field	Reset	Description

7.8.5 CAL_MRR32_DATA[0..3] (0x0B0 + N×4, R only)

First captured DFI read-data beat for MRR MR32 (DQ-cal pattern A). Four 32-bit slices of the full DFI read width (128 bits at the default geometry); `hw = w`, harvested from `mc_lpddr2_dq_cal`. Held until the next `cal_start` run captures a new beat.

7.8.6 CAL_MRR40_DATA[0..3] (0x0E0 + N×4, R only)

First captured DFI read-data beat for MRR MR40 (DQ-cal pattern B). Same layout as `CAL_MRR32_DATA`. The array starts at 0x0E0 because the retired 0x0C0–0x0DC hole below is never reused.

7.8.7 CAL_STATUS (0x0F0, R only)

Bits	Field	Description
0	`zq_busy`	ZQ calibration busy (post-grant hold window)
1	`cal_busy`	DQ calibration busy (sequence in progress)
2	`cal_done`	DQ calibration done — sticky, cleared only by soft reset
3	`cal_err`	DQ calibration readout timeout — sticky, cleared by soft reset
4	`zq_overdue`	ZQ interval expired without grant (see the `demand_i` note above)
15:5	`RSVD_15_5`	Reserved
31:16	`zqcs_total`	ZQCS/ZQCL commands issued since reset

7.9 Retired observation registers (0x0C0 – 0x1E0)

Twenty-seven `OBS_*` registers were removed on 2026-09-28. They were in the register map, in this book and in the generated docs, and **nothing drove them**: `pumice_top` carried no `hwif_in.OBS_*` assignment, so every one read zero forever. A 99.4%-row-hit board workload reporting zero hits is what finally surfaced it (pumice BUG-020). Their addresses are left as holes so the rest of the map does not shift; the calibration registers above were allocated around them (0x0B0–0x0BC before the hole, 0x0E0 / 0x0F0 after it), never through it.

Range	Was	Why it went
0x0C0..0x0DC	`OBS_REF_LATENCY[8]`	never driven, no consumer. Refresh remains observable via `REF_STATS_REF` / `REF_STATS_REF_BUSY`, which are driven
0x100..0x104	`OBS_TXN_QUEUE_DEPTH_{MAX,AVG}`	AXI-side occupancy; belongs to an external meter
0x108..0x118	`OBS_REFRESH_PENDING_MAX`, `OBS_REFRESH_DEFER_HIST_0..3`	never driven, no consumer
0x120	`OBS_PAGE_PRED_ACCURACY`	retired earlier with the HAPPY predictor
0x130..0x138	`OBS_AXI_{R,W}_LATENCY_*`	AXI latency is a property of the interface and is measured by `axi_bus_meter` + `axi_perf_latency_hist`, which the characterisation harness already instantiates. Measuring it from inside the controller is the wrong side of the interface
0x1C0..0x1E0	`OBS_WORDS[9]`	the aggregator that packed them was deleted in e8908eebf (2026-07-21, “retire three dead code trees”); the window outlived its source by two months and nobody noticed, because nothing read it

`OBS_ROW_HIT[8]` at 0x080 was **kept and wired**: it is the only sound row-hit count, and ISSUE-014 exists because the arithmetic used in its absence is not.

7.10 Module Identification (0xFF0 – 0xFFC)

7.10.1 ID (0xFF0, R only, reset = 0xD2020001)

Bits	Field	Reset	Description
7:0	version	0x01	Build version
15:8	memtype	0x00	0 = DDR2, 1 = LPDDR2
23:16	n_phases	0x02	Gear ratio (1, 2, or 4)
31:24	module_id	0xD2	Fixed 0xD2

7.10.2 BUILD (0xFF4, R only)

Build hash word. VAL[31:0], reset = 0.

7.11 Notes on This Revision

- `ADDR_MAP_TUNING` (with `scheme_or` / `synth_mask_obs`) has been **retired** and replaced by `ADDR_MAP` at 0x04C. No scheme-selector field exists in RTL.
- `TIMINGS_RTP_RTW` (0x054), `INIT_TIMING0` (0x058), `INIT_TIMING1` (0x05C), `DFI_PHASE` (0x060), and `PHY_TIMING` (0x064) expose knobs that were formerly hardcoded in the FUBs; they are now runtime CSRs.
- `PHY_TIMING.memtype` and `PHY_TIMING.refresh_burst` are new runtime fields.
- The observation block is single-rank as generated; the multi-rank per-rank layout described in earlier revisions is not present in the current RDL.

8 Document Information

8.1 DDR2 / LPDDR2 Family Controller Hardware Architecture Specification

Document Number: DDR2-LPDDR2-HAS-001 **Version:** 1.2 **Status:** Draft - reconciled with rearchitected RTL **Classification:** Open Source - MIT License

8.2 Document Purpose

This Hardware Architecture Specification (HAS) provides a high-level architectural overview of the DDR2 / LPDDR2 unified memory controller family. It describes the system-level design, external interfaces, characterization parameters, and integration requirements without detailing internal implementation specifics.

Target Audience:

- System architects evaluating the controller for SoC integration
- Hardware engineers planning RTL implementation
- Verification engineers planning system-level testing
- Software engineers developing low-level memory configuration

Companion Documents:

- DDR2/LPDDR2 Family Controller Pre-Architecture Spec (`../pre-aspec.md`) — bullet-form architecture rationale
- DDR2/LPDDR2 Paging and Refresh Notes (`../paging-refresh-notes.md`) — research-derived parameter sources

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8.3 Revision History

Version	Date	Author	Notes
0.1	2026-06-13	RTL Design Sherpa	First sketch — module decomposition
0.2	2026-06-14	RTL Design Sherpa	Top-level interface port list (§2.4), FUB SWAG (§2.5), family-wide config-bit registry (§5.4), multi-rank support across §2 / §3.3 / §3.4 / §3.6 / §5.2 / §6.3
0.3	2026-07-07	RTL Design Sherpa	Narrow-device (x16) support: distinguish DRAM beat vs physical device word (`DRAM_DEVICE_WIDTH`); device-word column granularity + burst-length scaling; `DFI_PHASE` (rd/wr phase) CSR. Fixes on-silicon DDR2 read failure (Nexys A7 x16 bring-up).
0.4	2026-07-12	RTL Design Sherpa	Full reconciliation with the rearchitected RTL: three-plane core (`pumice_axi4_plane` / `pumice_scheduler_plane` / `pumice_dfi_plane`); FSM-free `bank_timer`; de-FSM'd CAMs; single `ADDR_MAP.bank_lsb` address knob (scheme selector retired); fully-functional LPDDR2 (bit-exact JESD209-2F CA + JEDEC MR init); host-width `pumice_top_geared` gearing; block/hierarchy diagrams regenerated.
0.5	2026-09-xx	RTL Design Sherpa	(No revision-history row was recorded when v0.5 was cut; its content is whatever the 2026-07-22 build captured. Noted rather than reconstructed.)
0.6	2026-09-26	RTL Design Sherpa	Paging modes 6/7 (`rbl_static`/`rbl_dyn`) RETIRED and the `pumice_rbl_table` FUB removed — measured on silicon on a workload built to suit them, the mechanism worked (thrash 100% -> 57.8%) and still lost: mode 6 -26% bandwidth, mode 7 bit-identical to plain open page. `PAGE_RBL_CFG` @ 0x07C left a documented HOLE, not reused. Newly documented: the bank timers' advisory lookahead (`safe_*_la_o`), the arbiter's in-flight shadow and

Version	Date	Author	Notes
			FINAL-STAGE timing authority, and the TASK-006 stall counters + `REF_STATS_REF_BUSY`, none of which appeared in any prior revision.
0.7	2026-09-26	RTL Design Sherpa	**Page-policy characterization; the default change is RECOMMENDED but BLOCKED (BUG-003).** Mode 3 (`fixed_open`) with `tr_init = 2` measured strictly dominant over open page on the board at txn_scale=1000: +41.2% `col_major_interleaved_bl4`, +8.6..11.6% `col_major`, exactly flat on `incremental`/`row_major`, no scenario regressing, zero integrity failures. The RDL resets were changed to mode 3 / TR=2 and REVERTED the same day: the component gate caught a read-return-ring assertion ("DFI return beat with NO ticket in flight") at `rd_gap >= 8`, proven by controlled A/B to be caused by the new reset – a short timeout fires inside a read's DFI return latency and precharges under an in-flight read (BUG-003, P1). The mechanism finding stands regardless: auto-precharge costs 4.9x the activations and 2x the read latency on identical traffic (160,006 vs 32,400 ACT) because it is uncancellable, whereas a background precharge gated on bank idle is implicitly cancellable. Mode 4 (`adapt_time`) documented as SUBSUMED by mode 3 – its TR decays to `tr_min` and it equals `fixed_open(tr_min)` exactly at three different floors – and `policy_scope=0`'s "per-bank TR" documented as a fiction (`r_mc` is one global counter). Mode 5 (`adapt_access`) documented as unproven and mis-plumbed. Two CSR traps now documented: `tr_init = 0` DISABLES the timeout, and `PAGE_ADAPT_CFG.check_interval = 0` gates the entire mode-4 adjustment off – which is why mode 4 measured inert on every campaign before

Version	Date	Author	Notes
			this one.
0.8	2026-09-27	RTL Design Sherpa	**BUG-003 FIXED and the adaptive paging modes REMOVED.** BUG-003 root-caused in `pumice_cmd_arbiter.sv`: `cmd_valid_o` was `r_pick_valid` with no `w_out_safe` term, so a pick the arbiter's own final safety gate had REJECTED was still pushed to the cmd FIFO and executed by the DRAM – while `w_fire_out` (which does carry the term) withheld `evt_*`, so the bank timers, the guard chains and the CAM retire all recorded a command the DRAM had already run. That produced BOTH an illegal column op to a just-closed bank AND, because the CAM entry was never retired, a second issue of the same read – the "extra read" chased for weeks as a DV read-return modelling defect. One line; `w_fire_out == cmd_valid_o && cmd_ready_i` now holds by construction. Measured at the scheduler plane on the board operating point with the fix reverted: 470 pushes, 10 with `w_out_safe` low, 9 illegal column ops, plus a leaked PRE in `static_close`; with the fix, 0 and 0. Paging modes 4 (`adapt_time`) and 5 (`adapt_access`) RETIRED with `pumice_row_pred_table.sv` and the `PAGE_ADAPT_CFG` register (TASK-014): mode 4 was measured to BE `fixed_open(tr_min)`, mode 5 drove auto-precharge at 4.9x the activations and was removed as a decision rather than a measurement. `policy_scope`, `tr_min`/`tr_max`/`tr_step` and the hybrid counter fields are RESERVED at their original bit positions; 0x078 is left a hole, and the generated register maps were diffed to prove it – 81 -> 80 registers, ZERO moved. Modes 4..7 fall through to the build default, a contract now regressed by

Version	Date	Author	Notes
			`test_page_predictor` and the scheduler matrix. New verification: 12 named DRAM operating points shared by the scheduler and top TBs; a JEDEC command-stream checker with armed-rule counts; a 12-config x 7-mode x 8-pattern scheduler matrix (672 arms, 1.38M requests, 3.15M commands checked); and the FIRST check of the DDR2 init sequence's JEDEC legality at any level (TASK-016, P1) – command order plus a differential proof that the sequencer honours its own wait registers.
0.9	2026-10-08	RTL Design Sherpa	**LPDDR2 training plane landed and documented (Ch 3.9).** The core gains its fourth plane, `pumice_training_plane` (`rtl/macro/pumice_training_plane.sv`), instantiated in `pumice_core` between the scheduler and DFI planes: `pumice_zq_ctrl` raises periodic ZQ calibration as maintenance traffic (LPDDR2 has no ZQCS/ZQCL command – both are MRW to MR10, OP = 0x56 / 0xAB, with Mode-C defer-under-demand and overdue-escalation-to-ZQCL in the FUB), and `pumice_lp_cal` runs the one-shot DQ-calibration sequence (digital drift compensation: MRR to MR32 pattern A, capture, tMRR, MRR to MR40 pattern B, capture, with readout timeout and sticky done/err). MRR rides the existing `OP_MRS` opcode as a new 1-bit `cmd_mrr` flag – `dram_op_e` is not widened – through a +1-bit scheduler cmd FIFO word `{mrr, ap, col, row, bank, rank, op}` to the formatter's MRR CA word (JESD209-2F §5.12, CA3r=H), and the DFI read aligner gains a cal-capture sideband (`cal_expect`/`cal_data`/`cal_valid`). The arbiter gains one maintenance client on a `trn_cmd_*` channel at priority init > refresh > trn > demand, granted only with all banks idle, nothing row-affecting in flight, and tRFC elapsed; the post-grant

Version	Date	Author	Notes
			hold (`cal_busy`) is folded into `w_out_safe`, never the pick cone, per the BUG-003 fire==valid discipline (`w_fire_out == cmd_valid_o && cmd_ready_i` by construction). New CSRs at 0x0A0-0x0F0 (`CAL_CTRL`, `CAL_ZQ_INTERVAL`, `CAL_ZQ_TIMING`, `CAL_TRAIN_TIMING`, `CAL_MRR32_DATA`, `CAL_MRR40_DATA`, `CAL_STATUS`); DDR2 builds inert (no ZQ pin, no MRR). Both FUBs plus the formatter/aligner/scheduler paths are FUB- and macro-tested; `pumice_core_tb_top` ties the training inputs off (datapath/paging TB, not calibration).
1.1	2026-10-20	RTL Design Sherpa	Block diagrams redrawn to the repo element key (v3): class fills, double-bordered registered blocks, green return paths and a generated key under each figure. No technical content changed.
1.2	2026-10-20	RTL Design Sherpa	Section (h2) headings now open a new page, the pipeline default. No technical content changed.

8.4 Scope of This Document

This HAS describes a single parameterized memory controller covering DDR2 and LPDDR2. It defines:

- External interfaces (AXI4 slave, DFI v2.1 master, APB CSR slave)
- Module-level decomposition and behavioral specification
- Build-time and run-time configuration parameters
- Initialization, power-state, and refresh policies
- Verification strategy and characterization plan

It does not define:

- Bit-level pin assignments (the SystemVerilog port list in `rtl/top/pumice_top.sv` is the canonical wire-level source)
- Detailed timing diagrams (deferred to v0.3)
- Verilog package skeletons or RTL stubs
- Floorplan or layout guidance

8.5 Out of Scope

The following features are explicitly out of scope for this version of the controller and will be addressed in higher-generation family controllers or future revisions:

- Inline ECC
- DFI training, frequency-change, and low-power sub-interfaces
- AXI exclusive access semantics
- Bank groups (not present in DDR2 / LPDDR2)
- Command/Address parity, CRC, Data Bus Inversion (not present in DDR2 / LPDDR2)

9 Purpose and Scope

9.1 Document Purpose

This Hardware Architecture Specification (HAS) defines the high-level architecture of the DDR2 / LPDDR2 Family Memory Controller. It serves as the primary reference for:

- **System Integration** — understanding the controller's external interfaces and system-level requirements
- **RTL Implementation Planning** — module decomposition, interface contracts, and parameter sweeps
- **Verification Planning** — system-level test scenarios and the characterization sweep matrix
- **Driver Development** — programming model and APB register interface

This document complements the upcoming Micro-Architecture Specification (MAS), which will provide detailed block-level implementation specifics, signal-level pinouts, and timing diagrams.

9.2 Scope

This HAS covers a single unified controller that supports both DDR2 and LPDDR2 memory. The memory type is a runtime CSR selection (`PHY_TIMING.memtype`: 0 = DDR2, 1 = LPDDR2). The controller is built as five planes under `pumice_core`, all of them common-ip macros instantiated directly (there is no pumice-local plane or wrapper):

- **`mc_axi4_plane`** — AXI4 slave host interface: dumb write/read intakes (each with an `mc_addr_mapper`), the write splitter and burst chopper, and the `mc_rd_return_ring` that returns read data in AR order.
- **`mc_storage_plane`** — the write-data CAM (`mc_wr_data_cam`: write buffer + read-your-write snarf source) and the read-command CAM (`mc_rd_cmd_cam`).
- **`mc_scheduler_plane`** — the command-scheduling plane: a single command arbiter (`mc_cmd_arbiter`, open-page decision inline), per-(rank,bank) FSM-free JEDEC "safe" timers (`mc_bank_timers` / `mc_bank_timer`), global turnaround timers (`mc_global_timers`: tFAW / tRRD / tWTR / tRTW / tCCD), and its default DDR2/LPDDR2 children: `mc_xddr2_refresh_ctrl`, `mc_xddr2_init_sequencer` (DDR2 and LPDDR2 JEDEC MR init), `mc_xddr2_mode_register` (CL / CWL / BL / AL decode) and `mc_page_policy` (runtime page policy).
- **`mc_training_plane`** — LPDDR2 maintenance clients `mc_lpddr2_zq_ctrl` (periodic ZQ calibration) and `mc_lpddr2_dq_cal` (MRR DQ calibration) behind a two-slot arbiter, plus the `mc_training_shell` calibration-data crossing. Inert for DDR2.
- **`mc_dfi_2p1_plane`** — a single controller-to-PHY clock crossing (`mc_dfi_cdc`, async FIFOs) plus the DFI-domain command path (`mc_dfi_cmd_path` around `mc_xddr2_cmd_formatter`, which drives the phase-packed DFI buses directly), write serializer, and read aligner, presenting the DFI v2.1 pin bus.

Configuration (timings, phases, page policy, address map, memtype) is delivered by name from a PeakRDL-generated CSR register block (`pumice_csr`) instantiated in `pumice_top`. An optional outer wrapper, `pumice_top_geared`, gives the host a free AXI data width by inserting the repository's formally-verified `axi4_dwidth_converter_wr` / `_rd` between a host-width slave and the fixed-width core.

The differences between DDR2 and LPDDR2 are the command encoding (DDR2 ras/cas/we vs the LPDDR2 10-bit CA bus, both in `mc_xddr2_cmd_formatter`), the init MR sequence in

`mc_xddr2_init_sequencer`, and a small number of mode-register decode differences. Both memtypes pass the full simulation suite; everything else is shared.

9.3 Design Philosophy

This controller is intended as a **characterization-first** design. Algorithmic choices that have meaningful research-backed alternatives — page policy, refresh policy, refresh deferral depth, address mapping — are exposed as CSR fields rather than hardcoded. The verification plan includes a benchmark sweep over representative AXI traffic patterns to pick defaults from data, not assumption.

The principal characterization knobs, all programmed by name through the CSR block, are:

- **Page policy** — static OPEN / CLOSE via `REFRESH_TUNING.page_policy_or`, plus the runtime page-policy engine (`mc_page_policy`, selected by `PAGE_POLICY_CFG.policy_mode`): `fixed_open` idle-timeout close. The adaptive modes are retired: `adapt_time` (the Ghasempour-2015 adapting-timeout scheme) and `adapt_access` (a per-row 2-bit close predictor) were removed 2026-09-27 — `adapt_time` measured as `fixed_open(tr_min)` and `adapt_access` drove auto-precharge at 4.9x the activations (TASK-014). The earlier HAPPY_HYBRID address-hash predictor had already been retired; it was never wired into the rearchitected core. The open-page decision itself is inline in `mc_cmd_arbiter`.
- **Refresh policy** (`REFRESH_TUNING.refpb_policy_or`: ROUND_ROBIN / OLDEST_FIRST / DARP) — where DARP is the dynamic access refresh parallelization scheme from Chang et al. (HPCA 2014).
- **Address map** (`ADDR_MAP.bank_lsb`, `hash_en`, `hash_seed`) — a single knob that slides the bank field within the word address, subsuming the classic ROW_MAJOR / BANK_INTERLEAVE schemes, with an optional bank XOR-hash.

These are described in detail in Chapter 5.

9.4 Audience

This document assumes the reader is familiar with:

- AXI4 protocol semantics (handshakes, burst types, ID-based ordering)
- DFI v2.1 specification (control / write-data / read-data sub-interfaces)
- JEDEC DDR2 (JESD79-2F) and LPDDR2 (JESD209-2F) command tables
- Basic memory-controller concepts (open vs closed page policy, refresh deadlines, bank state machines)

Readers new to these topics should first consult the companion `paging-refresh-notes.md` in the family working area for an entry-point bibliography.

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10 Document Conventions

10.1 Typographic Conventions

Style	Meaning
`monospace`	Module names, parameter names, signal names, register names
bold	Defined terms on first introduction; important warnings
italic	Emphasis; placeholder values to be replaced
`NUM_RANKS`	Top-level build-time parameter (all caps)
`bank_timer`	RTL module name (lowercase with underscores)
`tRC`, `tRCD`	JEDEC timing parameters (as in JESD79-2 / JESD209-2)

10.2 Numbering Conventions

- **Decimal** numbers have no prefix: `16`, `256`.
- **Hexadecimal** numbers are prefixed with `0x`: `0x800`, `0xFF`.
- **Binary** literals use Verilog notation when needed: `8'b1011_0010`.
- Bit ranges use the convention `[MSB:LSB]`, e.g., `dfi_address[19:0]`.

10.3 Signal Direction Conventions

For DFI signals, direction is from the **memory controller's** perspective:

- **MC → PHY** — driven by the controller
- **PHY → MC** — driven by the PHY (sampled by controller)

For AXI signals, direction follows the standard AXI4 convention with the controller as the **slave**.

10.4 Module Diagram Conventions

ASCII block diagrams in this document use the following conventions:

All diagrams in this document are rendered from Mermaid source files in `assets/mermaid/`. The Mermaid source files (`.mmd`) are linked alongside each rendered image so the diagrams can be edited and re-rendered.

- **Rectangular boxes with sharp corners** denote RTL modules.
- **Subgraph rectangles** group modules by functional plane.
- **Solid arrows** show primary data or control flow.
- **Dashed arrows** show configuration or observation paths (typically from the CSR slave to other modules).
- **Trapezoidal nodes** denote external interface ports (AXI, DFI, APB).

10.5 Cross-Reference Conventions

- References to other chapters: “see Chapter 4”
- References to specific sections: “see §4.1” or “see Section 4.1”
- References to other documents: “see `pre-aspec.md`” (relative path)
- References to external standards: “per JESD79-2F §x.y”

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11 Definitions and Acronyms

11.1 Acronyms

Acronym	Expansion
AHB	Advanced High-performance Bus (ARM AMBA)
APB	Advanced Peripheral Bus (ARM AMBA)
AP	Auto-Precharge (DRAM column command modifier)
APD	Active Power Down
AXI	Advanced eXtensible Interface (ARM AMBA)
AxLEN	AXI burst length — the `AWLEN` / `ARLEN` signals (beats = `AxLEN`+1). NOT the DRAM BL: one AXI burst at core width is `(AxLEN+1)/DFI_RATE` DFI words and splits into fixed-BL DRAM commands. Use “AxLEN” whenever burst length refers to the AXI side, to avoid confusion with the JEDEC BL below.
BL	Burst Length — JEDEC/DRAM burst (MR0 device beats: BL4/BL8). Distinct from AxLEN.
CA	Command/Address (LPDDR2/3/4 multiplexed bus)
CDC	Clock Domain Crossing
CL	CAS Latency
CMD	Command
CSR	Control/Status Register
CWL	CAS Write Latency
DARP	Dynamic Access Refresh Parallelization
DBI	Data Bus Inversion
DDR	Double Data Rate
DDR2	Double Data Rate 2 (JESD79-2)
DFI	DDR PHY Interface
DPD	Deep Power Down (LPDDR2)
FSM	Finite State Machine
FR-FCFS	First-Ready, First-Come-First-Served
HAPPY	Hybrid Address-based Page Policy in DRAMs (retired predictor; see

Acronym	Expansion
	`mc_page_policy`)
HAS	Hardware Architecture Specification
LPDDR2	Low Power DDR 2 (JESD209-2)
MAS	Micro-Architecture Specification
MC	Memory Controller
MR / MRS	Mode Register / Mode Register Set command
EMR / EMRS	Extended Mode Register / Extended MRS
NOP	No Operation
ODT	On-Die Termination
OOO	Out-of-Order
PASR	Partial Array Self-Refresh (LPDDR2 feature)
PHY	Physical Plane (DRAM electrical interface)
PRE	Precharge command
PREA	Precharge All Banks
RD	Read column command
RDA	Read with Auto-Precharge
REF	Refresh command
REFab	All-Bank Refresh (LPDDR2)
REFpb	Per-Bank Refresh (LPDDR2)
SDRAM	Synchronous Dynamic RAM
SoC	System on Chip
SR	Self-Refresh
TCSR	Temperature-Compensated Self-Refresh (LPDDR2)
WR	Write column command
WRA	Write with Auto-Precharge
ZQ	Impedance Calibration (ZQ Calibration)

11.2 Defined Terms

Bank Timer — A per-bank JEDEC "safe" timing tracker (`bank_timer`). It is *not* a finite state machine: it is a set of preset/decrement countdown timers (tRCD, tRAS, tRC, tRP, precharge-block) plus a small open-row register and a single auto-precharge bit. The per-command `safe_*` outputs are combinational off the timers (one register stage). `mc_bank_timers` stamps one instance per (rank, bank).

Characterization Knob — A build-time or runtime CSR field intentionally exposed for performance sweeping during system characterization. Page policy, refresh policy, and address map are the principal knobs in this controller.

Command Formatter — `mc_xddr2_cmd_formatter` (inside `mc_dfi_cmd_path`): the module that translates the controller's abstract command record into DFI wire signals. A single module with a memtype branch — DDR2 drives ras/cas/we; LPDDR2 packs the 10-bit CA bus (two edges) onto `dfi_address`. It drives the phase-packed DFI buses directly; there is no separate packing stage.

Command Arbiter — `mc_cmd_arbiter`: the single command-pick core. It scans CAM readiness against the bank and global timers, applies the page policy (the open-page decision is inline here), and emits one abstract DRAM command per cycle.

FR-FCFS — First-Ready, First-Come-First-Served. The baseline DRAM scheduling policy: ready commands beat unready, row-hits beat row-misses, older entries beat younger on ties. Established by Rixner et al. (ISCA 2000).

Gearing (two senses) — (1) The internal DFI-rate split: one AXI/DFI word is spread across `DFI_RATE` DRAM beats and packed to per-phase buses by the DFI plane (`mc_xddr2_cmd_formatter` for commands, `mc_dfi_wr_serializer` / `mc_dfi_rd_aligner` for data). (2) Host-width gearing: `pumice_top_geared` inserts formally-verified AXI data-width converters so the host AXI data width can differ from the core width.

Init Sequencer — `init_sequencer`: the cold-boot engine that issues the memtype-specific JEDEC MR/init sequence, driving MR loads and commands and holding off traffic until init completes.

Page Hit / Row Hit — An access whose target row matches the currently open row in the target bank. Avoids the tRP + tRCD penalty.

Page Conflict / Row Miss — An access whose target row differs from the currently open row in the target bank. Pays tRP + tRCD before the access can begin.

Page Policy — The strategy for when to close an open row. Static OPEN / CLOSE via `REFRESH_TUNING.page_policy_or`; the runtime idle-timeout mode (`fixed_open`) via `PAGE_POLICY_CFG.policy_mode` in `mc_page_policy`. The per-command decision is inline in `mc_cmd_arbiter`.

Command CAMs — The two content-addressable buffers that hold in-flight requests between the AXI interface and the scheduler: `mc_wr_data_cam` (write data buffer + snarf source) and

`mc_rd_cmd_cam` (read reorder buffer). Both store burst data in an SRAM with FIFO-fed / oldest-pick streaming read engines and no active-slot state latch.

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12 Scope and Goals

12.1 High-Level Goals

This controller provides a single parameterized RTL implementation supporting both DDR2 and LPDDR2 memory devices via DFI v2.1. Primary design goals:

1. **Unified engine, one command formatter with a memtype branch.** DDR2 and LPDDR2 differ only at the wire-encoding plane (ras/cas/we vs the 10-bit CA bus, both in `mc_xddr2_cmd_formatter`) and in the init MR sequence; the command arbiter, bank/global timers, refresh policy, init sequencer, address mapper, and AXI interface are shared between both targets. Memtype is a runtime CSR selection (`PHY_TIMING.memtype`).
2. **Embedded / mobile / mid-range SoC targets.** Not optimized for high-end server DRAM (DDR5, RDIMM, multi-rank). Optimized for power efficiency and area economy.
3. **Characterization-first parameterization.** Algorithmic choices with research-backed alternatives are exposed as build-time parameters so the design can be swept on real workloads before defaults are locked.
4. **Verification-friendly.** Mirrors the data structures and conventions of the DDR2 / LPDDR2 DFI BFM in the DV repository so the same address mapping, timing parameters, and command encodings can drive both RTL and simulation.

12.2 In Scope

The following features are explicitly within scope for this controller:

- **AXI4 slave** at the fixed core data width `DW = DRAM_BEAT_WIDTH * DFI_RATE`; an optional `pumice_top_geared` wrapper gives the host a free AXI data width (verified host ∈ {64, 128, 256}) via formally-verified data-width converters
- **DFI v2.1 master** driving the command, write-data, and read-data sub-interfaces
- **CSR register block** (PeakRDL-generated `pumice_csr`, passthrough cpuif) for configuration and observation, programmed by name
- **DDR2 and LPDDR2 device support** selected at runtime by the `PHY_TIMING.memtype` CSR field
- **DFI frequency ratio** of 1, 2, or 4 phases via the `DFI_RATE` parameter
- **Multi-rank operation** — 1, 2, or 4 ranks via `NUM_RANKS` parameter; per-rank `CS_n`, `CKE`, `ODT`, and refresh tracking; rank-aware ODT cross-termination rules per JEDEC
- **Page policies:** OPEN, CLOSE, plus the runtime page-policy engine (`fixed_open` idle-timeout; the adaptive modes retired 2026-09-27, TASK-014; CSR-selected via `PAGE_POLICY_CFG.policy_mode`; open-page decision inline in `pumice_cmd_arbiter`)
- **Per-bank refresh** for LPDDR2 with DARP-style scheduling (parameterized)
- **All-bank refresh** for DDR2 (and as LPDDR2 fallback)
- **Elastic refresh deferral** up to JEDEC ceiling of 8× tREFI (parameterized)
- **Partial-array self-refresh (PASR)** for LPDDR2 with CSR-configurable mask
- **Temperature-compensated refresh** (LPDDR2 MR4 derating)
- **Cold-boot initialization** via microprogram step tables
- **Power state management** (Active, Active Power-Down, Self-Refresh, Deep Power Down)
- **Bring-up observability** via CSR-exposed counters (row-hit rate, refresh latency, queue depth)

12.3 Out of Scope

Explicitly **not** addressed in this controller version:

Feature	Reason
Inline ECC	Out-of-controller responsibility (SoC sideband)
DFI training sub-interface	Not required for v2.1 DDR2 / LPDDR2 operation
DFI frequency-change protocol	No use case for this generation
DFI low-power sub-interface	Power-state FSM uses CKE only
AXI exclusive accesses	Embedded targets don’t require
Bank groups	Not present in DDR2 / LPDDR2 (introduced in DDR4 / LPDDR4)
Command/Address parity	Not present in DDR2 / LPDDR2 (introduced in DDR4 / LPDDR4)
Write CRC	Not present in DDR2 / LPDDR2
Data Bus Inversion (DBI)	Not present in DDR2 / LPDDR2
DDR3 write leveling	Not present in DDR2 / LPDDR2 (LPDDR3 introduces)
DDR3 ZQCS / ZQCL on column	DDR2 uses OCD calibration; LPDDR2 uses MR10 ZQ
Higher DDR / LPDDR generations	Will be addressed in DDR3-LPDDR3 and DDR4-LPDDR4 controllers

12.4 Target Performance Envelope

Metric	Target
DRAM data rate	Up to DDR2-800 / LPDDR2-1066
AXI clock	Up to 200 MHz (typical embedded SoC)
Sustained read bandwidth	≥ 70% peak on stream workloads
99th-percentile read latency	≤ 200 ns (typical)
Refresh blocking overhead	< 5% (LPDDR2 with DARP); < 8% (DDR2)
Area	< 50K gates

These are target envelopes for v1; not guarantees. They will be validated and refined during the characterization sweep.

12.5 Differentiating Features

This controller is informed by published memory-controller literature and the open-source memory-controller ecosystem (LiteDRAM in particular — used as the cross-validation reference; see §6.4), but it is not a re-implementation of any existing controller. The design's distinguishing features:

1. **Centralized FR-FCFS scheduler with full out-of-order issue.** A single scheduler scans the full transaction queue every cycle, considering all banks' readiness and row-hit state in one decision. This is closer to the commercial-vendor pattern (Synopsys, Cadence, Xilinx MIG) than to most open-source RTL controllers — open-source controllers typically expose OoO only at the cross-bank level (a per-bank FIFO model where bank parallelism is the only source of reordering). True per-port within-stream reordering — where a younger row-hit request bypasses an older row-conflict request — is uncommon outside vendor IP. AXI4 per-ID ordering is honored at the response side regardless. **For real-time / safety-critical / formally-verifiable systems**, OoO reordering can be disabled at runtime via the `SCHED_TUNING.force_inorder` CSR field (§3.2); the scheduler degrades to a first-ready FIFO that preserves arrival order at the cost of 10–25% sustained bandwidth on streaming workloads.

2. **Multi-rank support with JEDEC-compliant cross-rank ODT.** Most open-source DDR2/3 controllers are single-rank only — LiteDRAM's default config, UberDDR3, and most academic prototypes assume one rank per channel. This controller supports 1, 2, or 4 ranks via the `NUM_RANKS` build parameter. Per-rank `CS_n`, `CKE`, and `ODT` signals are driven; the rank-aware refresh manager tracks `last_ref_age` per (rank, bank); and the cross-rank ODT termination rules (read from rank N → ODT-high on rank ≠N during the read window; write to rank N → ODT-high on rank N during the write window) are handled inside `dfi_cmd_formatter` / `mode_register` (no standalone ODT block) per JESD79-2 §x.y. Multi-rank is what makes the controller usable on real DIMM-class platforms rather than just on-board point-to-point DRAM.

3. **Unified DDR2 + LPDDR2 engine with one command formatter.** A single controller core supports both memtypes via the runtime `PHY_TIMING.memtype` CSR field. The LPDDR2 path includes the 10-bit CA bus encoding (JESD209-2F Table 60; packed two edges onto `dfi_address` in `dfi_cmd_formatter`), PASR mask handling (MR16/17), per-bank refresh (REFpb), temperature-compensated refresh (MR4 derating), and Deep Power Down — none of which are present in the open-source DDR2/3/4 reference controllers. LPDDR2 is fully functional (reads and writes) and passes the full simulation suite.

4. **DARP-style per-bank refresh scheduling** — picks the *idle* bank with the longest time since last refresh, falling back to oldest-first when no banks are idle. Based on Chang et al. (HPCA 2014); not present in available open-source controllers.

5. **Two-stage page management** — exact lookahead first, history-based fallback second.

- Stage 1: a lookahead window (`SCHED_TUNING.lookahead_active`, 0 disables) provides exact auto-precharge decisions when the next same-bank request is already visible to the arbiter.
- Stage 2: when lookahead is inconclusive (shallow queue or bursty traffic), the adaptive-timeout page policy (Ghasempour et al. 2015) uses address-bit-hashed saturating counters to predict the next access. Open-source controllers typically have either lookahead OR a fixed page policy. The two-stage combination is original to this design.

6. **Characterization-first parameterization.** Page policy, lookahead depth, refresh policy, refresh deferral, and address mapping are all runtime CSR fields with a defined sweep matrix (see §5 and §6.4). Most published controllers expose only a subset of these; few expose them all.

7. **Closed-page policy as an explicit option.** `REFRESH_TUNING.page_policy_or ∈ {OPEN, CLOSE}` (the HYBRID encoding is retired; adaptive policies are `PAGE_POLICY_CFG.policy_mode`). Most open-source controllers expose only on/off auto-precharge (effectively OPEN with optional lookahead). The CLOSE option is useful for adversarial / security-sensitive workloads where row-state side channels matter.

8. **Single-knob address mapping.** `ADDR_MAP.bank_lsb` slides the bank field within the byte-offset-stripped word address; the column auto-splits below (`col_lo`) and above (`col_hi`) it while the row/rank positions stay invariant. The classic schemes are just settings of this one knob (`bank_lsb == COL_WIDTH` = ROW_MAJOR; `bank_lsb == log2(DRAM_BL)` = maximum bank interleave), and an optional bank XOR-hash (`ADDR_MAP.hash_en` / `hash_seed`) folds row bits into the bank index to defeat pathological row-conflict strides. There is no separate scheme selector.

9. **AXI4-first interface** with ID-aware out-of-order completion. The AXI integration is a first-class design constraint rather than a wrapper on a native protocol.

10. **Mirrored verification address-mapping object.** The `addr_mapper` mirrors a Python address-mapping class used by the DFI BFM, so the same mapping function drives both RTL and simulation. This eliminates a common source of address-decode bugs between RTL and verification.

12.5.1 Features Adopted from Standard Memory-Controller Patterns

These design patterns are widespread in the memory-controller literature and the open-source controllers we examined. Adopting them is the right call; they are noted here so the design rationale is transparent:

- **Per-bank FSM-free JEDEC “safe” timers** (`bank_timer`, stamped per (rank,bank) by `mc_bank_timers`) + a central cross-bank/rank timer pool (`mc_global_timers`: tRRD / tFAW / tWTR / tRTW / tCCD) — the standard bank/global split, here without a per-bank state machine.
- **FR-FCFS scheduler baseline** (Rixner et al. ISCA 2000) — the academic baseline; realized in `mc_cmd_arbiter`.
- **Elastic Refresh deferral** (Stuecheli et al. MICRO 2010) — postponer with deterministic back-to-back drain, in `refresh_ctrl`.
- **Refresh arbitrated against bank readiness** — the arbiter waits for the addressed banks to be safe before issuing REF; established pattern.
- **Periodic ZQCS piggybacked on the refresh window** — established pattern for systems with thermal drift.
- **Init sequencer** issuing the memtype-specific JEDEC MR/init sequence (`init_sequencer`) — established pattern.

The design rationale for adopting versus extending each is documented in the relevant section.

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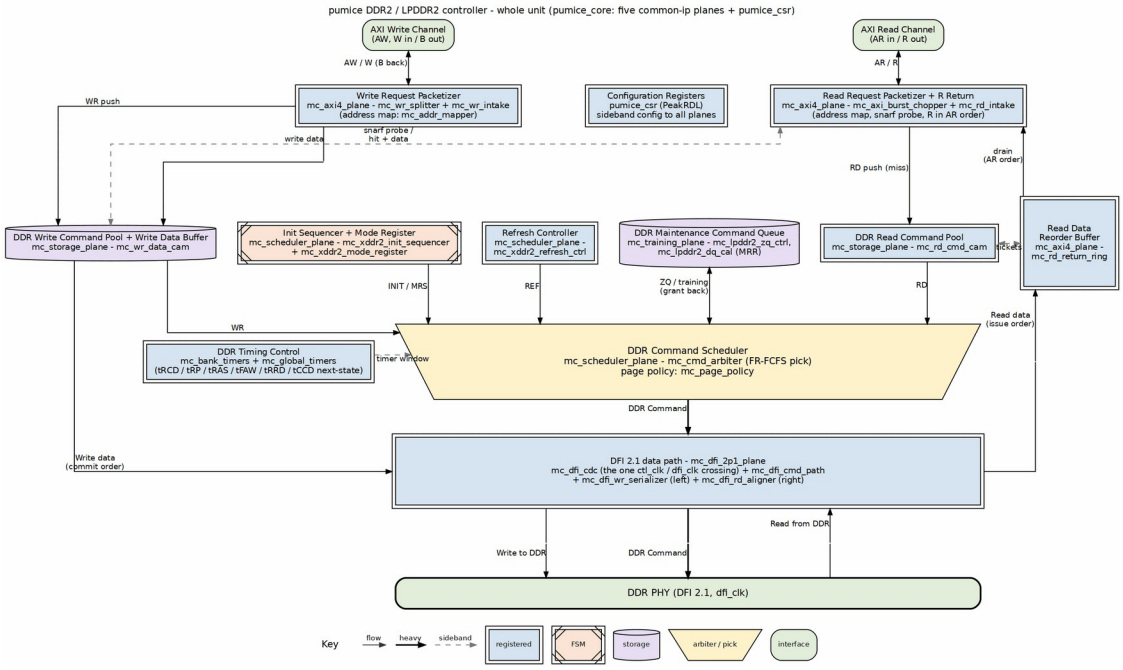
13 Block Diagram

13.1 Top-Level Block Diagram

The controller's top-level data and control flow is shown below. AXI4 traffic enters at top-left into `mc_axi4_plane` (common-ip), which accepts host transactions, maps addresses, and pushes per-direction descriptors to `mc_storage_plane` (common-ip). The storage plane holds the write-data and read-command CAMs and presents the same schedulable-entry view to the scheduler.

`pumice_core` instantiates the common `mc_scheduler_plane` directly, with its default children built in: `mc_xddr2_init_sequencer`, `mc_xddr2_mode_register`, `mc_xddr2_refresh_ctrl` and `mc_page_policy` around the `mc_cmd_arbiter` pick and the bank / global timers. It instantiates the common `mc_training_plane` with both LPDDR2 clients enabled: `mc_lpddr2_zq_ctrl` (periodic ZQ calibration, slot 0) and `mc_lpddr2_dq_cal` (MRR DQ calibration, slot 1) (`rtl/top/pumice_core.sv:601-611`, `:756-764`). There is no pumice-side scheduler or training wrapper any more. The `mc_dfi_2p1_plane` (common-ip) crosses the single controller-to-PHY clock boundary and formats the chosen command into DFI v2.1 wires, while its write serializer and read aligner move write beats out and return read beats in alignment with the scheduled commands. Returned read data lands in `mc_rd_return_ring` by ticket in issue order and drains to the AXI R channel in AR order. `pumice_top` instantiates `pumice_core` (these five planes) plus the PeakRDL `pumice_csr` block that drives all configuration by name.

The family architecture — plane contract, descriptor seam, and doctrines — is documented in common_has. Per-block microarchitecture for the common-ip planes is in common_mas; the pumice-specific wrappers and children are covered in the pumice MAS (scheduler macro, training macro).



PUMICE DDR2 / LPDDR2 controller whole-unit block diagram

13.2 Data Flow Summary

Write path: 1. AXI master issues an AW/W transaction; `mc_axi4_plane` accepts it and splits the host burst at DRAM-burst boundaries, producing mapped (rank, bank, row, col) descriptors. 2. `mc_storage_plane` fills the write burst into its write-data CAM SRAM and records the command; the fill-complete flag gates schedulability and snarf. 3. `mc_scheduler_plane` reads the CAMs' schedulable-entry views each cycle and picks the next abstract command (`mc_cmd_arbiter` against the bank and global timers, with `mc_page_policy` supplying the open-page decision). 4. In `mc_dfi_2p1_plane`, the command crosses the CDC and the command formatter encodes it into DFI cs_n / ras_n / cas_n / we_n / address / bank wires (DDR2) or the packed CA bus (LPDDR2); closed-page uses WRA (auto-precharge). 5. On the write-fire strobe, the DFI write serializer commit-drains the burst from the storage-plane write-data CAM and drives `dfi_wrdata` / `dfi_wrdata_en` / `dfi_wrdata_mask` with PHY alignment. 6. On retire, the write CAM slot is freed and `mc_axi4_plane` returns the B-response to the AXI master.

Read path: 1. AXI master issues an AR transaction; `mc_axi4_plane` accepts and splits the burst, producing the mapped descriptor. 2. `mc_storage_plane` records the read command. `mc_axi4_plane` probes the write-data CAM: on a snarf hit (unscheduled, same-id, same-BL) the read is served directly from the write-data CAM SRAM without going to DRAM (read-your-write forwarding). 3. `mc_scheduler_plane` issues ACT, then RD / RDA via `mc_dfi_2p1_plane`. 4. The DFI read aligner drives `dfi_rddata_en` `t_rddata_en` cycles after the RD command and captures `dfi_rddata` beats for the burst. 5. The read-command CAM entry was freed at issue; the read's ticket (a `mc_rd_return_ring` slot handed out at insert, in AR order) follows it. Returned beats land in the ticket's slot in issue order, and the ring drains its head in AR order into `mc_rd_intake`, which drives the R channel with the original AXI ID.

Refresh path: 1. `refresh_ctrl`'s tREFI counter elapses, incrementing the `refresh_pending` accumulator (JEDEC max 8 postponed). 2. When `refresh_pending` exceeds the soft threshold, refresh becomes highest-priority in the arbiter. 3. For REFab: the scheduler core waits for the addressed banks to be safe, then issues REF; the affected timers reload. 4. LPDDR2 per-bank refresh (REFpb) selection is driven by `REFRESH_TUNING.refpb_policy_or`.

Init / power path: 1. On cold reset, `init_sequencer` holds off AXI traffic. 2. `init_sequencer` executes the memtype-specific JEDEC MR/init sequence, driving CKE / RESET_N and issuing MR-write strobes into `mode_register`. `mode_register` propagates live CL / CWL / BL / AL to the DFI plane. 3. On completion, `init_done_o` asserts and the scheduler core begins servicing AXI traffic. 4. Power-down / self-refresh entry (Active / APD / SR / DPD) is not implemented; the optional `powerdown_ctrl` was removed on 2026-10-11 (Ch 3.5).

Post-Phase-2 note (2026-10-09). Until 2026-10-09 the pumice core used local `pumice_axi4_plane`, `pumice_dfi_plane`, and a monolithic `pumice_scheduler_plane`, with the CAMs grouped inside the AXI plane. The Phase-2 common-IP campaign extracted `mc_axi4_plane`, `mc_storage_plane`, `mc_scheduler_plane`, `mc_training_plane`, and `mc_dfi_2p1_plane` into `common-ip`. Pumice first kept thin rock wrappers (`pumice_scheduler_plane`, `pumice_training_plane`) for its generation-specific children;

those children have since moved into common-ip too (`mc_xddr2_*`, `mc_lpddr2_*`), and `pumice_core` now instantiates `mc_scheduler_plane` and `mc_training_plane` directly.

Detailed per-module behavior follows in Chapter 3.

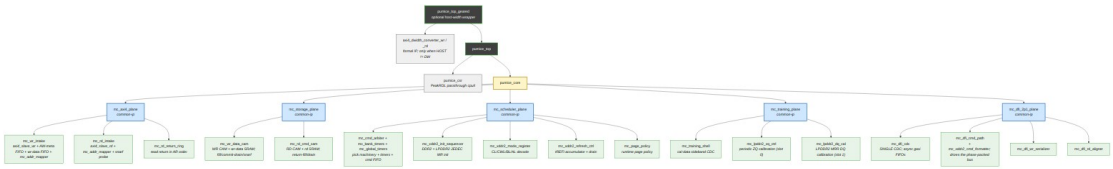
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14 Module Hierarchy

The controller is decomposed into **five planes under `pumice_core`**: `mc_axi4_plane`, `mc_storage_plane`, `mc_scheduler_plane`, `mc_training_plane`, and `mc_dfi_2p1_plane`. All five are shared common-ip planes that `pumice_core` instantiates directly; there is no pumice-local plane or wrapper. The pumice-specific choices are parameters: the scheduler plane builds its default DDR2/LPDDR2 children, and the training plane is built with both LPDDR2 clients enabled (`rtl/top/pumice_core.sv:601`, `:756`). Each FUB is independently verified at the unit level; each plane is verified as an integration ("macro") unit. See FUB Breakdown for the per-FUB role descriptions and Chapter 3 for the behavioral details.

14.1 Hierarchy Tree



Module Hierarchy

Source: 02_module_hierarchy.mmd

14.2 Plane Organization

```
SoC level
+-- pumice_top_geared                <- OPTIONAL host-width wrapper
    |                                   (axi4_dwidth_converter_wr/_rd
when HOST != DW)
    +-- pumice_top                   <- instantiated by the SoC
        +-- pumice_csr               <- PeakRDL passthrough cpuif
register block
        +-- pumice_core              <- the controller proper
            +-- mc_axi4_plane        <- common-ip; host AXI +
descriptor push
            |   +-- mc_wr_splitter, mc_axi_burst_chopper
            |   +-- mc_wr_intake, mc_rd_intake      (each with
mc_addr_mapper)
            |   +-- mc_rd_return_ring
            +-- mc_storage_plane     <- common-ip; wr/rd CAMs
            |   +-- mc_wr_data_cam
            |   +-- mc_rd_cmd_cam
            +-- mc_scheduler_plane   <- common-ip; default
DDR2/LPDDR2 children built
            |   +-- mc_xddr2_init_sequencer
            |   +-- mc_xddr2_mode_register
            |   +-- mc_xddr2_refresh_ctrl
            |   +-- mc_page_policy
            |   +-- mc_bank_timers    (one mc_bank_timer per
rank/bank)
            |   +-- mc_global_timers
            |   +-- mc_cmd_arbiter
            |   +-- gaxi_fifo_sync    (command FIFO to the DFI plane)
            +-- mc_training_plane    <- common-ip; HAS_LP_ZQ =
HAS_LP_DQ_CAL = 1
            |   +-- mc_lpddr2_zq_ctrl  (slot 0)
            |   +-- mc_lpddr2_dq_cal   (slot 1)
            |   +-- mc_training_shell  (cal-data sideband CDC)
            +-- mc_dfi_2p1_plane     <- common-ip; single CDC + DFI
v2.1 datapath
                +-- mc_dfi_cdc
                +-- mc_dfi_cmd_path
                |   +-- mc_xddr2_cmd_formatter
                |   +-- mc_cmd_history_checker  (only when
CMD_HISTORY_EN != 0)
                +-- mc_dfi_wr_serializer
                +-- mc_dfi_rd_aligner
```

The single controller-to-PHY clock crossing for the datapaths lives inside `mc_dfi_2p1_plane` (common-ip async FIFOs only). The training plane owns one small sideband crossing of its own (`cal_expect` down, captured MRR data up) for the read aligner's calibration capture — see

Training. The host AXI interface, CAMs, and command scheduler all run on `aclk`; the DFI command path, write serializer, and read aligner run on `dfi_clk`.

14.3 Plane Groupings

Plane / macro	Ownership	FUBs
mc_axi4_plane	common-ip	AXI4 slave intakes, shallow FIFOs, burst chop/split, final coalescing, read-data return endpoint (common_mas ch02)
mc_storage_plane	common-ip	wr/rd CAMs, read snooping, per-bank open-row image (common_mas ch03)
mc_scheduler_plane	common-ip	mc_cmd_arbiter, mc_bank_timers, mc_global_timers, command FIFO; default children mc_xddr2_init_sequencer, mc_xddr2_mode_register, mc_xddr2_refresh_ctrl, mc_page_policy (common_mas ch04)
mc_training_plane	common-ip	mc_lpddr2_zq_ctrl, mc_lpddr2_dq_cal, two-slot arbiter, mc_training_shell (common_mas ch05)
mc_dfi_2p1_plane	common-ip	single CDC, command formatter/pipeline, write serializer, read aligner (common_mas ch07)
pumice_core	pumice	wraps the five planes above

There are no optional pumice FUBs outside this tree: `page_predictor` and `powerdown_ctrl` were removed (see FUB Breakdown), and the pumice `rtl/` directory now holds only the three tops and `pumice_pkg`.

Chapter 3 follows the plane ordering (one section per architectural plane, with module names hyperlinked to their MAS chapters).

14.4 Memtype-Conditional Logic

Memtype (DDR2 vs LPDDR2) is a runtime CSR selection (`PHY_TIMING.memtype`), not an elaboration parameter. The memtype-dependent logic lives in:

- **`mc_dfi_2p1_plane`** — its command formatter contains a memtype branch: DDR2 drives ras/cas/we; LPDDR2 packs the 10-bit JESD209-2F CA-bus command (two edges) onto `dfi_address` (common_mas ch07).
- **`mc_xddr2_init_sequencer`** (child of `mc_scheduler_plane`) — runs the DDR2 or LPDDR2 JEDEC MR/init sequence.
- **`mc_xddr2_mode_register`** (child of `mc_scheduler_plane`) — CL / CWL / BL / AL decode differs by memtype.
- **`mc_training_plane`** — both LPDDR2 FUBs gate on `memtype == MEMTYPE_LPDDR2`; DDR2 has no ZQ pin and no MRR, so the plane is inert for DDR2 builds.

Both memtypes pass the full simulation suite.

14.5 Clock-Domain Crossings

The datapath clock-domain crossing is exactly one: the CDC inside `mc_dfi_2p1_plane`, built from asynchronous gaxi FIFOs. One FIFO word is one DFI cycle, so the command / write-data / read-data datapaths are bubble-free. Everything up to the CDC is on `aclk`; everything past it is on `dfi_clk`.

The training plane adds one narrow sideband crossing (`sync_pulse` / `cdc_synchronizer`, not FIFOs): `cal_expect` down to `dfi_clk` to arm the read aligner's calibration capture, and the captured MRR beat back up to `aclk`. It carries no datapath traffic — see Chapter 3.9.

14.6 What Changed vs the Earlier Architecture

The controller was rearchitected from an earlier FSM-based, elaboration-time decomposition into a three-plane, CSR-driven design. Until 2026-10-09 that design was implemented with pumice-local planes (`pumice_axi4_plane`, `pumice_scheduler_plane`, `pumice_dfi_plane`) and did not separate the CAM pool into a distinct storage plane. The 2026-10-09 Phase-2 common-IP campaign extracted the shared memory-controller logic into `common-ip` (`mc_axi4_plane`, `mc_storage_plane`, `mc_scheduler_plane`, `mc_training_plane`, and `mc_dfi_2p1_plane`). Pumice first retained thin rock wrappers (`pumice_scheduler_plane`, `pumice_training_plane`) for its generation-specific children; on 2026-10-11 those children moved into common-ip as well (`mc_xddr2_*`, `mc_lpddr2_*`, `mc_page_policy`) and the wrappers were removed. See the family architecture in common_has.

Notable changes across both transitions:

- **Retired**: `txn_queue`, `bank_machine`, `xbank_timers`, `cmd_encoder`, `odt_ctrl`, `scheme selector`, the `*_macro`-as-architecture names, `axi_intake`/`axi_frontend`, and the pumice-local plane names `pumice_axi4_plane` and `pumice_dfi_plane`, the rock wrappers `pumice_scheduler_plane` and `pumice_training_plane`, `dfi_signal_pack`, `page_predictor` and `powerdown_ctrl`.
- **Replaced by / moved to common-ip**: the two CAMs (`mc_wr_data_cam` + `mc_rd_cmd_cam`, now in `mc_storage_plane`), FSM-free `bank_timer` (stamped by `mc_scheduler_plane`), `mc_global_timers`, `dfi_cmd_formatter` (now `mc_xddr2_cmd_formatter` inside `mc_dfi_cmd_path`, which drives the phase-packed bus directly), and the single `ADDR_MAP.bank_lsb` knob.
- **Introduced**: the distinct `mc_storage_plane`, the common single-CDC `mc_dfi_2p1_plane`, the common `mc_training_plane` (with its `mc_training_shell` crossing), the PeakRDL `pumice_csr` block (configuration by name), and the optional `pumice_top_geared` host-width wrapper.

15 Top-Level Interfaces

This section is the integrator's quick-reference for the controller's top-level module port list. Detailed protocol semantics for each interface are in Chapter 4 (Interfaces); this is the **wire-level** view — names, directions, widths, parameter dependencies.

The top module is `pumice_top` and has four external interfaces:

1. AXI4 slave (host request path), signals prefixed `s_axi_*`
2. DFI v2.1 master (DRAM PHY path), signals suffixed `_o` / `_i`
3. CSR register cpuif (configuration and observation) — a PeakRDL passthrough bus, **not** APB
4. Clocks and resets (two domains) plus `init_done_o`

All four are exposed at the top level as flat per-signal ports. SystemVerilog interfaces are **not** used at the top boundary because the framework's interface convention (see CocoTBFramework BFM components) drives flat-port pin-level handshakes. Internal sub-blocks use interface objects, but the top-of-design is flat for synthesis and BFM compatibility.

An optional outer wrapper, `pumice_top_geared`, adds a `HOST_AXI_DATA_WIDTH` parameter and inserts formally-verified AXI data-width converters ahead of `pumice_top`; when `HOST_AXI_DATA_WIDTH == DW` the converters are bypassed and the build is bit-identical. This section describes `pumice_top` itself.

15.1 Parameter-Dependent Widths

The core data width is fixed by the DFI geometry: `DW = DRAM_BEAT_WIDTH * DFI_RATE`. There is no separate `AXI_DATA_WIDTH` parameter on `pumice_top` — the host AXI data width equals `DW`. Throughout this section:

- `AW` = `AXI_ADDR_WIDTH` (default 32)
- `DFI_RATE` = DFI phase count / frequency ratio (default 2)
- `DRAM_BEAT_WIDTH` = per-beat DRAM data width (default 64)
- `DW` = `DRAM_BEAT_WIDTH * DFI_RATE` (host AXI + DFI word width; default 128)
- `SW` = `DW/8` (byte-strobe width)
- `IW` = `AXI_ID_WIDTH` (default 8)
- `NR` = `NUM_RANKS` (default 1)
- `NB` = `NUM_BANKS` (default 8)
- `RW` = `ROW_WIDTH` (default 14)
- `CW` = `COL_WIDTH` (default 10)
- `BKW` = `$clog2(NUM_BANKS)`
- `CSR_ADDR_W` = CSR address width (default 12)

15.2 1. AXI4 Slave (Host-side)

All host AXI ports carry the `s_axi_` prefix.

15.2.1 Write Address Channel (AW)

Signal	Direction	Width	Description
`s_axi_awid`	input	`IW`	Write transaction ID
`s_axi_awaddr`	input	`AW`	Write start address
`s_axi_awlen`	input	8	Burst length – 1
`s_axi_awsize`	input	3	Beat size encoding
`s_axi_awburst`	input	2	Burst type (INCR mandatory; FIXED / WRAP optional)
`s_axi_awlock`	input	1	Lock (ignored — exclusives not supported)
`s_axi_awcache`	input	4	Cache hint (observed; no behavior)
`s_axi_awprot`	input	3	Protection bits (observed)
`s_axi_awqos`	input	4	QoS hint (boosts scheduler priority — see §3.2)
`s_axi_awregion`	input	4	Region (observed)
`s_axi_awuser`	input	1	User sideband (observed)
`s_axi_awvalid`	input	1	Address-valid
`s_axi_awready`	output	1	Address-ready

15.2.2 Write Data Channel (W)

Signal	Direction	Width	Description
`s_axi_wdata`	input	`DW`	Write data
`s_axi_wstrb`	input	`SW`	Byte enables
`s_axi_wlast`	input	1	Last beat of burst
`s_axi_wuser`	input	1	User sideband
`s_axi_wvalid`	input	1	Data-valid
`s_axi_wready`	output	1	Data-ready

15.2.3 Write Response Channel (B)

Signal	Direction	Width	Description
s_axi_bid	output	IW	Response ID
s_axi_bresp	output	2	Response code (OKAY / SLVERR)
s_axi_buser	output	1	User sideband
s_axi_bvalid	output	1	Response-valid
s_axi_bready	input	1	Response-ready

15.2.4 Read Address Channel (AR)

Signal	Direction	Width	Description
s_axi_arid	input	IW	Read transaction ID
s_axi_araddr	input	AW	Read start address
s_axi_arlen	input	8	Burst length – 1
s_axi_arsize	input	3	Beat size encoding
s_axi_arburst	input	2	Burst type
s_axi_arlock	input	1	Lock (ignored)
s_axi_arcache	input	4	Cache hint
s_axi_arprot	input	3	Protection
s_axi_arqos	input	4	QoS hint
s_axi_arregion	input	4	Region
s_axi_aruser	input	1	User sideband
s_axi_arvalid	input	1	Address-valid
s_axi_arready	output	1	Address-ready

15.2.5 Read Data Channel (R)

Signal	Direction	Width	Description
s_axi_rid	output	IW	Read data ID
s_axi_rdata	output	DW	Read data
s_axi_rresp	output	2	Response code
s_axi_rlast	output	1	Last beat
s_axi_ruser	output	1	User sideband
s_axi_rvalid	output	1	Data-valid

Signal	Direction	Width	Description
s_axi_rready	input	1	Data-ready

15.3 2. DFI v2.1 Master (PHY-side)

The DFI v2.1 sub-interfaces present on this controller are limited to **command**, **write-data**, **read-data**, and the **init handshake**. Training, frequency-change, low-power, update, and CRC sub-interfaces are not driven — see §2.1 Out of Scope.

The DFI ports are flat, per-phase-packed buses (the `DFI_RATE` phases are concatenated into one vector) rather than SystemVerilog arrays. All DFI ports carry the `_o` (output) or `_i` (input) suffix. Command formatting runs on `dfi_clk`, past the single CDC in `mc_dfi_2p1_plane`.

15.3.1 DFI Command Bus

Signal	Direction	Width	Description
`dfi_address_o`	output	`RW * DFI_RATE`	Address operand (row / column); LPDDR2 CA-bus command is packed here
`dfi_bank_o`	output	`BKW * DFI_RATE`	Bank operand
`dfi_cs_n_o`	output	`NR * DFI_RATE`	Active-low chip-select, per rank per phase
`dfi_ras_n_o`	output	`DFI_RATE`	RAS strobe per phase (DDR2; idle for LPDDR2)
`dfi_cas_n_o`	output	`DFI_RATE`	CAS strobe per phase (DDR2; idle for LPDDR2)
`dfi_we_n_o`	output	`DFI_RATE`	WE strobe per phase (DDR2; idle for LPDDR2)
`dfi_odt_o`	output	`NR * DFI_RATE`	ODT, per rank per phase

15.3.2 DFI Write-Data Bus

Signal	Direction	Width	Description
`dfi_wrdata_o`	output	`DW`	Write data (all phases packed; = one DFI word)
`dfi_wrdata_en_o`	output	`DFI_RATE`	Per-phase write-data enable
`dfi_wrdata_mask_o`	output	`SW`	Per-byte write mask (DFI polarity: 1 = do-not-write)

15.3.3 DFI Read-Data Bus

Signal	Direction	Width	Description
dfi_rddata_en_o	output	DFI_RATE	Per-phase read-data enable
dfi_rddata_i	input	DW	Read data (all phases packed)
dfi_rddata_valid_i	input	DFI_RATE	Per-phase read-data valid

15.3.4 DFI Init Handshake

Signal	Direction	Width	Description
dfi_init_start_o	output	1	Init handshake — start of DRAM init
dfi_init_complete_i	input	1	PHY signals init complete

15.4 3. CSR Register Interface (Configuration / Observation)

Configuration is driven by name from a PeakRDL-generated register block (`pumice_csr`) via a passthrough **cpuif** request bus (not APB). Address width is `CSR_ADDR_W` (default 12). Detailed register map in §6.3.

Signal	Direction	Width	Description
`s_cpuif_req`	input	1	Request valid
`s_cpuif_req_is_wr`	input	1	1 = write, 0 = read
`s_cpuif_addr`	input	`CSR_ADDR_W`	Register address
`s_cpuif_wr_data`	input	32	Write data
`s_cpuif_wr_biten`	input	32	Per-bit write enable
`s_cpuif_req_stall_wr`	output	1	Write back-pressure
`s_cpuif_req_stall_rd`	output	1	Read back-pressure
`s_cpuif_rd_ack`	output	1	Read data valid
`s_cpuif_rd_err`	output	1	Read error
`s_cpuif_rd_data`	output	32	Read data
`s_cpuif_wr_ack`	output	1	Write accepted
`s_cpuif_wr_err`	output	1	Write error

15.5 4. Clocks, Resets, and Status

The design has two clock domains — the controller/AXI domain (`aclk`) and the DFI/PHY domain (`dfi_clk`) — with the single crossing inside `mc_dfi_2p1_plane`.

Signal	Direction	Width	Description
`aclk`	input	1	Controller clock (host AXI, CAMs, scheduler)
`aresetn`	input	1	Controller reset, active low
`dfi_clk`	input	1	DFI/PHY clock
`dfi_rstn`	input	1	DFI reset, active low
`init_done_o`	output	1	Asserted when DRAM init completes

15.6 Hierarchical Top-Level View

The top-level pinout in shorthand:

```
pumice_top
    ├── AXI4 slave        (host)         s_axi_awid,
s_axi_awaddr, ..., s_axi_rvalid, s_axi_rready, ...
    ├── DFI v2.1 master   (PHY)          dfi_address_o, dfi_bank_o,
dfi_cs_n_o, ..., dfi_rddata_i, ...
    ├── CSR cpuif         (config)       s_cpuif_req,
s_cpuif_addr, ..., s_cpuif_rd_data, ...
    └── Clocks + status                  aclk, aresetn, dfi_clk,
dfi_rstn, init_done_o
```

The canonical SystemVerilog port list is `rtl/top/pumice_top.sv`. This section is the **integrator's overview**, not the build-time canonical list.

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16 AXI4 Frontend

The AXI4 frontend is the host-facing boundary of the pumice controller. In `pumice_core` it is instantiated as `mc_axi4_plane` from the common memory-controller stack, the first plane on the controller clock (`aclk`). It terminates the host AXI4 channels, chops and splits host bursts into DFI-burst-sized descriptors, and hands those descriptors across the descriptor seam to `mc_storage_plane`. The descriptor seam contract is documented in the family architecture specification (`common_has` chapter 2); the plane's internals are documented in the family microarchitecture specification (`common_mas` chapter 2).

Until 2026-10-09 this functionality lived in the pumice-local `pumice_axi4_plane` (`rtl/macro/pumice_axi4_plane.sv`), which was removed during the Phase-2 common-stack extraction.

16.1 What pumice consumes from the common stack

`mc_axi4_plane` provides:

- host AXI4 slave termination and burst chop/split (`mc_wr_splitter`, `mc_axi_burst_chopper`);
- write and read intakes (`mc_wr_intake`, `mc_rd_intake`) that push decoded descriptors into the storage plane;
- the read-return ring (`mc_rd_return_ring`) that decouples in-flight read count from CAM depth and enforces AXI per-ID ordering;
- the snarf probe path that lets reads hit unscheduled writes in the write-data CAM.

Pumice instantiates this plane with `DRAM_BEAT_WIDTH = DW` (the DFI word width) and `AXI_BEATS_PER_BURST = BURST_WORDS`, so one AXI beat equals one DFI word and the front end remains DFI-word granular. Sub-DFI-word burst expansion is handled downstream in `mc_dfi_2p1_plane`.

16.2 What pumice configures

16.2.1 Address mapping (`mc_addr_mapper`)

The flat-AXI-address to `{rank, bank, row, col}` decode is performed inside `mc_axi4_plane`: each intake (`mc_wr_intake`, `mc_rd_intake`) instantiates the common `mc_addr_mapper` (`common-ip/rtl/fub/mc_addr_mapper.sv`) and pushes the decoded coordinates across the descriptor seam. There is no pumice-local mapper. What pumice owns is the configuration: the mapping knobs are runtime CSRs in the `ADDR_MAP` register:

Field	Width	Role
`ADDR_MAP.bank_lsb`	5	Bank-field LSB in the byte-offset-stripped word address
`ADDR_MAP.hash_en`	1	Enable bank XOR-hash
`ADDR_MAP.hash_seed`	8	XOR-hash seed

16.2.1.1 Mapping function (single knob: `bank_lsb`)

There is **no scheme selector** anymore. The mapping is driven by one runtime knob, `ADDR_MAP.bank_lsb`, which places the bank field within the byte-offset-stripped word address (`word = axi_addr >> BYTE_OFFSET_WIDTH`). The column auto-splits around the bank; row and rank stack above the column region and their positions are **invariant**:

```
word address, low -> high:
[ col_lo(bank_lsb) | bank(BW) | col_hi(CW - bank_lsb) | row(RW) |
rank(KW) ]
col = { col_hi, col_lo }
row LSB is always at CW + BW (invariant — only the bank slides)
```

The RTL clamps `bank_lsb` to `[0, COL_WIDTH]` so the `col_hi` width stays non-negative and the row/rank slices land where the geometry expects.

16.2.1.2 Classic schemes as `bank_lsb` settings

Effect	Setting	Notes
`ROW_MAJOR`	`bank_lsb == COL_WIDTH`	Bank above the whole column; default (`0x0A`)
max `BANK_INTERLEAVE`	`bank_lsb == log2(DRAM_BL)`	Minimal `col_lo` keeps a DRAM burst inside one bank
partial	any value in between	Column splits around the bank

Effect	Setting	Notes
interleave		

Software keeps `log2(DRAM_BL) <= bank_lsb <= COL_WIDTH` so a DRAM burst's column walk stays inside one bank. (`DRAM_BL`, not `cols/burst` and not `BL/DFI_RATE` — the mapper indexes device words, so a burst spans `DRAM_BL` columns here; measured in `dv/tests/fub/test_addr_mapper.py::minimum_bank_lsb_is_measured`. pumice BUG-022.)

16.2.1.3 Optional bank XOR-hash

When `hash_en` is set, the bank index is XOR-folded with row bits and the seed to defeat power-of-two-stride hot-banking (the former `XOR_HASH` scheme, now an orthogonal overlay on any `bank_lsb` placement):

```
bank[i] ^= row[i] ^ row[i + BW] ^ hash_seed[i]
```

The mid-slice index is clamped so it never runs past `ROW_WIDTH`.

16.2.2 AXI gearing wrapper (`pumice_top_geared`)

`pumice_top_geared` (`rtl/top/pumice_top_geared.sv`) is a pumice-specific wrapper around `pumice_top` that allows the host AXI data width to differ from the controller's fixed DFI-word width. It inserts the repository's formally-verified AXI data-width converters (`axi4_dwidth_converter_wr` / `axi4_dwidth_converter_rd`) between a host-width AXI slave and the `pumice_top` AXI interface. When `HOST_AXI_DATA_WIDTH == DW` the converters are bypassed by generate so the build is bit-identical to `pumice_top`. The wrapper enforces a compile-time power-of-two ratio between host width and DFI word width.

16.2.3 Observed status registers

`pumice_top` captures controller-level status into the PeakRDL-generated CSR. AXI-frontend-relevant observable registers include:

- `STATUS.init_done` — initialization complete;
- `ADDR_MAP.bank_lsb`, `ADDR_MAP.hash_en`, `ADDR_MAP.hash_seed` — address-map controls.

The per-bank/page statistics, stall counters, and training-plane status are produced by downstream planes and are described in the data-paths, scheduler, and training chapters.

17 Command Scheduler

The command-scheduling plane is the common `mc_scheduler_plane` (`common-ip/rtl/macro/mc_scheduler_plane.sv`), instantiated directly by `pumice_core` as `u_sched_plane` (`rtl/top/pumice_core.sv:601`). It runs on the single controller clock (`aclk`), builds its default DDR2/LPDDR2 bring-up and policy children, and emits one abstract DRAM command stream `{op, rank, bank, row, col, ap, mrr}` into the scheduler-to-DFI command interface.

Until 2026-10-09 this block was a pumice-local plane that contained `pumice_cmd_arbiter`, `pumice_bank_timers`, and `global_timers`; the pick machinery and timing FUBs were extracted into the common core during the Phase-2 common-IP campaign, and a thin `pumice_scheduler_plane` wrapper kept the DDR2/LPDDR2 children. On 2026-10-11 those children moved into common-ip as well and the wrapper was removed. Pumice now selects its scheduler purely by parameter.

The plane does not hold the transaction queue. The schedulable-entry vectors (valid, bank, row, col, age, oldest, qos) and the commit/issue handshakes are external because the CAMs live in `mc_storage_plane`. The family plane-stack and descriptor-seam contract are in the family HAS (Plane Stack, Descriptor Seam). The scheduler plane itself is documented in the family MAS (`mc_scheduler_plane`); this chapter only describes how pumice configures and connects it, and the measured history that remains pumice-owned.

17.1 Composition

`pumice_core` leaves the `HAS_*` child selectors at their defaults, so `mc_scheduler_plane` builds:

- `mc_xddr2_init_sequencer` (`HAS_INIT_SEQUENCER = 1`) — JEDEC MRS cold-boot init; gates traffic until done (see `05_init_power.md`).
- `mc_xddr2_mode_register` (`HAS_MODE_REGISTER = 1`) — CL / CWL / BL decode shadow, updated by init MRS writes.
- `mc_xddr2_refresh_ctrl` (`HAS_REFRESH_CTRL = 1`) — tREFI tracking + postpone/pull-in accumulator (see `04_refresh.md`).
- `mc_page_policy` (`HAS_PAGE_POLICY = 1`) — runtime auto-precharge engine.
- `mc_bank_timers`, `mc_global_timers` and `mc_cmd_arbiter` — the pick machinery and JEDEC timers.
- `gaxi_fifo_sync` — the scheduler-to-DFI command FIFO.

`HAS_CMD_HISTORY` is left at 0, so the scheduler-side `mc_cmd_history_checker` is not built in pumice; the `CMD_HISTORY_EN` knob arms the DFI-wire checker inside `mc_dfi_cmd_path` instead (Ch 3.6). Because the children are built in, `pumice_core` ties the plane's external init / refresh / page-policy override inputs off (`init_done_i = 1`, `refresh_req_i = 0`, `ap_mode_en_i = 0`, …); they exist for products that supply their own children. The timing and scheduler CSR inputs come straight from `pumice_top`'s CSR block.

17.2 Mapping the training channel onto the generic maintenance slot

The scheduler plane exposes one generic maintenance slot: `maint_req_i`/`maint_grant_o`, `maint_op_i`, `maint_bank_i`, `maint_row_i`, `maint_col_i`, `maint_extra_i`, `maint_busy_i`. `pumice_core` wires the `mc_training_plane` command channel onto that slot:

Training plane signal	Scheduler maintenance slot
`trn_cmd_req_o`	`maint_req_i`
`trn_cmd_grant_i`	`maint_grant_o`
`trn_cmd_op_o`	`maint_op_i`
`trn_cmd_bank_o`	`maint_bank_i`
`trn_cmd_row_o`	`maint_row_i`
(tied `'0`)	`maint_col_i`
`trn_cmd_mrr_o`	`maint_extra_i[0]`
`cal_busy_o`	`maint_busy_i`

The single-bit `mrr` flag is carried in `maint_extra_i[0]` and recovered on the command output as `cmd_extra_o`, which `pumice_core` packs into the command word as `w_cmd_mrr`. No training logic is implemented inside the scheduler plane; the slot is a request/grant pipe that `mc_training_plane` drives (see `09_training.md`).

17.3 DDR2/LPDDR2 children and CSR surface

mc_xddr2_init_sequencer, mc_xddr2_mode_register, and mc_xddr2_refresh_ctrl are the DDR2/LPDDR2-generation children (xddr2); other family products disable them with the HAS_* parameters and supply their own. mc_page_policy is the runtime page-policy engine; it consumes the page_mode_i/page_tr_init_i CSRs and produces the auto-precharge controls consumed by mc_cmd_arbiter.

The static page-policy override page_policy_i (OPEN / CLOSE / HAPPY_HYBRID) is still accepted at the plane's top level; the arbiter honors it when the runtime engine is disabled. The HAPPY_HYBRID enum encoding and the standalone page_predictor.sv are retired; the encoding maps to the build default. The adaptive modes adapt_time and adapt_access were RETIRED 2026-09-27, TASK-014.

17.4 Measured history preserved in the common pick machinery

Pumice does not alter the common `mc_cmd_arbiter` pick core. The following measured bug-history sentences from the merge inventory (mc_scheduler_knobs.md §6.4) are preserved verbatim because they govern the fire-stage authority that pumice relies on:

- **pumice BUG-021**: dropping an ACT whose only obstacle was the rank-global tFAW/tRRD window regressed close-page throughput from the accepted 30.77% calibration to 28.57%; holding recovered to 30.14%.
- **ISSUE-002**: moving tFAW/tRRD to the classify mask drained the 3-stage pipeline and dropped close-page throughput from 30.77% to ~18%.

The common MAS chapter lists the remaining fire-stage invariants (scoria BUG-001, BUG-002 / tZQCS, the one-cycle blind spot, and the parked-victim PRE guard) and their line-level RTL evidence.

17.5 v1 Scope Notes

The scheduler plane is built with `pumice_core`'s `NUM_RANKS` (default 1); multi-rank pick remains a future exercise. Write-batch watermarking is passed through the `sched_wr_high_wm_i`/`sched_wr_batch_max_i`/`sched_wr_low_wm_i` CSRs into the plane. Power-down coordination is not implemented; it is documented as a TODO in `05_init_power.md`.

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18 Bank Timers and Cross-Bank Timers

Per-bank JEDEC timing is tracked by an **FSM-free** `mc_bank_timer` per (rank, bank); cross-bank / bus-turnaround timing is tracked by `mc_global_timers`. There is no multi-state bank machine in this design.

18.1 mc_bank_timer

18.1.1 Purpose

Track one DRAM bank's JEDEC "safe" timing. RTL: `common-ip/rtl/fub/mc_bank_timer.sv`. It is **not a state machine** — it is a set of preset/decrement countdown timers plus a trivial row-open register and a single auto-precharge bit. The arbiter drives one `set_*` strobe per issued command; each timer loads its config value on its trigger, free-runs decrementing (saturating at 0), and reports its constraint as "safe" when the count is 0.

Because the per-command `safe_*` outputs are a purely combinational AND of the relevant timers behind a **single register stage** (the counter), a just-issued command is reflected one cycle later with no multi-stage lag. This was the point of retiring the old 3-state FSM, whose double-registered state was the root cause of the refresh-vs-ACT and column-vs-PRE hazards.

18.1.2 Instantiation

`mc_bank_timers` (`common-ip/rtl/fub/mc_bank_timers.sv`, inside `mc_scheduler_plane`) stamps one `mc_bank_timer` per `(rank, bank)` pair — for the default (`NUM_RANKS=1`, `NUM_BANKS=8`) that is 8 instances. It routes the arbiter's `evt_*` event strobes to the addressed bank and fans the per-bank readiness back out to the arbiter.

18.1.3 Timers

Each timer loads on its event, counts down, and saturates at 0:

Timer	JEDEC	Loaded on	Gates
`r_rcd`	tRCD	ACT	`safe_rd`/`safe_wr`
`r_ras`	tRAS	ACT	`safe_pre`
`r_rc`	tRC	ACT	`safe_act`
`r_rp`	tRP	explicit PRE or auto-PRE fire	`safe_act`
`r_preblk`	tRTP (RD) / tWR (WR)	RD / WR	`safe_pre` only

The timer config inputs (`t_rcd_i`, `t_rp_i`, `t_ras_i`, `t_rc_i`, `t_wr_i`, `t_rtp_i`) are controller-cycle, command-relative values supplied by the scheduler plane from CSRs.

18.1.4 Row State (minimal — not an FSM)

Two elements track row occupancy:

- `r_row_valid` + `r_open_row` — set on ACT (captures `row_i`), cleared on an explicit PRE or when an auto-precharge completes.

- `r_ap_pending` — a single bit set from the `set_ap_i` qualifier on a RD/WR. When both `r_preblk` and `r_ras` reach 0, the auto-precharge "fires" (`w_ap_fire`): the row closes internally and tRP loads — with no scheduler PRE and no extra states.

18.1.5 Combinational Safe Outputs

The readiness outputs are combinational off the timers (single register stage behind):

- `safe_act_o = !r_row_valid && (r_rp == 0) && (r_rc == 0)` — bank precharged, tRP since PRE, tRC since last ACT.
- `safe_rd_o = r_row_valid && (r_rcd == 0) && !r_ap_pending` — row open, tRCD met, not auto-precharging.
- `safe_wr_o = safe_rd_o` — same condition.
- `safe_pre_o = r_row_valid && (r_ras == 0) && (r_preblk == 0) && !r_ap_pending` — row open, tRAS and tRTP/tWR met, not auto-precharging.

Open-page behavior falls out naturally: RD/WR only load `r_preblk` and (optionally) the auto-PRE bit, so column commands stream to an open row. The tCCD / tWTR / tRTW turnaround constraints are enforced globally in `mc_global_timers` and ANDed by the arbiter, not here.

18.1.6 Observability State

For debug only, `state_o` is derived combinationally from the timers into a `bank_state_e` value (`BANK_IDLE`, `BANK_ACTIVATING`, `BANK_ACTIVE`, `BANK_PRECHARGING`). No downstream logic depends on it — it is purely an observation output alongside the `obs_*_nz` timer-nonzero flags.

18.1.7 Refresh Interaction

There is no per-bank refresh handshake. The arbiter serializes refresh: on a refresh request it precharges active banks one per cycle (using `safe_pre` / `bank_row_active`), then issues `REF` once no bank has an open row AND `w_ref_safe` holds (no in-flight/guarded row op; prior tRFC recovery elapsed — see the scheduler chapter). REF recovery (tRFC) is an arbiter-side down-counter; the `bank_timer` never sees REF events. The `bank_timer` participates only through its standard row-open and `safe_pre` outputs.

18.2 mc_global_timers

18.2.1 Purpose

Enforce the cross-bank and shared-bus turnaround constraints that a single bank cannot see locally. RTL: `common-ip/rtl/fub/mc_global_timers.sv`. Implemented once (not per bank) because these constraints are inherently global.

18.2.2 Tracked Constraints

Constraint	Description	Scope
`tFAW`	At most 4 ACTs in any rolling tFAW window	per-rank
`tRRD`	Minimum gap between any two ACT commands	per-rank
`tWTR`	Write-to-read data-bus turnaround	global
`tRTW`	Read-to-write data-bus turnaround	global
`tCCD`	Column-to-column spacing	global

18.2.3 Interface

The timers consume the arbiter's `evt_act` (with `evt_act_rank`), `evt_rd`, and `evt_wr` strobes and produce the readiness window signals the arbiter ANDs into its pick:

- `tfaw_window_ok_o` / `trrd_window_ok_o` — per-rank vectors, gate `ACT`.
- `twtr_global_ok_o` / `trtw_window_ok_o` / `tccd_window_ok_o` — gate column commands.

tFAW / tRRD are kept per-rank because they are device-local activate-stagger limits; the data-bus turnaround windows (tWTR / tRTW / tCCD) are global because the DQ bus is shared across ranks.

All five are registered from the **next** state of their counters rather than the current one, so the arbiter may AND them into its pick directly and every JEDEC window holds. They were previously flopped from the current state, which left each gate open for one cycle past the command that should have closed it – see pumice ISSUE-018, and `formal/mc_common/mc_global_timers` for the proof that the published outputs are now sufficient on their own.

19 Refresh Controller

`mc_xddr2_refresh_ctrl` (`common-ip/rtl/fub/mc_xddr2_refresh_ctrl.sv`, a default child of `mc_scheduler_plane`) owns tREFI timing and refresh request accounting. It does not issue commands itself — it raises a request to the command arbiter, which serializes the precharge-then-REF sequence and pulses a grant once REF reaches the DFI bus.

19.1 refresh_ctrl

19.1.1 Purpose

Track the interval between mandatory refreshes and tell the scheduler when refresh is owed. JEDEC permits up to 8 postponed refreshes; `refresh_ctrl` uses an 8-deep accumulator plus a drain-quota mechanism so a batch of owed refreshes can be granted back-to-back.

19.1.2 Interface

Signal	Direction	Purpose
`t_refi_i[15:0]`	input	Refresh interval in MC cycles (CSR-backed)
`refresh_burst_i[3:0]`	input	1..8 REFs to drain per request cycle (CSR-backed)
`refpb_mode_i`	input	0 = REFab, 1 = REFpb (LPDDR2) bank-rotor select
`enable_i`	input	Gates tREFI counting; driven by `init_done`
`refresh_req_o`	output	High while pending refreshes remain
`refresh_grant_i`	input	Pulsed by the arbiter when REF issues on the bus
`refresh_drain_active_o`	output	High during a back-to-back drain burst
`refresh_kind_o`	output	Registered REFab/REFpb selector
`refresh_bank_o`	output	Bank rotor value (valid in REFpb mode)
`pending_refreshes_o`	output	Current accumulator value
`obs_*`	output	Internal state harvested for CSR readout

19.1.3 tREFI Counter and Pending Accumulator

`r_refi_cnt` counts down from `t_refi_i`. It only ticks while `enable_i` is high (i.e. after init completes); before that it is held reloaded at `t_refi_i`. On expiry it reloads and the pending accumulator `r_pending` increments by 1 (saturating at the JEDEC maximum of 8). Each accepted grant decrements `r_pending`. A simultaneous expiry and grant is a net-zero change. `refresh_req_o` stays high whenever `r_pending > 0`.

Saturating at 8 is JEDEC-conformant: the controller is expected to keep at most 8 postponed refreshes outstanding before it must catch up. If the system stays bandwidth-starved past 8 x tREFI, the counter saturates (a DRAM retention violation looming) rather than growing unbounded.

19.1.4 Drain Quota

`r_burst_remaining` implements back-to-back draining. When the previous burst is fully drained and pending work exists, it (re)loads `min(refresh_burst_i, r_pending)`. Each grant decrements it. While `r_burst_remaining > 0` and `r_pending > 0`, `refresh_drain_active_o` is asserted, which the arbiter reads as “keep granting REF back-to-back without yielding to reads/writes”. Setting `refresh_burst_i = 1` disables batching (one REF per tREFI).

19.1.5 REFpb Bank Rotor

When `refpb_mode_i` is set (LPDDR2 per-bank refresh), `r_bank_rotor` advances 0..`NUM_BANKS-1` on each grant and is exposed on `refresh_bank_o`. In REFab mode it stays at 0. The current default build wires `refpb_mode_i = 0` (REFab) in the scheduler.

19.2 Arbiter-Side Refresh Sequence

`refresh_ctrl` only raises `refresh_req` / `refresh_drain`. The precharge-then-REF sequence is performed by `mc_cmd_arbiter` at refresh priority (second only to init):

1. While any bank on the target rank has an open row, precharge the active banks one per cycle (lowest ready bank first, honoring the ACT/PRE guard).
2. Once no bank has an open row **and `w_ref_safe` holds** — nothing row-affecting in flight or inside its 2-cycle guard window (the registered bank view alone is 2-3 cycles stale, which once let a REFab collide with a just-issued ACT), and the previous REF's tRFC recovery elapsed — issue `OP_REF` and assert `refresh_grant_o` to `refresh_ctrl`, decrementing the accumulator / drain quota. On each fired REF the arbiter loads a **tRFC down-counter** from `TIMINGS_RFC_REFI.tRFC` (`t_rfc_i`); while non-zero, ACT picks and further REFs are blocked (mission-mode refresh recovery — init-time refreshes wait `INIT_TIMING1.t_rfc_wait` separately).
3. Repeat back-to-back while `refresh_drain_active` is high (each REF spaced by tRFC), until the accumulator is drained.

This "wait until all banks are precharged" behavior is implicit in the arbiter's readiness gating rather than an explicit bank-grant handshake.

19.3 Notes and Scope

- **Self-refresh** / **power-down** are not implemented; the earlier optional `powerdown_ctrl` block was removed (see `05_init_power.md`).
- **PASR, DARP idle-bank-first selection, temperature-compensated tREFI, and periodic ZQCS** are not part of the current `refresh_ctrl` RTL. REFpb selection is a simple bank rotor; the richer selection policies remain future work.
- **REFab vs REFpb** for LPDDR2 is selectable via `refpb_mode_i` and the bank rotor, but the default build uses REFab.

---

20 Init Sequencer and Power-Down

Cold-boot bring-up is handled by `mc_xddr2_init_sequencer`, a default child of the common `mc_scheduler_plane` (Ch 3.2). The controller has no power-down block: the earlier optional `powerdown_ctrl` was removed on 2026-10-11 (see the last section).

20.1 init_sequencer

20.1.1 Purpose

Sequence post-reset DRAM bring-up: the DFI init handshake, JEDEC mode-register loads, precharge, and refresh. RTL: `common-ip/rtl/fub/mc_xddr2_init_sequencer.sv`. It is a compact hard-coded FSM (not a microprogram ROM). Its commands are **issued to the DRAM**, not merely shadowed — this was the fix for the on-board bring-up failure where an earlier shadow-only walk left the read DLL unlocked.

20.1.2 Command Path

While `init_busy_o` is high, the scheduler stays idle and forwards the sequencer's single-cycle command pulses (`init_cmd_valid_o` / `init_cmd_op_o` / `init_cmd_bank_o` / `init_cmd_row_o`) straight to the command formatter (`mc_xddr2_cmd_formatter`). Each command state is occupied for exactly one cycle, then the FSM parks in `S_WAIT` for the JEDEC inter-command delay before advancing. Because the formatter is always `cmd_ready`, a single-cycle pulse issues exactly one command with no grant handshake.

The `mode_register` shadow is updated in lockstep (`mr_seq_we_o` / `mr_seq_index_o` / `mr_seq_data_o`) so the live CL/CWL/BL decode tracks what was programmed.

20.1.3 CSR-Backed Waits

The inter-command delays are CSR-backed (INIT_TIMING registers), zero-extended into a 16-bit countdown, rather than hardcoded:

Input	JEDEC	Default
`t_init_wait_i`	CKE / tINIT	512
`t_dll_wait_i`	DLL lock tDLLK	256
`t_mrd_wait_i`	tMRD	8
`t_rp_wait_i`	tRP	8
`t_rfc_wait_i`	tRFC	8

20.1.4 DDR2 Sequence

Mirrors LiteDRAM's DDR2 PHY init (the reference proven on the Nexys A7 board):

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i` (the PHY runs its own DLL-lock / IO training), then wait tINIT.
2. Precharge All.
3. EMRS in JEDEC order EMRS(2) then EMRS(3) then EMRS(1) — all defaults 0.

4. MRS(0) + DLL reset (`MR0.VAL | 0x100`; reset default `0x533` = BL8/CL3/tWR3/DLL_RESET); wait tDLLK.
5. Precharge All.
6. Auto Refresh x2 (each followed by tRFC).
7. MRS(0) without DLL reset (`MR0.VAL`, reset default `0x433`) — clears the reset bit.
8. EMRS(1) + OCD default (`0x380`) then EMRS(1) + OCD exit (`0x000`).
9. `init_done_o = 1`.

The DDR2 MR values are CSR-backed (`MR0..MR3.VAL`; MR0 reset `0x433`, MR1 `0x0000`, MR2/MR3 `0x0000`) with the OCD default (`0x380`) ORed in by the sequencer; software may retune them before an `init_restart`.

20.1.5 LPDDR2 Sequence (fully functional)

Per JESD209-2F power-up and mode-register configuration:

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i`; tINIT settle.
2. MRW(MR63) = Reset (OP don't-care).
3. MRW(MR10) = `0xFF` ZQ Init Calibration.
4. MRW(MR1) = `0x23` (BL8 / nWR3), MRW(MR2) = `0x01` (RL3 / WL1), MRW(MR3) = `0x02` (DS 40 ohm).
5. `init_done_o = 1`.

Only MR1/MR2/MR3 update the CL/CWL/BL decode shadow; MR63/MR10 are issued to the DRAM but not shadowed (they exceed the 5-bit shadow index). The MR index (MA, up to MR63) and data (OP) are carried packed as `{MA[5:0], OP[7:0]}` in the ROW request field, so `mc_xddr2_cmd_formatter` can build the full LPDDR2 CA MRW word — a 3-bit bank port alone could not reach MR10/MR63.

20.1.6 Memtype Selection

`memtype_i` (from the PHY_TIMING CSR: 0 = DDR2, 1 = LPDDR2) chooses the sequence at runtime. After the shared DFI-init step, DDR2 branches to `S_PREA1` and LPDDR2 to `S_L_RESET`.

20.1.7 Status Outputs

Signal	Purpose
`dfi_init_start_o`	High after leaving reset
`init_busy_o`	High until the FSM reaches `S_DONE`
`init_done_o`	Asserted at `S_DONE`

Signal	Purpose
zqcl_req_o	Tied off — DDR2 has no ZQCL

20.2 Power-down (removed)

The optional `powerdown_ctrl` FUB (idle-detect into precharge power-down or self-refresh) was never wired into the default `pumice_core` build and was removed on 2026-10-11 with no replacement block. No module in the current hierarchy drives CKE low for power-down or issues SRE/SRX; low-power entry is a scheduler maintenance concern still to be designed.

20.2.1 Scope / TODO

Precharge power-down, self-refresh, Deep Power Down (LPDDR2), per-rank power-down, and `dfi_dram_clk_disable` control are all open TODOs. When they land they belong in the scheduler plane's maintenance path, not in a standalone FUB.

21 Command Formatter and Command Path

Two common-ip modules in the DFI plane translate the scheduler's abstract command into DFI wire signals: `mc_xddr2_cmd_formatter` (memtype-specific encoding and phase placement, with flopped outputs) and `mc_dfi_cmd_path`, which pops the command FIFO, instantiates the formatter (one per packed sub-command), and drives the phase-packed bus to the PHY directly. There is no separate output-packing stage; the earlier `dfi_signal_pack` register stage was removed on 2026-10-11.

21.1 mc_xddr2_cmd_formatter

21.1.1 Purpose

Translate the scheduler's chosen `dram_op_e` plus `(rank, bank, row, col)` into the multi-phase DFI v2.1 control bus. RTL: `common-ip/rtl/fub/mc_xddr2_cmd_formatter.sv`. It is the only place where DDR2 and LPDDR2 differ at the wire plane, and its outputs are strict-flopped.

21.1.2 Input

The command arrives on `cmd_valid_i` / `cmd_ready_o` (always ready) with `cmd_op_i` (`dram_op_e`), `cmd_rank_i`, `cmd_bank_i`, `cmd_row_i`, `cmd_col_i`, `cmd_len_i`, `cmd_mrr_i` (MRR vs MRW select on the LPDDR2 `OP_MRS` CA word), and the runtime phase-placement knobs `rd_phase_i` / `wr_phase_i`.

21.1.3 Multi-Phase Output

For `DFI_RATE = N`, every control signal is packed as N phases side by side on the bus (`dfi_address_o`, `dfi_bank_o`, `dfi_cas_n_o`, `dfi_ras_n_o`, `dfi_we_n_o`, `dfi_cs_n_o`, `dfi_odt_o`). The decoded command is placed on its target phase and the other phases emit selected/deselected NOP. The target phase is:

- `rd_phase_i` for `OP_RD` / `OP_RDA`,
- `wr_phase_i` for `OP_WR` / `OP_WRA`,
- phase 0 for everything else (ACT/PRE/REF/MRS have no data-phase contract).

The R/W phase knobs match the PHY's rdphase/wrphase contract (e.g. a7ddrphy DDR2 CL3 nphases=2). Defaults of 0/0 preserve the legacy "everything on phase 0" behavior. `cs_n` is per-rank: `cs_n[r]=0` selects rank `r`; all-ones is a deselected NOP.

21.1.4 DDR2 Encoding

Combinational truth table per JESD79-2 Table 49:

op	cs_n	ras_n	cas_n	we_n	bank	addr
NOP	sel	1	1	1	-	-
ACT	sel	0	1	1	BA	row
RD	sel	1	0	1	BA	col (A10=0)
RDA	sel	1	0	1	BA	col + (1<<10)
WR	sel	1	0	0	BA	col

op	cs_n	ras_n	cas_n	we_n	bank	addr
WRA	sel	1	0	0	BA	col + (1<<10)
PRE	sel	0	1	0	BA	A10=0
PREA	sel	0	1	0	-	(1<<10)
REF	sel	0	0	1	-	-
MRS	sel	0	0	0	MR	MR data

The auto-precharge / all-bank bit is A10. MRS data rides `cmd_row_i` (ROW_WIDTH), not `cmd_col_i` — MR0 = `0x533` needs bit 10, which a 10-bit column field would truncate.

21.1.5 LPDDR2 Encoding (bit-exact CA bus)

For LPDDR2 the command rides the multiplexed 10-bit CA bus over 2 edges, packed as a flat 20-bit word carried on `dfi_address` (low bits); `ras_n`/`cas_n`/`we_n` stay idle and `cs_n` asserts for the target rank. The two CA edges are already inside the word, so there is no per-DFI-phase command placement.

The CA word is built **bit-exact to JESD209-2F Table 60**, matching the DV BFM's `lpddr_ca` encoder. Layout: `w_lpddr2_ca[i] = CA{i}` rising edge (i = 0..9), `w_lpddr2_ca[10+i] = CA{i}` falling edge. Encoded commands include ACT, RD/RDA, WR/WRA, PRE, PREA, REF (all-bank), REFPB (per-bank), MRW, and MRR; NOP/Deselect drives CA0r..CA3r high. Column bit C0 is implied 0 and never transmitted; the auto-precharge flag lands on CA0f. The transcription reference is `docs/uarch/LPDDR2_CA_ENCODING.md`.

For MRW, `MA0..MA5` map to `CA4r..CA9r`, `MA6/MA7` to `CA0f/CA1f`, and `OP0..OP7` to `CA2f..CA9f`. The init sequencer supplies the full MR index by packing `{MA[5:0], OP[7:0]}` into the ROW field, so MR10/MR63 are reachable.

MRR rides the same `OP_MRS` branch as a mode: `cmd_mrr_i` drives CA3r (`1` for MRR, `0` for MRW), so MRR is MRW with CA3r inverted, bit-exact to JESD209-2F §5.12 — the MA select field packing is unchanged and the OP field is don't-care-zero for MRR. LPDDR2 has no ZQCS/ZQCL opcodes, so those stay on the default NOP branch: ZQ calibration is pure MRW (see `09_training.md`).

> Note: the module's top-of-file header comment still carries a stale "LPDDR2 (TODO)" line from an earlier revision. The body implements the full bit-exact CA encoding above; LPDDR2 reads and writes are functional and pass the sim suite.

21.1.6 Idle Defaults

When not issuing, every phase is driven to its deselected idle: `cs_n=1, ras_n=1, cas_n=1, we_n=1, bank=0, address=0, odt=0` (DFI v2.1 Table 2 defaults). ODT is driven from this module — there is no standalone `odt_ctrl` block.

21.2 mc_dfi_cmd_path

21.2.1 Purpose

The DFI-clock command path. RTL: `common-ip/rtl/fub/mc_dfi_cmd_path.sv`. It pops the abstract command stream from the CDC command FIFO, unpacks `{op, rank, bank, row, col, ap, mrr}`, drives the formatter, and emits the `wr_fire` / `rd_fire` strobes that the write serializer and read aligner use to place their data phases.

21.2.2 Behavior

- The formatter's outputs are already registered, so the command path adds no pipeline stage of its own. When one DFI word packs `N_SUBCMD > 1` JEDEC bursts, the path instantiates one formatter per sub-command and merges their phase buses combinationally (address/bank OR, active-low controls AND) before the PHY.
- Bus widths are `DFI_*_WIDTH * DFI_RATE`, so the multi-phase content leaves the formatter in its final layout. The internal `DW = DRAM_BEAT_WIDTH * DFI_RATE` write and read words are split across phases by `mc_dfi_wr_serializer` and `mc_dfi_rd_aligner`.
- `dfi_dram_clk_disable` is not driven by the controller; clock-disable and power-down are documented TODOs (Ch 3.5).
- With `CMD_HISTORY_EN != 0` the path also builds `mc_cmd_history_checker` on the DFI wires (simulation only).

21.2.3 Multi-Cycle Commands

LPDDR2's 2-edge CA command is not split here. The 20-bit CA word is already packed onto `dfi_address` by `mc_xddr2_cmd_formatter`; the PHY splits it into two DRAM cycles per DFI v2.1.

21.2.4 Verification

The RTL output round-trips against the DFI BFM: the DDR2 truth table against the JESD79-2 reference, and the LPDDR2 CA bus against the shared `lpddr_ca` encoder/decoder (the same layout the RTL encodes).

22 Write and Read Data Paths

The write/read data path in pumice is the common storage plane, `mc_storage_plane`, instantiated in `pumice_core` between `mc_axi4_plane` and the scheduler. It holds the write-data CAM (write buffer plus snarf source) and the read-command CAM (read scheduling window), presents the scheduler-facing bank/row/col image, and accepts the read-snoop probe from the upstream AXI4 plane. The descriptor seam it exposes to the scheduler is the same contract documented in the family architecture specification (`common_has` chapter 2); the plane's internals are documented in the family microarchitecture specification (`common_mas` chapter 3).

Until 2026-10-09 these CAMs lived in the pumice-local `pumice_axi4_plane` as `pumice_wr_data_cam` and `pumice_rd_cmd_cam`; they were extracted into `mc_storage_plane` during the Phase-2 common-stack campaign.

22.1 What pumice consumes from the common stack

`mc_storage_plane` provides:

- `mc_wr_data_cam` — the write-command/write-data CAM with fill, commit-drain, and snarf (read-your-write) movers;
- `mc_rd_cmd_cam` — the read-command scheduling window;
- read snooping (snarf) against unscheduled writes;
- per-entry scheduling vectors (validity, bank, row, column, age, age-order matrix, QoS) across the descriptor seam.

Pumice instantiates the plane with `AXI_BEATS_PER_BURST = BURST_WORDS` and wires it to the pumice-local scheduler. Legacy `oldest_*` and `sched_lu_*` CAM ports are tied off inside `mc_storage_plane`.

22.2 What remains pumice-specific

22.2.1 Observed status registers

`pumice_top` captures storage-plane-adjacent telemetry from the scheduler and training plane into the CSR. Observable registers relevant to the data path include:

- `PAGE_STATS_HIT` — every column command issued (READ/WRITE, with or without AP); this is a lower-bound hit count, not a true hit count (do not derive hits as `PAGE_STATS_HIT - SCHED_STATS_ACT`; see pumice ISSUE-014);
- `PAGE_STATS_MISS`, `PAGE_STATS_EMPTY` — page-state counters;
- `OBS_ROW_HIT[0..7]` — per-bank row-hit counts (the only sound row-hit count; pumice BUG-020);
- `SCHED_STATS_ACT`, `SCHED_STATS_PRE` — activate/precharge counts;
- `REF_STATS_REF`, `REF_STATS_REF_BUSY` — refresh commands issued and refresh commands that deferred useful work (TASK-012);
- `STALL_BP`, `STALL_REFRESH`, `STALL_TURNAROUND`, `STALL_TCCD`, `STALL_ACTLIMIT`, `STALL_BANKTIMER`, `STALL_NOREQ` — scheduler stall attribution.

These counters are free-running and clear only on `aresetn`; software brackets a workload and reads deltas.

22.2.2 DFI-side data movement

The controller-side CAMs are only half of the end-to-end data path. The DFI-side write serialization and read alignment are delegated to the common `mc_dfi_2p1_plane` (common_mas chapter 7); the pumice-local `mc_dfi_wr_serializer` and `mc_dfi_rd_aligner` blocks are no longer instantiated in the active build. Pumice passes runtime PHY timing offsets (`t_phy_wrlat`, `t_rddata_en`) and DFI phase selects (`rd_phase`, `wr_phase`) from the `PHY_TIMING` and `DFI_PHASE` CSRs into that plane.

23 DFI Plane and CSR

The bottom of the controller is the DFI v2.1 plane. Until 2026-10-09 this plane was pumice-local (`pumice_dfi_plane`); it is now the common `mc_dfi_2p1_plane` consumed from `common-ip`. `pumice_top` instantiates it and wires the scheduler command stream, write-data stream, read-data return, and init tokens across the single controller-to-PHY clock-domain crossing. The family architecture and generation split are documented in the family HAS (Plane Stack, DFI Generations). The block microarchitecture is in the family MAS (`mc_dfi_2p1_plane`); this chapter only describes the pumice integration and the pumice-owned CSR surface.

23.1 mc_dfi_2p1_plane in pumice

23.1.1 Purpose

Present the controller-clock command / write-data / read-data streams on one side and the DFI v2.1 pin bus on the other, with exactly one clock-domain crossing between them. RTL in the build is `common-ip/rtl/macro/mc_dfi_2p1_plane.sv`. The controller runs on `ctl_clk`; the PHY-facing datapath runs on `dfi_clk`.

23.1.2 The Single CDC

`mc_dfi_cdc` is the only clock-domain crossing in the whole controller. It is built from asynchronous `gaxi` FIFOs — three streams cross it:

- **cmd** (`ctl_clk -> dfi_clk`): the abstract command stream from the scheduler.
- **wrdata** (`ctl_clk -> dfi_clk`): DFI-word-granular `{last, strb, data}` drained from the write CAM.
- **rddata** (`dfi_clk -> ctl_clk`): `{last, resp, data}` captured from the PHY.
- plus the init handshake (`init_start` across, `init_complete` back).

The internal datapath unit is the DFI word (`dfi_wrdata` width, all `DFI_RATE` phases), so one FIFO word equals one DFI cycle and the crossing is bubble-free. Keeping the CDC in one place is what makes the rest of the controller a single-clock design.

23.1.3 DFI-Domain Datapath

On the `dfi_clk` side the common plane instantiates three mechanical stages:

- `mc_dfi_cmd_path` — pops the command FIFO and drives the multi-phase DFI command bus (see `06_encoder_output.md`). It emits `wr_fire` / `rd_fire` strobes to time the data movers.
- `mc_dfi_wr_serializer` — drives `dfi_wrdata` at `t_phy_wrlat` (see `07_data_paths.md`).
- `mc_dfi_rd_aligner` — captures `dfi_rddata` at `t_rddata_en` (see `07_data_paths.md`).

The command path performs **no JEDEC timing enforcement**; all delays come from the scheduler. This policy followed pumice BUG-017 (was `PUMICE-042`): holding columns for `tRTW`/`tWTR` in the DFI path compressed REF→ACT spacing in the in-order FIFO and caused 180 consecutive reads to return an undriven DQ bus.

23.1.4 DFI Sub-Interfaces

The plane drives the three core DFI v2.1 sub-interfaces needed for DDR2 / LPDDR2 operation: the command bus, the write-data interface (`wrdata` / `wrdata_en` / `wrdata_mask`), and the read-data interface (`rddata_en` / `rddata` / `rddata_valid`), plus the DFI init handshake. Update / training / low-power / status sub-interfaces beyond `init_complete` are not driven; the controller manages power via CKE directly.

23.1.5 Runtime PHY Knobs

Rather than hard-coded PHY latencies, the plane takes runtime inputs so one build ports across PHYs: `rd_phase_i` / `wr_phase_i` (per-command DFI sub-phase placement), `t_phy_wrlat_i`, `t_rddata_en_i`, plus `memtype_i`. These are driven from the CSR (see below). The board-validated Nexys A7 tuple (2026-07-21) is `rd_phase=0`, `t_rddata_en=6`, `t_phy_wrlat=1`, plus harness `DFI_TUNING.rddata_delay=7` (see Runtime CSR Fields and the Register Map).

23.2 pumice_csr

23.2.1 Purpose

Hold all runtime configuration and observation registers. The register block is **generated by PeakRDL** from `rtl/macro/pumice_csr.rdl` (via `bin/peakrdl_generate.py`), with a **passthrough cpuif** — it is not a hand-written APB3 slave.

23.2.2 Interface

The generated block exposes a simple request/response passthrough cpuif rather than a full bus protocol:

Signal	Direction	Purpose
`s_cpuif_req`	input	Request valid
`s_cpuif_req_is_wr`	input	Write (1) vs read (0)
`s_cpuif_addr`	input	Register address
`s_cpuif_wr_data[31:0]`	input	Write data
`s_cpuif_wr_biten[31:0]`	input	Per-bit write enable
`s_cpuif_rd_data[31:0]`	output	Read data
`s_cpuif_*_ack` / `*_err` / `*_stall_*`	output	Response / flow-control

Registers are `regwidth = 32` / `accesswidth = 32`. `pumice_top` connects the cpuif and fans the decoded `hwif_out` struct fields to the core; the core's status feeds back through `hwif_in`. Any external bus (APB, AXI-Lite) is adapted to this cpuif outside the block.

23.2.3 Access by Name

DV never uses hard-coded offsets. Configuration is programmed by field name through the generated `dv/tbclasses/pumice_regmap.py`, which stays in sync with the RDL. This is split-proof — if a register moves, the by-name access still resolves.

23.2.4 Key DFI / PHY Configuration Registers

The RDL is authoritative; the registers that drive the DFI plane and PHY integration are:

Register	Offset	Notable fields
`DFI_PHASE`	0x060	`rd_phase[2:0]`, `wr_phase[6:4]` — DFI sub-phase placement (default 0/0)

Register	Offset	Notable fields
PHY_TIMING	0x064	t_phy_wrlat[7:0] (reset 0; Nexys A7 tuple programs 1), t_rddata_en[15:8] (reset 6), memtype[16] (0=DDR2, 1=LPDDR2), refresh_burst[23:20] (1..8)
CAL_ZQ_TIMING	0x0A8	t_zqcs[15:0] / t_zqcl[31:16] — ZQ post-grant holds consumed by the training plane (see 09_training.md)
CAL_TRAIN_TIMING	0x0AC	t_mrr[15:0], t_readout[31:16] — MRR spacing and readout timeout

memtype selects DDR2 vs LPDDR2 at runtime and is consumed by init_sequencer, mc_dfi_cmd_path, and mode_register. Confirm every register and field against the RDL and the generated regmap. The full register map (JEDEC timing, address mapping, refresh/policy modes, calibration, and observation registers) is in the pumice MAS Register Map.

23.2.5 Narrow-Device (x16) Width Granularities

The controller keeps two width concepts distinct: the **DRAM beat** (one DFI phase's data, DRAM_BEAT_WIDTH) and the **physical device word** (the DQ width, DRAM_DEVICE_WIDTH, e.g. 16 for a x16 device). When a beat is wider than the device, burst length (beats per command) and the column-address stride are device-word quantities. The RTL scales the MR0 burst length by DRAM_BEAT_WIDTH / DRAM_DEVICE_WIDTH and sizes the column granularity to the device word; missing either produced the on-silicon DDR2 read failure on the Nexys A7 x16 bring-up (over-read plus column-overlap scramble). PHY read-path integration (data-vs-rddata_valid skew and per-command read/write phase) is handled by the runtime DFI_PHASE / PHY_TIMING knobs above rather than hard-coded PHY latencies, so one controller build ports across PHYs.

24 AXI4 Slave Interface

The controller exposes a standard AXI4 slave (`pumice_top` / `pumice_core`) for SoC connectivity. All slave ports use the `s_axi_*` prefix with no `i_`/`o_` decoration (e.g. `s_axi_awid`, `s_axi_awaddr`, `s_axi_wdata`, `s_axi_rdata`).

The AXI side runs on `aclk` / `aresetn`.

24.1 Data Width Is Fixed at DW

The core's AXI data width is **not** a free parameter. It is fixed at

```
DW = DRAM_BEAT_WIDTH * DFI_RATE
```

i.e. one AXI beat carries exactly one DFI word (= `DFI_RATE` DRAM beats). There is no independent `AXI_DATA_WIDTH` parameter on `pumice_top`; `s_axi_wdata` / `s_axi_rdata` / `s_axi_wstrb` are always `DW` / `DW` / `DW/8` bits.

To attach a host AXI bus of a *different* width, use the optional wrapper `pumice_top_geared` (see Host-Width Gearing below).

24.1.1 Hard width rule (compile-enforced)

`pumice_top_geared`'s `HOST_AXI_DATA_WIDTH` and the DFI word `DW` **MUST** relate by an **exact power-of-two ratio** — `AXI:DFI = G:1` or `1:G`, `G ∈ {1,2,4,8,…}`. An `initial assert … $fatal` in `pumice_top_geared` **fails elaboration** / **Vivado synthesis** on any other pairing, so a bad width choice is a **compile error**, never a silent broken build. Rationale (identical to LiteDRAM, whose AXI frontend is 1:1 with its native port and routes all width change through a power-of-two stride converter): the DFI word is the atomic memory-side transfer, so an AXI beat must be a whole power-of-two number of DFI words. `DW` and `HOST_AXI_DATA_WIDTH` are the **only** compile-time width parameters — gear (active phases) and burst length are runtime CSRs (build-for-max).

24.2 Channels

Standard AXI4 — all five channels (AW, W, B, AR, R) are mandatory. No exclusive access support; `s_axi_awlock` / `s_axi_arlock` are present but ignored.

24.2.1 AW (Write Address) — SoC master → controller slave

Signal	Width	Notes
`s_axi_awid`	`AXI_ID_WIDTH`	Transaction ID
`s_axi_awaddr`	`AXI_ADDR_WIDTH`	Flat byte address
`s_axi_awlen`	8	Burst length minus 1
`s_axi_awsize`	3	Beat size; up to bus width
`s_axi_awburst`	2	`INCR` mandatory; `FIXED`/`WRAP` optional
`s_axi_awlock`	1	Present; ignored (no exclusive access)
`s_axi_awcache`	4	Present; ignored
`s_axi_awprot`	3	Present; ignored
`s_axi_awqos`	4	Present; ignored
`s_axi_awregion`	4	Present; ignored
`s_axi_awuser`	1	Present; ignored
`s_axi_awvalid`	1	Master asserts
`s_axi_awready`	1	Controller asserts when the WR intake / CAM can accept

24.2.2 W (Write Data)

Signal	Width	Notes
`s_axi_wdata`	`DW`	Beat data (`DW = DRAM_BEAT_WIDTH*DFI_RATE`)
`s_axi_wstrb`	`DW/8`	Byte enables; inverted to `dfi_wrdata_mask`
`s_axi_wlast`	1	Final beat of burst
`s_axi_wuser`	1	Present; ignored
`s_axi_wvalid`	1	Master asserts
`s_axi_wready`	1	Controller asserts while the WR data path can accept

Note: AXI4 W channel has no `wid`; beats are presumed in the order their AW transactions were issued. The WR intake associates write data with its AW by arrival order.

24.2.3 B (Write Response)

Signal	Width	Notes
s_axi_bid	AXI_ID_WIDTH	Matches awid
s_axi_bresp	2	OKAY or SLVERR
s_axi_buser	1	Present; ignored
s_axi_bvalid	1	Controller asserts on response ready
s_axi_bready	1	Master asserts

24.2.4 AR (Read Address)

Signal	Width	Notes
s_axi_arid	AXI_ID_WIDTH	Transaction ID
s_axi_araddr	AXI_ADDR_WIDTH	Flat byte address
s_axi_arlen	8	Burst length minus 1
s_axi_arsize	3	Beat size
s_axi_arburst	2	INCR mandatory; others optional
s_axi_arlock	1	Present; ignored
s_axi_arcache	4	Present; ignored
s_axi_arprot	3	Present; ignored
s_axi_arqos	4	Present; ignored
s_axi_arregion	4	Present; ignored
s_axi_aruser	1	Present; ignored
s_axi_arvalid	1	Master asserts
s_axi_arready	1	Controller asserts when the RD intake / CAM can accept

24.2.5 R (Read Data)

Signal	Width	Notes
s_axi_rdata	DW	Beat data
s_axi_rid	AXI_ID_WIDTH	Matches arid
s_axi_rresp	2	OKAY (v1 always)
s_axi_rlast	1	Final beat of burst
s_axi_ruser	1	Present; ignored

Signal	Width	Notes
s_axi_rvalid	1	Controller asserts when beat ready
s_axi_rready	1	Master asserts

24.3 Ordering Rules

- **Within an ID:** AXI4 ordering preserved. Reads from `arid = X` return in AR order; writes to `awid = X` commit in AW order.
- **Across IDs:** Out-of-order completion permitted by default (`AXI_OOO_ACROSS_IDS = true`). Disabling forces in-order across IDs at the cost of bandwidth.

24.4 Burst Type Support

Burst	Mandatory	Notes
FIXED	optional	Enable with SUPPORT_FIXED_BURST; rarely useful for DRAM
INCR	mandatory	Standard incrementing burst
WRAP	optional	Enable with SUPPORT_WRAP_BURST

24.5 Burst Length and DRAM-Burst Geometry

AXI4 allows up to 256 beats. The frontend `mc_axi4_plane` splits every host burst at DRAM-burst-byte boundaries (`axi_master_wr_splitter` / `axi_master_rd_splitter`) so each split request maps to exactly one DRAM burst.

At the `DW` side the required geometry is:

```
(awlen + 1) * DFI_RATE == BL
```

i.e. one AXI burst at `DW` equals one DRAM burst of `BL` beats. (The older "`AXI_DATA_WIDTH == DRAM_BEAT_WIDTH`" coupling no longer applies — the AXI beat is a full DFI word of `DFI_RATE` DRAM beats.)

24.6 Narrow Transfers

AXI4 supports `awsize < bus width`. The narrow case is handled via `wstrb` — the unused byte lanes are masked. `dfi_wrdata_mask = ~wstrb`, so the DRAM commits only the unmasked bytes.

24.7 Write Response Timing

The B response is **commit-driven**: it is returned once the write is accepted and committed by the controller (`mc_wr_intake` raises B from the downstream commit-done handshake). A `SLVERR` B is self-generated by the intake for a malformed request; otherwise `OKAY`.

24.8 Frontend and Backpressure

The AXI frontend is `mc_axi4_plane`: dumb 1:1 write and read intakes (`mc_wr_intake`, `mc_rd_intake`) feeding a write-data CAM (`mc_wr_data_cam`, the write buffer / snarf source) and a read reorder CAM (`mc_rd_cmd_cam`), with in-order commit. There is no monolithic `txn_queue`.

- AW / AR `.ready` deassert when the corresponding intake or CAM cannot accept a new request (insertion `ready` low).
- W `.ready` deasserts when the write-data path (CAM buffer) is full.
- R / B `.valid` follow standard protocol; consumers may stall indefinitely without controller misbehavior.

24.9 Host-Width Gearing

To use a host AXI data width different from DW, instantiate the optional wrapper pumice_top_geared (parameter HOST_AXI_DATA_WIDTH, verified at {64, 128, 256}). It inserts the repo's formally-verified axi4_dwidth_converter_wr / axi4_dwidth_converter_rd between the host-width slave and the DW-width core. When HOST_AXI_DATA_WIDTH == DW, a generate branch bypasses the converters entirely, so the build is bit-identical to pumice_top (no converter, no added latency). See docs/AXI_DRAM_GEARING_SCOPE.md.

24.10 AXI4 Reset

`aresetn` is the AXI reset (active-low). When asserted, all AXI-side state is cleared. `aclk` / `aresetn` clock and reset the controller/AXI domain; the DFI side uses the independent `dfi_clk` / `dfi_rstn` (see the DFI chapter).

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25 DFI v2.1 Master Interface

The controller drives DFI v2.1 master signals to the PHY. DFI v2.1 is the first DFI revision that supports LPDDR2; this controller targets that revision.

25.1 Gearing and Per-Phase Packing

The gearing / phase count is set by the parameter `DFI_RATE` (default 2), **not** `N_PHASES`. Rather than the DFI `_pN` suffix convention, this controller packs the `DFI_RATE` sub-phases **flat** into each bus. All DFI pins carry the `_o`/`_i` direction suffix (e.g. `dfi_address_o`, `dfi_rddata_i`).

Widths (top-level `pumice_top`, derived from geometry):

- `dfi_address_o` — `DFI_ADDR_BUS_W = ROW_WIDTH * DFI_RATE`
- `dfi_bank_o` — `$clog2(NUM_BANKS) * DFI_RATE`
- `dfi_cas_n_o` / `dfi_ras_n_o` / `dfi_we_n_o` — `1 * DFI_RATE`
- `dfi_cs_n_o` / `dfi_odt_o` — `NUM_RANKS * DFI_RATE`
- `dfi_wrdata_o` / `dfi_rddata_i` — `DFI_DATA_WIDTH = DRAM_BEAT_WIDTH * DFI_RATE`
- `dfi_wrdata_mask_o` — `DFI_DATA_WIDTH/8`
- `dfi_wrdata_en_o` / `dfi_rddata_en_o` — `DFI_RATE`
- `dfi_rddata_valid_i` — `DFI_RATE`

Per-phase slice `p` of a bus occupies `[p*W_phase +: W_phase]`.

Command placement (see `mc_xddr2_cmd_formatter.sv`): the decoded DDR2 command is placed on a target sub-phase — READ on `rd_phase`, WRITE on `wr_phase` (both CSR-driven via the `DFI_PHASE` register), and all other commands (ACT / PRE / REF / MRS) on phase 0. Non-driven phases emit NOP. This aligns R/W commands with the PHY's `rdphase` / `wrphase` contract.

25.2 Command Interface (MC → PHY)

Signal	Width	Notes
dfi_address_o	ROW_WIDTH*DFI_RATE	DDR2: row / column per phase; LPDDR2: 20-bit packed CA word on the low bits
dfi_bank_o	clog2(NUM_BANKS)*DFI_RATE	DDR2: bank; LPDDR2: held idle
dfi_cas_n_o	DFI_RATE	DDR2: CAS encoding; LPDDR2: held idle
dfi_ras_n_o	DFI_RATE	DDR2: RAS encoding; LPDDR2: held idle
dfi_we_n_o	DFI_RATE	DDR2: WE encoding; LPDDR2: held idle
dfi_cs_n_o	NUM_RANKS*DFI_RATE	Per-rank chip select; cs_n[r]=0 selects rank r
dfi_odt_o	NUM_RANKS*DFI_RATE	On-die termination

There is no dfi_cke or dfi_reset_n port on the controller top; CKE / reset sequencing is not part of this pin bus.

25.2.1 DDR2 Command Encoding (JESD79-2)

The DDR2 truth table implemented by mc_xddr2_cmd_formatter.sv:

op	cs_n	ras_n	cas_n	we_n	bank	addr[10]	addr[..0]
NOP	0	1	1	1	X	X	X
ACT	0	0	1	1	BA	row	row
RD	0	1	0	1	BA	0	col
RDA	0	1	0	1	BA	1 (auto-PRE)	col
WR	0	1	0	0	BA	0	col
WRA	0	1	0	0	BA	1 (auto-PRE)	col
PRE	0	0	1	0	BA	0 (single bank)	X
PREA	0	0	1	0	X	1 (all banks)	X
REF	0	0	0	1	X	X	X
MRS	0	0	0	0	MR idx	MR data	MR data

cs_n is per-rank: cs_n[r]=0 selects rank r; all-ones deselects (NOP). A10 (addr[10]) is the auto-precharge bit for RD/WR and the all-bank bit for PRE. MRS carries its mode-register data on the

row field (`cmd_row_i`, `ROW_WIDTH`), not the column field, so the full DDR2 MR bit range is preserved.

25.2.2 LPDDR2 Command Encoding (JESD209-2F)

LPDDR2 is fully functional (reads and writes). LPDDR2 has no RAS/CAS/WE strobes: the command and a scrambled address are multiplexed onto the 10-bit CA bus over two clock edges. `mc_xddr2_cmd_formatter.sv` builds the bit-exact JESD209-2F Table 60 encoding as a flat **20-bit** word (10 bits rising + 10 bits falling) placed on the low 20 bits of `dfi_address_o`; `dfi_ras_n_o` / `dfi_cas_n_o` / `dfi_we_n_o` are held idle and `dfi_cs_n_o` is asserted for the target rank. The full pin layout is the locked spec in `docs/uarch/LPDDR2_CA_ENCODING.md`, which both the RTL and the DV BFM encode against.

25.3 Write Data Interface (MC → PHY)

Signal	Width	Notes
dfi_wrdata_o	DFI_DATA_WIDTH	Write data (one DFI word = all phases)
dfi_wrdata_en_o	DFI_RATE	High during valid write-data phases
dfi_wrdata_mask_o	DFI_DATA_WIDTH/8	mask = ~wstrb (1 = mask byte)

Write data is driven by mc_dfi_wr_serializer: on a WR command it waits t_phy_wrlat DFI cycles, then streams one DFI word per cycle from the write-data FIFO until the burst's last word (bubble-free, back-to-back bursts).

25.4 Read Data Interface (PHY ↔️ MC)

Signal	Width	Direction	Notes
dfi_rddata_i	DFI_DATA_WIDTH	PHY → MC	Read data (one DFI word)
dfi_rddata_valid_i	DFI_RATE	PHY → MC	High when a read word is valid
dfi_rddata_en_o	DFI_RATE	MC → PHY	Read capture window

Read alignment is done by mc_dfi_rd_aligner: on a RD command it asserts dfi_rddata_en_o for the burst's word count starting t_rddata_en DFI cycles later, and captures dfi_rddata_i on each dfi_rddata_valid_i. There is no dfi_rddata_dnv port.

25.5 Status / Init Interface

Signal	Width	Direction	Notes
dfi_init_start_o	1	MC → PHY	Controller requests PHY init start
dfi_init_complete_i	1	PHY → MC	High when the PHY has finished init

These are the only status handshake pins on the controller top. Update and training sub-interfaces are not present on this pin bus.

25.6 Timing Parameters Honored

The DFI datapath (`mc_dfi_2p1_plane`) honors these CSR-driven knobs:

- `t_phy_wrlat` — delay from the WR command to `dfi_wrdata_en` (write serializer)
- `t_rddata_en` — delay from the RD command to the `dfi_rddata_en` capture window (read aligner)
- `rd_phase` / `wr_phase` — the DFI sub-phase carrying the READ / WRITE command (`DFI_PHASE` register)

25.7 Clock Domains and CDC

The controller/AXI side runs on `aclk`; the PHY/DFI side runs on the independent input clock `dfi_clk` (with `dfi_rstn`). The two are **not** related by a fixed frequency ratio and there is no combinational phase replication.

The single clock-domain crossing lives inside `mc_dfi_2p1_plane` (in `mc_dfi_cdc`) and is built from **asynchronous gaxi FIFOs** — one each for the command, write-data, and read-data streams (plus the init handshake). The internal datapath unit is one DFI word (= one DFI cycle), so one FIFO word maps to one DFI cycle and the crossing is bubble-free.

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26 CSR Register Interface (PeakRDL cpuif)

Configuration and observation registers are exposed through a **PeakRDL-generated register block** (`pumice_csr`), not a hand-written APB slave. The register map is authored in `rtl/macro/pumice_csr.rdl` and generated with the passthrough cpuif (`bin/peakrdl_generate.py`); `pumice_top` instantiates the generated `pumice_csr` and drives the core by name from `hwif_out.*` (see §6.3 for the full map). The DV mirror used for by-name access is `dv/tbclasses/pumice_regmap.py`.

The SoC-facing bus is therefore the PeakRDL **passthrough cpuif**, a simple request/ack register interface. An SoC that natively speaks APB / AXI-Lite places a thin protocol adapter in front of this cpuif; the controller itself does not embed one in this generation.

26.1 cpuif Signals

`CSR_ADDR_W = 12` (4 KB, 32-bit registers). The block runs on `aclk` / `~aresetn` (the controller clock/reset — see below).

Signal	Width	Direction	Notes
`s_cpuif_req`	1	input	Request valid
`s_cpuif_req_is_wr`	1	input	1 = write, 0 = read
`s_cpuif_addr`	12	input	Register byte address (4 KB space)
`s_cpuif_wr_data`	32	input	Write data
`s_cpuif_wr_biten`	32	input	Per-bit write enable
`s_cpuif_req_stall_wr`	1	output	Back-pressure a write request
`s_cpuif_req_stall_rd`	1	output	Back-pressure a read request
`s_cpuif_rd_ack`	1	output	Read data valid
`s_cpuif_rd_err`	1	output	Read error (e.g. unmapped address)
`s_cpuif_rd_data`	32	output	Read data
`s_cpuif_wr_ack`	1	output	Write accepted
`s_cpuif_wr_err`	1	output	Write error

26.2 Access Semantics

Standard PeakRDL passthrough behavior:

- Single-cycle accepts for the typical case; the `*_stall_*` outputs allow the block to hold a request when needed.
- `s_cpuif_rd_err` / `s_cpuif_wr_err` are raised for unmapped addresses; the error terminates the access and does not disturb controller state.
- Read-only fields ignore writes; write-only / self-modifying fields (e.g. `CTRL.init_start`, `CTRL.soft_reset`) pulse `swmod`.

26.3 Clock Domain

The register block is clocked by `aclk` (the controller / host-AXI clock) and reset by `~aresetn`. It is **not** on a separate management clock in this generation — there is no independent `apb_pclk`. Config fields (`hwif_out.*`, `hw = r`) are read combinationally by `pumice_top` and fanned out to the core; the single clock-domain crossing to the PHY side (`dfi_clk`) lives inside `mc_dfi_2p1_plane` (async gaxi FIFOs), not in the register block. Observation fields (`hwif_in.*`, `hw = w`) are written from the core clock domain.

26.4 Register Map

The full address map is documented in §6.3, reconciled against `rtl/macro/pumice_csr.rdl` and `dv/tbclasses/pumice_regmap.py`.

26.5 Configuration Sequencing

Recommended bring-up order (see §6.1 for reset details):

1. Hold `aresetn` asserted.
2. Program timing parameters (`TIMINGS_*`), MR overrides, and `PHY_TIMING` (including `memtype` and `t_phy_wrlat` / `t_rddata_en`).
3. Program `ADDR_MAP` (`bank_lsb` / `hash_en` / `hash_seed`) and `DFI_PHASE`.
4. Configure PASR mask (LPDDR2 only).
5. Release `aresetn`.
6. Wait for `STATUS.init_done`.
7. Begin AXI traffic.

Runtime tuning writes (scheduler / refresh / page policy) take effect at the next configuration quiet point; the SoC owns the drain — see §5.4.

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27 Build-Time vs. Runtime Configuration

This controller distinguishes between **build-time parameters** that set the silicon footprint and **runtime CSR registers** that select behavior within that footprint. The two planes serve different purposes and should not be confused. This chapter applies the principle to every configurable knob in the design.

27.1 Principle

Plane	What it controls	Frozen at	Cost of changing
Build-time parameter	Synthesized silicon footprint: signal widths, table depths, queue sizes, instantiation counts, encoder selection, MAX bounds of runtime values	Elaboration time	Full RTL rebuild + place-and-route + tape-out (silicon) or bitstream rebuild (FPGA)
Runtime CSR register	Behavior selection *within* the synthesized footprint: policy choices, threshold values, on/off toggles, active counts up to the build-time MAX	Programmable any time the configuration quiet point is reached (typically during init or under SoC orchestration)	A single APB write

The architectural rule is straightforward: **if a knob is changing silicon area, it must be build-time. If a knob is only changing behavior of already-synthesized hardware, it must be runtime.**

27.2 Why This Distinction Matters

A controller designed without this distinction tends to collapse one of two ways:

- **Too much build-time** — every algorithmic choice is a parameter that requires a new bitstream / silicon revision to evaluate. Characterization becomes impossibly slow; post-silicon debug becomes impossible. Most academic prototype controllers fall here.
- **Too much runtime** — every option is always synthesized, including ones that will rarely be used. Area balloons; timing closure suffers; the design becomes harder to verify because every code path is live. This is the failure mode of "ultra-configurable" commercial IP.

This controller targets the middle: **build-time picks what's physically present; runtime picks what's active.**

27.3 The Hybrid Pattern: "Build-Time MAX + Runtime Active"

Many of this controller's knobs follow a hybrid pattern where the build-time parameter sets a synthesized MAX, and a paired runtime CSR field selects the actually-used value within that MAX.

Concrete example for LOOKAHEAD_DEPTH:

```
Build-time:    LOOKAHEAD_DEPTH_MAX = 4
               → synthesizes 4 row-address comparators per bank
machine

Runtime CSR:   sched_tuning.lookahead_active [3:0]
               → scheduler peeks min(LOOKAHEAD_DEPTH_MAX,
lookahead_active)
                 same-bank requests per cycle

For debug:     Write lookahead_active = 0 to force pure page-policy
mode
               Write lookahead_active = 4 to use the full synthesized
window
               No rebuild required.
```

The same pattern applies to TXN_QUEUE_DEPTH, AGE_MAX, REFRESH_DEFER_MAX, INIT_ZQ_RETRIES, ZQCS_FREQ_HZ, and others — area is paid once at synthesis for the MAX; runtime selects how much of the synthesized hardware is actively gated.

27.4 The “Build-Time + Runtime Mux” Pattern

For algorithmic choices where multiple alternatives have similar area cost, the build-time parameter selects **which** code paths are synthesized, and a runtime CSR field picks **which** of the synthesized paths is active.

Concrete example for PAGE_POLICY:

```
Build-time:    (page policy has no build-time parameter -- the engine
is
                always present; a mode is a CSR value, not a build)

Runtime CSR:   refresh_tuning.page_policy_or [1:0]
               00 = use build-time default (HAPPY_HYBRID here)
               01 = force OPEN
               10 = force CLOSE
               11 = force HAPPY_HYBRID

For debug:     Sweep policies on real workloads with no rebuilds.
```

The build-time choice PAGE_POLICY = OPEN would have omitted the HAPPY predictor table from the synthesized design entirely, saving area but losing the ability to switch to HAPPY at runtime. The build-time choice PAGE_POLICY = HAPPY_HYBRID synthesizes everything, paying the area cost for the predictor table, in exchange for the ability to switch among any of the three at runtime.

The same pattern applies to REFPB_POLICY, SCHEDULER_MODE, and HAPPY enable/disable.

Note that address mapping is **not** an example of this pattern. Earlier revisions exposed an ADDR_MAP_SCHEME build-time set plus an ADDR_MAP_TUNING.scheme_or runtime mux; both have been retired. Address mapping is now a single runtime placement knob — see “The ‘Runtime Placement’ Pattern” below.

27.5 The “Runtime Placement” Pattern

Some knobs neither gate synthesized alternatives nor cap a synthesized MAX — they simply reposition a field within already-present combinational logic. There is no mux and no area choice; the same decode hardware produces different behavior depending on a runtime value.

Address mapping is the canonical case. `addr_mapper.sv` decodes a flat AXI address into `{rank, bank, row, col}` from **one** runtime CSR field, `ADDR_MAP.bank_lsb`, which selects where the bank field sits in the byte-offset-stripped word address. The column fills below (`col_lo`) and above (`col_hi`) the bank; row/rank stack above at invariant positions. The classic “schemes” are just settings of `bank_lsb`:

```
Runtime CSR:   ADDR_MAP.bank_lsb [4:0]   (reset = 0x0A = COL_WIDTH =
ROW_MAJOR)
               = COL_WIDTH               -> bank above whole column =
ROW_MAJOR
               = log2(DRAM_BL)           -> minimal col_lo          =
BANK_INTERLEAVE
               in between                -> partial interleave

Runtime CSR:   ADDR_MAP.hash_en [8], ADDR_MAP.hash_seed [23:16]
               folds row bits + seed into the bank index (= XOR_HASH),
               defeating power-of-two-stride hot-banking.
```

There is no build-time parameter and no scheme selector: `ROW_MAJOR`, `BANK_INTERLEAVE`, and `XOR_HASH` are all reachable by writing `bank_lsb` (and optionally `hash_en`/`hash_seed`) at runtime. The address-decode logic is combinational and always present.

`MEMTYPE` is likewise a runtime knob in this generation, not a build-time string. The core decodes `memtype_e` from the CSR field `PHY_TIMING.memtype` (`0 = DDR2`, `1 = LPDDR2`) — see `pumice_top.sv` `w_memtype`. There is no `MEMTYPE` build parameter that gates an encoder or step table; the DDR2 vs LPDDR2 behavior is selected at runtime within the synthesized datapath.

27.6 The “Runtime Only” Pattern

A small number of knobs have zero area cost and are runtime-only: simple counter compare values, timeout thresholds, retry counts. Examples include `ZQCS_FREQ_HZ`, `INIT_ZQ_RETRIES`, and the various observation-counter clear-on-read controls.

These have no build-time presence at all.

27.7 Configuration Quiet Points

Runtime CSR writes that change scheduler or refresh-manager behavior take effect at the next **configuration quiet point** — defined as: no DRAM commands in flight, no refresh sequence in progress, and the relevant FSM in its idle state. The CSR slave does not enforce quiet-point writes; the SoC firmware is responsible for sequencing. Writing during normal operation is well-defined but the change does not apply until the next quiet point. There is **no** `config_apply` strobe in this generation — the SoC owns the drain and must quiesce host AXI traffic (and, if needed, wait out any in-flight refresh) before the change is guaranteed to take effect.

Writes to register-file CSRs that have no FSM impact (timing parameter overrides, MR override values, observation counters) take effect immediately and have no quiet-point requirement.

27.8 Summary

The following two chapters document this controller's complete configuration interface:

- **§5.2 Top-Level Parameter Table** — every build-time parameter, with its type, range, governing section, and (where applicable) the paired runtime CSR field.
- **§5.3 Characterization Knobs** — the subset of build-time + runtime knobs intentionally exposed for performance sweeping during silicon / FPGA characterization.

The full runtime CSR register map is in §6.3.

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28 Top-Level Parameter Table

All build-time parameters for the controller, with type, default, valid range, the section that governs their use, and a one-line purpose summary.

28.1 Parameter Table

The parameters below are the actual elaboration parameters of `pumice_top` / `pumice_core` (with the host-width parameter added by `pumice_top_geared`). Several knobs that earlier revisions listed as build-time — `MEMTYPE`, `WRPHASE`, `RDPHASE`, and the address-map scheme set — are runtime CSR fields in this generation and are called out under “Runtime CSR knobs (not build parameters)” below.

Parameter	Type	Default	Range / valid set	Section	Purpose
`AXI_ID_WIDTH`	int	8	1-16	§3.1	AXI ID tag width (`IW`).
`AXI_ADDR_WIDTH`	int	32	24-40	§3.1	AXI flat address width (`AW`).
`NUM_RANKS`	int	1	`{1, 2, 4}`	§3.3	Number of physical ranks (per-rank `CS_n`, `CKE`, `ODT`). 1 = single-rank point-to-point; 2 / 4 = DIMM-class. Rank bits stack at the top of the address map (§3.1) and refresh state is tracked per (rank, bank).
`NUM_BANKS`	int	8	`{4, 8}`	§3.3	Per-device bank count (per rank). Sets the bank-machine / lookup count.
`ROW_WIDTH`	int	14	12-16	§3.3	Row-address width.
`COL_WIDTH`	int	10	9-12	§3.3	Column-address width. Also the `ROW_MAJOR` setting of `ADDR_MAP.bank_lsb`.
`DFI_RATE`	int	2	`{1, 2, 4}`	§3.6	DFI frequency-ratio gear (phase count). Drives all DFI bus widths and the internal data-unit width. This is the gear parameter formerly called

Parameter	Type	Default	Range / valid set	Section	Purpose
					N_PHASES.
DRAM_BEAT_WIDTH	int	64	{16, 32, 64}	§3.6	Width of one DRAM data beat (per DFI phase).
BL	int	8	{4, 8}	§3.6	DRAM burst length in beats. The AXI intake splits host bursts into fixed-BL commands; one AXI burst at the core (DW) side is (awlen+1)*DFI_RATE == BL.
NUM_ENTRIES	int	8	4-32 (power of 2)	§3.2	Depth of the read/write command CAMs (in-flight transaction slots). Pointer width is clog2(NUM_ENTRIES).
N_SRAM_SLOTS	int	8	4-32 (power of 2)	§3.2	Write-data SRAM slot count (buffered write payloads awaiting commit).
BYTE_OFFSET_WIDTH	int	3	clog2(beat byte size)	§3.1	Low byte-offset bits stripped before address decode (log2 of beat byte size).
AGE_WIDTH	int	16	8-24	§3.2	Width of the per-transaction age counter used for FR-FCFS anti-starvation tie-breaking.
HOST_AXI_DATA_WIDTH	int	128	{64, 128, 256} (verified)	§3.1	**pumice_top_geared only.** Free host-side AXI data width. Formally-verified axi4_dwidth_converter_wr/_rd shims bridge the host width to the fixed core width DW. HOST_AXI_DATA_WIDTH == DW is a generate-bypassed, bit-identical direct connection (no converter, no added latency).

Derived parameters (computed at elaboration, not overridable independently):

- `DW = DRAM_BEAT_WIDTH * DFI_RATE` — the fixed core AXI data width **and** the DFI word width (default `64 * 2 = 128`). `DFI_DATA_WIDTH = DW`. One core AXI beat == one DFI word == `DFI_RATE` DRAM beats.
- `SW = DW / 8` — core AXI / DFI strobe width.
- `PHW = clog2(DFI_RATE)` (min 1) — DFI sub-phase index width.
- `DFI_STRB_WIDTH = DW/8`, `DFI_EN_WIDTH = DFI_VALID_WIDTH = DFI_RATE`, `DFI_ADDR_BUS_W = ROW_WIDTH*DFI_RATE`, `DFI_BANK_BUS_W = clog2(NUM_BANKS)*DFI_RATE`, `DFI_CS_BUS_W = NUM_RANKS*DFI_RATE` — the DFI 2.1 pin-bus widths, all scaled by `DFI_RATE`.

Runtime CSR knobs (not build parameters):

These select behavior at runtime and have no build-time parameter in this generation. Full field detail is in §6.3.

- `PHY_TIMING.memtype` (1-bit) — `0 = DDR2`, `1 = LPDDR2`. The core decodes `memtype_e` from this field (`pumice_top.sv` `w_memtype`); there is no build-time `MEMTYPE` string.
- `DFI_PHASE.wr_phase` / `DFI_PHASE.rd_phase` (3-bit each, sliced to `PHW` downstream) — which DFI sub-phase carries the WRITE / READ command. Defaults 0/0. These replace the former build-time `WRPHASE` / `RDPHASE`.
- `ADDR_MAP.bank_lsb` (5-bit, reset `0x0A` = `COL_WIDTH` = ROW_MAJOR), `ADDR_MAP.hash_en` (1-bit), `ADDR_MAP.hash_seed` (8-bit) — the single address-map placement knob (+ optional bank XOR-hash). The classic ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH “schemes” are just settings of these fields; there is no `ADDR_MAP_SCHEMES_SYNTH` / `ADDR_MAP_SCHEME_DEFAULT` build parameter and no scheme mux. See `addr_mapper.sv`.
- `REFRESH_TUNING.page_policy_or` (2-bit) — `00 = build-time default`, `01 = OPEN`, `10 = CLOSE`, `11 = reserved` (was HYBRID — retired; maps to build default). Adaptive paging is `PAGE_POLICY_CFG.policy_mode`.
- `REFRESH_TUNING.refpb_policy_or` (2-bit), `REFRESH_TUNING.refresh_defer_active` (4-bit), `REFRESH_TUNING.zqcs_freq_hz` (16-bit, reset 1 Hz).
- `SCHED_TUNING.lookahead_active` (4-bit), `force_inorder` (1-bit), `age_max_runtime` (8-bit), `txn_queue_high_water` (8-bit); `lookahead_max_obs` (4-bit, RO echo of the build MAX).

- PAGE_POLICY_CFG.policy_mode, PAGE_TIMEOUT_CFG.tr_init (the only live fields after TASK-014; policy_scope, tr_min/tr_max/tr_step and the whole PAGE_ADAPT_CFG register retired 2026-09-27) — the runtime page-policy engine (replaced the retired PAGE_PRED_TUNING). **policy_mode RESETS TO 0.** Mode 3 (fixed_open) with tr_init = 2 is the measured best default and is RECOMMENDED, but the reset change is BLOCKED on BUG-003 (a short timeout precharges under an in-flight read). Measured 2026-09-26: measured strictly dominant over open page on the board (+41.2% col_major_interleaved_bl4, +8.6..11.6% col_major, exactly flat on incremental/row_major, nothing regressing) (pre-sdpram-fix 2026-10-02; off-datapath fix, reproduced 2026-10-03 — FINDINGS_litedram_ab_2026-10-03.md). Beware two traps on these fields — tr_init = 0 DISABLES the timeout rather than selecting a build default, and check_interval = 0 gates the whole adapt_time TR adjustment off, which is why mode 4 measured inert for every campaign before 2026-09-26. See TASK-013.
- Timing CSRs — TIMINGS_RC_RCD_RP_RAS (tRC/tRCD/tRP/tRAS), TIMINGS_RFC_REFI (tRFC/tREFI), TIMINGS_RRD_FAW_WTR_CCD (tRRD/tFAW/tWTR/tCCD), TIMINGS_CL_CWL_WR (CL/CWL/tWR/tRFCpb), TIMINGS_RTP_RTW (tRTP/tRTW).
- PHY_TIMING.refresh_burst (4-bit, 1..8), PHY_TIMING.t_phy_wrlat (8-bit), PHY_TIMING.t_rddata_en (8-bit).
- Init timings — INIT_TIMING0 (t_init_wait/t_dll_wait), INIT_TIMING1 (t_mrd_wait/t_rp_wait/t_rfc_wait), INIT_TUNING.zq_retries (4-bit, reset 3), INIT_TUNING.init_timeout_ms (8-bit, reset 10).

These cost zero silicon area beyond the register flops; making them build-time parameters would have been wrong (and was, in earlier revisions of this document).

28.2 Memtype-Dependent Constraints

MEMTYPE is a runtime CSR field (PHY_TIMING.memtype) rather than an elaboration parameter, so the DDR2-vs-LPDDR2 selection does not gate elaboration-time asserts. The memtype-dependent behavior is instead enforced at run time within the synthesized datapath:

- PHY_TIMING.memtype == 1 (LPDDR2) enables the LPDDR2-only features (PASR bank/segment masks, DPD request, temperature-derate readback, per-bank tRFC via TIMINGS_CL_CWL_WR.tRFCpb).
- PHY_TIMING.memtype == 0 (DDR2) uses the DDR2 command encoding and all-bank refresh.
- The DFI address bus is ROW_WIDTH * DFI_RATE wide (DFI_ADDR_BUS_W) and carries the LPDDR2 CA packing when memtype == 1; it must be wide enough for whichever memtype is programmed, which is guaranteed by ROW_WIDTH >= 14.

28.3 Timing-Configuration Sanity Checks

The controller expects the following relationships to hold on the loaded timing CSR values. Because timings are runtime CSRs, these are boot-time programming contracts the SoC firmware must honor (they are documented here so the sweep tool and bring-up scripts can assert them):

- `tREFI_cycles >= 100` — the MC clock must be fast enough relative to tREFI that the refresh timer is well-resolved.
- `tRCD_cycles >= 2`, `tRP_cycles >= 2`, `tRC_cycles >= tRAS + tRP` — basic JEDEC consistency checks.
- `tRFC_cycles > tRP_cycles` — the refresh-cycle time must exceed the precharge time.

These are configuration bugs that should be caught at boot time before traffic starts, not preventable run-time faults.

28.4 Parameter Categories

Category	Parameters
Geometry	NUM_RANKS, NUM_BANKS, ROW_WIDTH, COL_WIDTH
Multi-rank	NUM_RANKS
Bus widths	AXI_ID_WIDTH, AXI_ADDR_WIDTH, DRAM_BEAT_WIDTH, BYTE_OFFSET_WIDTH, HOST_AXI_DATA_WIDTH; derived DW/DFI_DATA_WIDTH
Gear / DFI	DFI_RATE, BL (with runtime DFI_PHASE.rd_phase / DFI_PHASE.wr_phase)
Capacity	NUM_ENTRIES, N_SRAM_SLOTS, AGE_WIDTH
Characterization	Runtime CSR knobs paired with the geometry above — SCHED_TUNING.lookahead_active, REFRESH_TUNING.page_policy_or / refpb_policy_or / refresh_defer_active, PAGE_POLICY_CFG.* / PAGE_TIMEOUT_CFG.* / PAGE_ADAPT_CFG.*, ADDR_MAP.bank_lsb / hash_en (see §5.3 and §6.3)

29 Characterization Knobs

A subset of the build-time parameters is intentionally left configurable for **characterization sweeping**. These are the algorithmic choices that have research-backed alternatives where the best choice depends on workload. Rather than baking in a one-size-fits-all default, we expose them as parameters so the verification plan can sweep them on representative AXI traffic and pick defaults from data.

29.1 Characterization Parameters

Parameter	Choices	Recommended default	Rationale
LOOKAHEAD_DEPTH	0 / 1 / 2 / 4	1	Same-bank lookahead window for **exact** auto-precharge decisions when the next same-bank request is already in the queue (typical for streaming / DMA). Falls back to PAGE_POLICY when the queue is shallow. 0 disables; 2 or 4 buys diminishing returns at modest comparator cost.
PAGE_POLICY	OPEN / CLOSE / HAPPY_HYBRID	HAPPY_HYBRID (LPDDR2), OPEN (DDR2)	Fallback for when lookahead is inconclusive. Streaming workloads favor OPEN; mobile-mixed favors HAPPY_HYBRID. Ghasempour 2015 reports near-optimal accuracy with small storage.
PAGE_PREDICTOR_TABLE_BITS	8 – 16	12	4 KB predictor is sufficient for typical mobile workloads. Larger tables marginal improvement.
REFPB_POLICY	ROUND_ROBIN / OLDEST_FIRST / DARP	DARP	Chang HPCA 2014: DARP wins over round-robin on uneven bank activity (real workloads).
REFRESH_DEFER_MAX	1 – 8	8 (JEDEC ceiling)	Stuecheli MICRO 2010: deferring up to 8 maximizes Elastic Refresh benefit. Set to 1 for real-time predictability.
TXN_QUEUE_DEPTH	4 – 64	16	Larger queues help bandwidth on multi-master AXI; smaller queues reduce area.
AGE_MAX	32 – 1024	256	Caps FR-FCFS starvation; larger values let row-hits dominate longer at the cost of tail latency.
DFI_RATE	1 / 2 / 4	depends	Build-time DFI gear (phase count). 4

Parameter	Choices	Recommended default	Rationale
		on PHY	typical for FPGA SoCs; 2 for slower mobile platforms; 1 for simulation-only. Formerly named `N_PHASES`.
`ADDR_MAP.bank_lsb` (+ `hash_en`)	`log2(DRAM_BL)` .. `COL_WIDTH`, hash off / on	`bank_lsb = COL_WIDTH` (ROW_MAJOR), hash off	Runtime address-map placement knob (not a build parameter). `bank_lsb = COL_WIDTH` = ROW_MAJOR (best for streaming); lowering `bank_lsb` toward `log2(DRAM_BL)` interleaves banks (best for random); `hash_en` folds row bits into the bank index to defeat power-of-two-stride hot-banking (adversarial). Sweep `bank_lsb` across the legal range with hash off, then repeat the best point with hash on. See `addr_mapper.sv`. (`DRAM_BL`, not `BL/DFI_RATE` — the mapper indexes device words; pumice BUG-022, measured in `dv/tests/fub/test_addr_mapper.py::minimum_bank_lsb_is_measured`.)

29.2 Sweep Plan

The verification plan (see §6.4) includes a characterization sweep that runs each parameter through its choices on a small benchmark suite:

29.2.1 Benchmark Suite

A key open question for the sweep is the **interaction between `LOOKAHEAD_DEPTH` and `PAGE_POLICY`**. With deep queues and conclusive lookahead, the choice of `PAGE_POLICY` barely matters because the fallback rarely fires. With shallow queues, `PAGE_POLICY` dominates. The sweep matrix should cover both extremes.

Workload	Pattern
`stream-read`	Sequential 4 KB read bursts
`stream-write`	Sequential 4 KB write bursts
`stream-mixed`	Interleaved sequential read and write
`random-narrow`	Random small AXI bursts (1–4 beats)
`random-wide`	Random large AXI bursts (16–64 beats)
`mobile-mixed`	Bursty traffic with periodic large writes (modeled on smartphone GPU + display)
`multi-master`	Two AXI masters issuing concurrent traffic with different patterns

29.2.2 Metrics Collected

For each (workload, parameter) combination, the sweep records:

- **Sustained read bandwidth** (% peak)
- **Sustained write bandwidth** (% peak)
- **Average read latency**
- **99th-percentile read latency**
- **Refresh blocking time** (% of cycles where the bus was blocked by refresh)
- **Row-hit rate** (per-bank, averaged)
- **Refresh deferral utilization** (avg `refresh_pending` over time)

29.2.3 Default Selection

After the sweep, each parameter's default is set to the value that:

1. Maximizes sustained bandwidth across the benchmark suite
2. Subject to: 99th-percentile latency does not regress by more than 20%
3. Subject to: area cost is within budget

The defaults shown in the parameter table above are the **expected** outcomes based on the cited research. The actual defaults will be confirmed by the sweep.

29.3 Runtime Overrides via CSR

Some characterization parameters are also accessible at runtime via CSR (see §6.3). This allows in-system tuning without rebuild:

- `PAGE_POLICY` runtime override (`REFRESH_TUNING.page_policy_or`)
- `REFPB_POLICY` runtime override (`REFRESH_TUNING.refpb_policy_or`)
- Address-map placement (`ADDR_MAP.bank_lsb`, `hash_en`, `hash_seed`) — this is a runtime knob outright, not an override of a build default
- Page predictor warm-up cycles and hysteresis (`PAGE_PRED_TUNING.warmup_cycles` / `hysteresis`)

Runtime overrides take effect at the next "quiet point" (no commands in flight). The SoC is responsible for ensuring no DRAM traffic during the transition.

29.4 Observability for Sweep

The CSR exposes the following observation counters (described in §6.3):

- `OBS_ROW_HIT[8]` — per-bank ROW HITS, free-running (subtract two reads).
- `PAGE_STATS_HIT` / `PAGE_STATS_MISS` / `PAGE_STATS_EMPTY` — global page-outcome counters.
- `SCHED_STATS_ACT` / `SCHED_STATS_PRE` — activations and closes.
- `REF_STATS_REF` / `REF_STATS_REF_BUSY` — refreshes, and those that fired with work pending.
- `STALL_*` — stall-cause attribution.

AXI latency and queue occupancy are **not** controller counters: they are measured by `rtl/amba/shared/axi_bus_meter.sv` and `axi_perf_latency_hist.sv`, which the characterisation harness instantiates directly (`char_engine_block.sv:608-712`). > **27 `OBS_*` registers were removed on 2026-09-28 (pumice BUG-020).** They were declared, documented and readable, and nothing drove them – every one read zero. `OBS_ROW_HIT[8]` was kept and is now driven; the rest went. AXI-side latency and occupancy are measured by `axi_bus_meter` and `axi_perf_latency_hist` outside the controller.

These feed back into the sweep tool for in-loop default selection.

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30 Family-Wide Config Bits

This section is the **family-wide config-bit registry** — a catalog of runtime configuration bits used across the DDR2 / LPDDR2, DDR3 / LPDDR3, and DDR4 / LPDDR4 memory controllers. Some bits apply to all generations; some are introduced in one generation and absent in earlier ones; some are *defined* across the family but ignored in flavors where the underlying mechanism does not exist. **That last case is intentional.** Allocating the bit family-wide — even when unused — keeps the CSR map mechanically compatible across controllers and lets software treat the family as a single product line.

30.1 Why a Family-Wide Registry

Each controller in the family has its own CSR map (per §6.3 of each generation's HAS). But the field encodings, bit positions within shared registers, and the *names* are aligned across the family. The intent:

- **Software portability**: a single device driver template understands the bit layout and only needs per-flavor capability checks.
- **Bring-up portability**: bring-up software written for DDR3 should mostly work on DDR2 with the same register reads.
- **Verification portability**: the cocotb test suite's CSR-poke helpers can be reused.
- **Debug consistency**: the same "config dump" script produces a parseable output regardless of which flavor is running.

A bit may be marked **N/A** in a flavor; reads return 0 and writes are ignored (no `pslverr`, just silently ignored — because software portability requires that writing a fully-portable config word does not error on flavors that don't implement every bit).

30.2 Applicability Notation

The Apply column uses this shorthand:

- **all** — applies to every flavor in the family
- **DDR-only** — applies to DDR2/3/4 only; LPDDR variants ignore
- **LP-only** — applies to LPDDR2/3/4 only; DDR variants ignore
- **DDR3+** — applies to DDR3, DDR4, and forward; DDR2 ignores
- **DDR4+** — applies to DDR4 only (introduced for bank groups); earlier ignore
- **LPDDR3+** — LPDDR3 and LPDDR4 only
- **LPDDR4-only** — LPDDR4 only

30.3 SCHED_TUNING (Scheduling Family Bits)

Register block at the same offset (0x040) in every controller. These bits govern the FR-FCFS scheduler's runtime behavior.

Bit Field	Width	Apply	Description
`lookahead_active`	4	all	Runtime lookahead window (0 .. build-time MAX). 0 disables lookahead.
`force_inorder`	1	all	Forces FR-FCFS into first-ready FIFO mode for debug / real-time.
`happy_enable`	1	all	Runtime kill-switch for the HAPPY page predictor. Reads as 0 in flavors built without HAPPY.
`age_max_runtime`	8	all	Anti-starvation age cap; runtime override of `AGE_MAX`.
`txn_queue_high_water`	8	all	Backpressure threshold; AXI `awready` deasserts when queue ≥ threshold.
`lookahead_max_obs`	4	all (R only)	Echo of build-time `LOOKAHEAD_DEPTH_MAX`. Software discovery.
`qos_high_priority`	1	all	**PLANNED — not in the RDL yet** (roadmap Axis 1, QoS step): when 1, scheduler boosts requests with `awqos`/`arqos` ≥ 8.
`bank_group_balance`	1	DDR4+	Round-robin across bank groups in scheduler tie-break (DDR4/LPDDR4 only); N/A elsewhere.

30.4 REFRESH_TUNING (Refresh Family Bits)

Bit Field	Width	Apply	Description
refpb_policy_or	2	LP-only	REFpb selection-policy override (LPDDR2/3/4); N/A for DDR-only flavors
page_policy_or	2	all	Runtime override for the LEGACY flat page policy (01=OPEN, 10=CLOSE). Only consulted when PAGE_POLICY_CFG.policy_mode == 0; since policy_mode now RESETS to 3, this field is inert out of reset. 10=CLOSE drives the legacy auto-precharge path, which measures 4.9x the activations of a background precharge – see MAS ch02 “08_page_policy”.
refresh_defer_active	4	all	Active deferral count (1 .. build-time MAX). 1 = no batching.
zqcs_freq_hz	16	all	Periodic ZQCS interval. 0 = init-only.
fgr_mode	2	DDR3+	Fine-Granularity Refresh mode (DDR3 1x/2x/4x; DDR4 fixed-rate, on-the-fly). N/A in DDR2.
refresh_per_rank	1	all	Force per-rank dispatch even when NUM_RANKS=1 (debug). Reset = 1 always when multi-rank.

30.5 PAGE_POLICY_CFG / PAGE_TIMEOUT_CFG (Axis 2)

The runtime page-policy engine (`mc_page_policy`). **This axis carries the largest measured runtime win on the controller**, and as of 2026-09-27 its resets SHIP that win: `policy_mode = 3` (`fixed_open`) and `tr_init = 2`.

Those two must move together. `tr_init = 0` disables the timeout outright, so mode 3 with the old `tr_init` reset of 0 would have been open page wearing another name – which is exactly the trap the field description records.

THE ADAPTIVE MODES WERE REMOVED 2026-09-27 (TASK-014). Paging modes 4 (`adapt_time`) and 5 (`adapt_access`) are gone, along with `pumice_row_pred_table.sv` and the whole `PAGE_ADAPT_CFG` register. Mode 4 had been MEASURED to be `fixed_open(tr_min)` – its mistake counter is dominated by the held-too-long case, so TR decayed monotonically to the floor and stayed there, landing exactly on the matching fixed point at three different floors. Mode 5 drove auto-precharge, which costs 4.9x the activations of a background precharge and double the read latency, and commits at the column op before it is known whether more same-row requests are coming – it fights the FR-FCFS reordering that justifies this design. Mode 5 was removed as a DECISION, not a measurement: it was unproven and mis-plumbed rather than disproven.

`policy_mode` remains a 3-bit field, so software can still WRITE 4..7. Those encodings fall through to the build default and never auto-precharge; that contract is regressed by `test_page_predictor` and by the scheduler matrix.

Field	Width	Reset	Notes
`PAGE_POLICY_CFG.policy_mode`	3	**3**	0=build default, 1=static_open, 2=static_close, 3=fixed_open. **4..7 RETIRED** and fall through to the build default (4/5 on 2026-09-27, TASK-014; 6/7 on 2026-09-26, TASK-011). **RESET IS 3** (`fixed_open`) since 2026-09-27: it is the measured optimum and BUG-003, which

Field	Width	Reset	Notes
			blocked the change, is fixed. Measured on the board 2026-09-27 (75 MHz, BL4 x16, 600 MB/s peak, 24 cells over four scenario families, every cell integrity-clean): +9.1% col_major (195.2 -> 212.9 MB/s), +35.1% col_major_interleaved (262.1 -> 354.1), exactly flat on incremental (561.1) and row_major (572.0), nothing regressing (pre-sdpram-fix 2026-10-02; off-datapath fix, reproduced 2026-10-03 — FINDINGS_litedram_ab_2026-10-03.md). TR=1 and TR=2 measure identical; the cliff is between 2 and 3.
`PAGE_POLICY_CFG.policy_scope`	1	0	**RESERVED** since 2026-09-27. Was documented as

Field	Width	Reset	Notes
			“per-bank TR” but could not diverge: `r_mc` was one global counter driving every `r_tr[b]`. Retired with mode 4.
`PAGE_POLICY_CFG.ctr_open_max` / `ctr_init`	4 each	0	**RESERVED** since 2026-09-27. Were the mode-5 predictor’s close threshold and counter init. Bit positions held so `policy_mode` does not move.
`PAGE_TIMEOUT_CFG.tr_init`	8	**2**	The ONLY live field in this register. **RESET IS 2** since 2026-09-27, moved together with policy_mode – `tr_init=0` DISABLES the timeout, so a reset of 0 would silently neuter mode 3 and the two must change together. Idle MC cycles before the background precharge fires. **`0` DISABLES the timeout** – it is not

Field	Width	Reset	Notes
			a build-default sentinel on this field. TR=1 and TR=2 measure identically; TR=4 already loses the plain `col_major` wins.
`PAGE_TIMEOUT_CFG.tr_min` / `tr_max` / `tr_step`	8 each	0	**RESERVED** since 2026-09-27. Were the mode-4 TR clamps and step. Bit positions held so `tr_init` does not move.

`PAGE_ADAPT_CFG` (0x078) is **deleted**. The address is left a HOLE rather than reused, exactly as 0x07C was when `PAGE_RBL_CFG` retired: every register in this map carries an explicit absolute offset, so removing one shifts nothing, and an old host reading 0x078 gets nothing back instead of another register's meaning. Verified by diffing the generated register maps before and after – 81 -> 80 registers, **zero moved**.

> **STALE ROWS ELSEWHERE IN THIS FILE.** `happy_enable` (SCHED_TUNING) and `zqcs_freq_hz` / `refresh_defer_active` / `refpb_policy_or` (REFRESH_TUNING) describe retired hardware: the HAPPY predictor and `page_predictor.sv` are gone, and the three REFRESH_TUNING fields were retired 2026-09-09 (refresh mode and the JEDEC postpone/pull-in credits are `REF_CTRL` now; no ZQCS engine ever consumed the interval). Left in place rather than silently rewritten because they predate this change and are not part of it – they need their own pass against the RDL.

30.6 ADDR_MAP (Address-Mapping Family Bits)

The address-map register at 0x04C is a single **placement knob**, not a scheme mux. The old `ADDR_MAP_TUNING` register (with a `scheme_or` selector and a `synth_mask_obs` synthesized-scheme bitmask) has been **retired** in this generation: the classic ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH schemes are just settings of `bank_lsb` plus the optional `hash_en` fold, so there is nothing to mux (see §5.4 §3/§4 and `common-ip/rtl/fub/mc_addr_mapper.sv`).

Bit Field	Width	Apply	Description
`bank_lsb`	5	all	Bank-field LSB in the word address. `= COL_WIDTH` -> ROW_MAJOR; lower -> interleave. RTL clamps to `[0, COL_WIDTH]`.
`hash_en`	1	all	Enable bank XOR-hash fold (`bank ^= fold(row) ^ hash_seed`) = the old XOR_HASH scheme.
`hash_seed`	8	all	XOR-hash seed. Runtime seed change without rebuild.

Family note: for DDR4+ the bank-group field is placed by extending the same `bank_lsb`-relative stack rather than by a separate `bg_field_position` selector; that generation's RDL documents the exact field split.

30.7 INIT_TUNING (Init Family Bits)

Bit Field	Width	Apply	Description
zq_retries	4	all	ZQ calibration retry count (DDR2 uses OCD; LPDDR2 uses MR10 ZQ).
init_timeout_ms	8	all	Per-step init timeout (ms; scaled by SIM_INIT_SCALE in sim).
wl_retries	4	DDR3+	Write-leveling retry count. N/A in DDR2 (no write-leveling).
mpr_enable	1	DDR3+	Multi-Purpose Register read-back during init. N/A pre-DDR3.
ca_train_enable	1	LPDDR3+	LPDDR3/4 CA-bus training. N/A in LPDDR2 (no CA training).
cbt_enable	1	LPDDR4-only	Command-Bus-Training. N/A pre-LPDDR4.

30.8 POWER_TUNING (Power-State Family Bits)

Bit Field	Width	Apply	Description
apd_idle_threshold	16	all	Cycles of bank-idleness before APD entry
srf_idle_threshold	24	all	Cycles before Self-Refresh entry
dpd_enable	1	LP-only	Deep-Power-Down enable. N/A for DDR-only flavors.
pasr_active	1	LP-only (R)	Reflects whether any rank has a non-zero PASR mask. R-only.
ckedis_after_sr	1	DDR3+	Drop CKE after SR-entry for sub-mW power floor. N/A in DDR2.
low_freq_mode	1	DDR3+	Reduced-frequency operation hint to PHY. N/A in DDR2.

30.9 ECC_TUNING (Future — Inline ECC)

This block is reserved for future inline-ECC controllers. The bits are **N/A in all current controllers** (DDR2/3/4 and LPDDR2/3/4 v1) because inline ECC is out of scope for the current generation. Future revisions of any flavor that adopt inline ECC will populate this block.

Bit Field	Width	Apply	Description
`ecc_enable`	1	(reserved)	Enable inline ECC. Reads as 0 in all v1 controllers.
`ecc_correct_mode`	2	(reserved)	SECDED / Chipkill / off

30.10 RANK_TUNING (Multi-Rank Family Bits)

These are present in any flavor with NUM_RANKS > 1. For NUM_RANKS = 1 builds, the bits read as 0 and writes are silently ignored.

Bit Field	Width	Apply	Description
rank_enable_mask	4	all	Per-rank enable; clearing a bit suppresses all commands to that rank. Up to 4 ranks.
odt_rule_or	2	all	Runtime override for ODT_RULE_MULTIRANK (00 = build-time default; 01 = JEDEC_DDR2; 10 = JEDEC_LPDDR2; 11 = OFF).
num_ranks_obs	3	all (R only)	Echo of build-time NUM_RANKS. Software discovery.
cs_assert_cycles	4	all	Number of cycles to hold CS_n asserted per command; tunable for PHY timing margin.

30.11 Quiet-Point Behavior

Runtime config bits take effect at the **next configuration quiet point** — no DRAM commands in flight, no refresh sequence in progress. The register block does not enforce quiet points automatically; SoC firmware is responsible for sequencing writes at a drain point (typically during init or an orchestrated idle window). This generation's RDL does not implement a `config_apply` strobe or a `config_settled` status bit — a future family revision may add one; today the SoC owns the drain.

The quiet-point requirement is family-wide because every flavor has hand-off state between scheduler / refresh / power-state that would be corrupted by mid-flight reconfiguration.

30.12 Bit Discovery via the ID Register

This generation does not implement a dedicated capability vector; software discovers the build via the `ID` register at `0xFF0` (see §6.3): `memtype`, `n_phases` (gear ratio), and `version` are readable there, and `module_id` is the fixed `0xD2` family tag. Echo fields embedded in the tuning registers (`SCHED_TUNING.lookahead_max_obs`, etc.) expose the remaining synthesized ceilings. A future family revision may add a packed capability vector along the lines below:

Bits	Field	Description
0	`cap_happy`	1 = HAPPY predictor synthesized
1	`cap_bg_balance`	1 = bank-group balance scheduler logic present (DDR4+)
2	`cap_fgr`	1 = Fine-Granularity Refresh supported (DDR3+)
3	`cap_pasr`	1 = PASR mask is implemented (LP-only)
4	`cap_dpd`	1 = Deep-Power-Down supported (LP-only)
5	`cap_wl`	1 = Write-leveling implemented (DDR3+)
6	`cap_cbt`	1 = CBT implemented (LPDDR4-only)
7	`cap_xor_hash`	1 = bank XOR-hash implemented
11:8	`cap_max_ranks`	Build-time `NUM_RANKS` (echo of geometry)
15:12	`cap_n_phases`	Build-time gear ratio (`DFI_RATE`)
19:16	`cap_memtype`	0 = DDR2, 1 = DDR3, 2 = DDR4, 4 = LPDDR2, 5 = LPDDR3, 6 = LPDDR4

30.13 Per-Flavor Applicability Matrix

Bit	DDR2	DDR3	DDR4	LPDDR2	LPDDR3	LPDDR4
`lookahead_active`	Y	Y	Y	Y	Y	Y
`force_inorder`	Y	Y	Y	Y	Y	Y
`happy_enable`	Y	Y	Y	Y	Y	Y
`bank_group_balance`	—	—	Y	—	—	Y
`refpb_policy_or`	—	—	—	Y	Y	Y
`fgr_mode`	—	Y	Y	—	—	—
`bg_field_position`	—	—	Y	—	—	Y
`wl_retries`	—	Y	Y	—	—	—
`ca_train_enable`	—	—	—	—	Y	Y
`cbt_enable`	—	—	—	—	—	Y
`dpd_enable`	—	—	—	Y	Y	Y
`ckedis_after_sr`	—	Y	Y	—	—	—
`rank_enable_mask`	Y	Y	Y	Y	Y	Y
`odt_rule_or`	Y	Y	Y	Y	Y	Y
`hash_en` / `hash_seed`	Y	Y	Y	Y	Y	Y

The "—" entries are present in the CSR map at the documented offsets but read 0 and ignore writes in that flavor. Software treats these as "soft-N/A" — not an error condition, just an absent feature.

31 Clocks and Reset

31.1 Clock Domains

The controller uses two independent clocks. The register block is a PeakRDL passthrough `cpuif` clocked by `aclk` — there is no separate management clock.

31.1.1 `aclk`

Controller clock. All host-AXI4 logic, the scheduler, the bank timers, the write/read CAMs, and the register block (`pumice_csr`) run on `aclk`. The controller-side of the command / write-data / read-data streams into the DFI plane is on `aclk`.

Typical frequency: 100 – 200 MHz on FPGA SoCs; higher on silicon.

> **Warning — `aclk`'s frequency is a CSR-visible property.** Every timing CSR in the register map (§6.3) is a count of `aclk` cycles. Change this clock's frequency and all of them — JEDEC timings, init waits, ZQ and calibration intervals, PHY round-trip latencies — must be re-derived at the new frequency and reprogrammed before traffic resumes. A stale value violates a real DRAM timing and presents as data corruption, not as an error; there is no hardware frequency-change protocol, so reprogramming is software's job.

31.1.2 `dfi_clk`

PHY / DFI-side clock. The DFI 2.1 command bus, write serializer, and read aligner run on `dfi_clk`; the PHY consumes and drives the DFI pin bus on this clock. The gear ratio (`DFI_RATE`) sets how many DRAM phases each `dfi_clk` cycle carries.

`aclk` and `dfi_clk` are asynchronous to each other. The single crossing between them lives inside `mc_dfi_2p1_plane` (see the CDC summary below).

31.2 Reset Signals

Both resets are active-low.

31.2.1 `aresetn` (active low, controller / AXI domain)

Controller reset, tied to the `aclk` domain. The register block resets on `~aresetn` (`pumice_csr .rst(~aresetn)`). Resets:

- Host AXI4 interface state (`mc_axi4_plane`)
- Write-data and read-command CAMs
- Command scheduler and bank timers (`mc_scheduler_plane`)
- Refresh manager and init sequencer
- Register block (`pumice_csr`) and observation state

31.2.2 `dfi_rstn` (active low, DFI / PHY domain)

DFI-domain reset, tied to the `dfi_clk` domain. Resets the DFI command path, the write serializer, the read aligner, and the DFI-side of the CDC FIFOs inside `mc_dfi_2p1_plane`.

> Note: `init_sequencer` names its ports `mc_clk` / `mc_rst_n` internally, but the top level (`pumice_top` → `pumice_core`) ties these to `aclk` / `aresetn`.

31.3 Reset Sequencing

31.3.1 Cold Boot

1. Both resets asserted (`aresetn`, `dfi_rstn`)
2. Clocks stable
3. `dfi_rstn` deasserted; `aresetn` deasserted (software programs the CSR timings/phases/policy via the register `cpuif`, or defaults are used)
4. The init sequencer asserts `dfi_init_start_o` and waits on `dfi_init_complete_i` (the PHY runs its own DLL-lock / IO training)
5. The init sequencer walks the JEDEC MRS / precharge / refresh sequence (see the Init Sequences chapter)
6. On `init_done_o`, the controller begins servicing AXI traffic

31.3.2 Warm Reset (Clock-Gated)

The SoC can warm-reset the controller without losing DRAM content:

1. SoC requests self-refresh entry (not yet implemented: the controller has no power-down block since `powerdown_ctrl` was removed; see Ch 3.5)
2. Controller acknowledges; DRAM is in self-refresh
3. SoC asserts `aresetn`; clocks may be gated
4. SoC deasserts `aresetn`; clocks resume
5. Controller re-runs the init sequence
6. Controller resumes normal operation

31.4 Clock Domain Crossing Summary

There is exactly one clock-domain crossing in the design: the `aclk` ↔ `dfi_clk` crossing inside `mc_dfi_2p1_plane` (`mc_dfi_cdc.sv`). It is built from asynchronous `gaxi` FIFOs only — command, write-data, and read-data each get their own async FIFO, plus the `init_start` / `init_complete` handshake. The register `cpuif` shares the `aclk` domain, so there is no CSR-clock CDC.

Crossing	Mechanism	Latency
`aclk` → `dfi_clk`	Async `gaxi` FIFOs (cmd / wrdata) in `mc_dfi_cdc`	FIFO + `N_FLOP_CROSS`-flop sync
`dfi_clk` → `aclk`	Async `gaxi` FIFO (rddata) in `mc_dfi_cdc`	FIFO + `N_FLOP_CROSS`-flop sync

31.5 Reset and Power Considerations

DFI v2.1 requires `dfi_init_start` to be asserted by the controller as part of init; the init sequencer drives `dfi_init_start_o` out of reset (once it leaves `S_RESET`) and waits on the PHY's `dfi_init_complete_i` before proceeding to the MRS / precharge / refresh walk. The `init_start` / `init_complete` handshake crosses the `aclk` ↔ `dfi_clk` boundary through the same CDC as the command and data streams.

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32 Init Sequences

The cold-boot init sequences for DDR2 and LPDDR2 are implemented as a single FSM in `common-ip/rtl/fub/mc_xddr2_init_sequencer.sv` (not separate step-package files). The FSM issues real MRS / precharge / refresh commands to the DRAM through the command scheduler while `init_busy_o` is high, and updates the mode-register shadow in lockstep so the live CL / CWL / BL decode tracks what was programmed. The `memtype_i` input selects the DDR2 or the LPDDR2 branch.

The DDR2 branch mirrors LiteDRAM's `get_ddr2_phy_init_sequence` and is proven on the Nexys A7 board.

32.1 DDR2 Cold Init

Per JESD79-2F. FSM states from `init_sequencer.sv`; MR data rides the ROW request field (`init_cmd_row_o`), and the MR / EMR number is carried in the bank index (`init_cmd_bank_o`).

Step	State	Command	MR data	Post-wait
1	`S_DFI_INIT`	assert `dfi_init_start`, wait `dfi_init_complete`, then tINIT	—	`t_init_wait`
2	`S_PREA1`	PRECHARGE all	—	`t_rp_wait` (tRP)
3	`S_EMR2`	MRS EMR(2)	`0x0000`	`t_mrd_wait` (tMRD)
4	`S_EMR3`	MRS EMR(3)	`0x0000`	`t_mrd_wait`
5	`S_EMR1`	MRS EMR(1)	`0x0000`	`t_mrd_wait`
6	`S_MR0_DLL`	MRS MR0 + DLL reset	`MR0.VAL | 0x100` (reset default `0x0533` = BL8 / CL3 / tWR3 / DLL_RESET)	`t_dll_wait` (tDLLK)
7	`S_PREA2`	PRECHARGE all	—	`t_rp_wait`
8	`S_REF1`	AUTO REFRESH	—	`t_rfc_wait` (tRFC)
9	`S_REF2`	AUTO REFRESH	—	`t_rfc_wait`
10	`S_MR0`	MRS MR0 (DLL reset cleared)	`MR0.VAL` (reset default `0x0433`)	`t_mrd_wait`
11	`S_OCD_DEF`	MRS EMR(1) + OCD default	`0x0380`	`t_mrd_wait`
12	`S_OCD_EXIT`	MRS EMR(1) + OCD exit	`0x0000`	`t_mrd_wait`
13	`S_DONE`	assert `init_done`	—	—

Note the JEDEC extended-mode-register order: EMR(2), then EMR(3), then EMR(1), before MR0. The OCD-default state (`S_OCD_DEF`) leaves the MR1 shadow unchanged; `S_OCD_EXIT` restores it to the final `0x0000` value so the live decode is stable.

32.2 LPDDR2 Cold Init

Per JESD209-2F. LPDDR2 uses the Mode Register Write (MRW) chain. Because the MR address (MA) can reach MR63 / MR10 — beyond a 3-bit bank port — the sequencer packs `{MA[5:0], OP[7:0]}` into the ROW request field, and `mc_xddr2_cmd_formatter.sv` unpacks it (row[13:8] = MA, row[7:0] = OP) to build the bit-exact LPDDR2 CA-bus MRW word. Only MR1 / MR2 / MR3 update the CL / CWL / BL decode shadow; MR63 and MR10 are issued to the DRAM but not shadowed.

Step	State	Command	OP data	Post-wait
1	`S_DFI_INIT`	assert `dfi_init_start`, wait `dfi_init_complete`, then tINIT	—	`t_init_wait`
2	`S_L_RESET`	MRW(MR63) Reset	`0x00`	`t_init_wait` (tINIT4)
3	`S_L_ZQ`	MRW(MR10) ZQ Init Calibration	`0xFF`	`t_dll_wait` (tZQINIT)
4	`S_L_MR1`	MRW(MR1) BL8 / nWR3	`0x23`	`t_mrd_wait` (tMRW)
5	`S_L_MR2`	MRW(MR2) RL3 / WL1	`0x01`	`t_mrd_wait`
6	`S_L_MR3`	MRW(MR3) DS 40 ohm	`0x02`	`t_mrd_wait`
7	`S_DONE`	assert `init_done`	—	—

The LPDDR2 branch reuses the same CSR waits as DDR2: `t_init_wait` covers both tINIT and tINIT4, `t_dll_wait` covers tZQINIT, and `t_mrd_wait` covers the post-MRW tMRW delay.

The MR1/2/3 OP data is CSR-backed (`LP_MR1_OP` 0x180, `LP_MR2_OP` 0x184, `LP_MR3_OP` 0x188; resets 0x23 / 0x01 / 0x02) — the values above are the JEDEC defaults. Software can retune BL/nWR/RL/WL/drive strength at bring-up, and sweep the values to defeat a scrambled CA/A-lane board mapping (the same escape hatch as the DDR2 `MRx.VAL` registers). A change takes effect on the next init: `CTRL.init_force_restart` re-runs the MRW chain with the current CSR values, or re-reset the controller. Verified end-to-end by the `lp_mr_op_csr_sweep_lpddr2` top test, which sweeps the CSRs and checks the swept values in the DFI slave's decoded CA MRW stream.

Since 2026-10-15 the sequencer and the mode-register shadow compose the common `mc_init_plane` (family MAS ch10); the sequencing documented here is unchanged.

LPDDR2 is now fully functional in sim: reads and writes, bit-exact JESD209-2F CA encoding, and the full MR init chain above.

32.3 Inter-Command Waits (CSR-Backed)

The JEDEC inter-command delays were previously hardcoded; they are now driven by CSR fields (zero-extended to the 16-bit internal countdown). All counts are in MC (`aclk`) cycles — **and are therefore MC-frequency-dependent: when the controller clock frequency changes (including the 75 MHz ↔ 100 MHz build switch), re-derive every `INIT_TIMING*` value from the JEDEC minimums for the new period.** The same caution applies to every timing CSR in this book.

CSR field	Purpose	Reset value
`INIT_TIMING0.t_init_wait`	CKE / tINIT settle	512
`INIT_TIMING0.t_dll_wait`	DLL lock (tDLLK)	256
`INIT_TIMING1.t_mrd_wait`	post mode-register (tMRD)	8
`INIT_TIMING1.t_rp_wait`	post precharge (tRP)	8
`INIT_TIMING1.t_rfc_wait`	post auto-refresh (tRFC)	16

Init is a one-time event, so generous margins are fine — the reset defaults comfortably cover the JEDEC minimums at the on-board clock rate. Simulation programs shorter `INIT_TIMING*` values to keep test runtimes practical; there is no separate scaling parameter — the shorter waits are just smaller CSR values.

32.4 Power-State Transitions

Self-refresh and deep-power-down transitions are not implemented in the current build, and they are not the init sequencer's job. The optional `powerdown_ctrl.sv` power-state FSM that once covered them was never wired into `pumice_core` and was removed on 2026-10-11 (Ch 3.5). For reference, the two entry modes it implemented were:

- Precharge Power Down (PDE) — CKE low, banks may stay active.
- Self Refresh (SR) — CKE low plus the SRE command; the DRAM self-refreshes internally and the controller stops issuing REFs.

Self-refresh entry requires all banks precharged first (the scheduler enforces this precondition before granting). Self-refresh exit timing (tXSDLL for DDR2 / tXSR for LPDDR2) is enforced by the scheduler; the power-state FSM itself only drops and restores CKE.

LPDDR2 Deep Power Down (DPD) is supported as a planned power-state entry (CKE low after precharge-all). DPD exit loses DRAM content, so the full LPDDR2 cold init sequence above must run again.

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33 Verification Strategy

A planeed verification approach with the existing DFI BFM as the primary reference.

33.1 Planeed Approach

33.1.1 Plane 1: Module-Level Unit Tests

Every FUB pumice instantiates is a common-ip `mc_*` module (`projects/components/mem-ctrl-ip/common-ip/rtl/`). Their unit tests belong with that RTL, not in the pumice area: a pumice test points only at a pumice DUT (owner ruling, 2026-10-13). The `mc_*` FUB tests that lived under `dv/tests/fub/` were removed for that reason. common-ip carries no tests of its own yet. The table lists the blocks whose pumice-area test was removed; `mc_global_timers` and `mc_wr_splitter` keep a directed test in the scoria and andesite areas, the rest now have none. `mc_addr_mapper`, `mc_cmd_arbiter`, `mc_page_policy`, `mc_dfi_cdc`, `mc_dfi_cmd_path`, `mc_dfi_wr_serializer` and `mc_dfi_rd_aligner` lost their pumice tests earlier, in the common-ip move (scoria/andesite cover the last two; `mc_bank_timers`' leaf `mc_bank_timer` is also tested there, the array wrapper is not):

Module	Test focus a common-ip unit test should cover
`mc_wr_intake`	AW-meta + wr-data FIFO handshakes; backpressure; burst splitting into fixed-BL commands
`mc_rd_intake`	AR channel handshakes; address mapping; snarf (read-your-write) probe
`mc_wr_splitter`	Host burst split / pad / ragged-tail into fixed-BL sub-commands
`mc_wr_data_cam`	WR CAM fill / commit-drain / snarf movers; `r_fdone` fill-complete gating
`mc_rd_cmd_cam`	RD CAM return-fill / drain movers; oldest-first, data-ready gating
`mc_rd_return_ring`	Read-return ring wrap and occupancy at each depth
`mc_bank_timers`	Per-(rank,bank) preset/decrement timers; combinational safe_act/rd/wr/pre readiness
`mc_global_timers`	tFAW / tRRD / tWTR / tRTW / tCCD against JESD79-2 / JESD209-2 truth tables
`mc_xddr2_refresh_ctrl`	tREFI counter; refresh-deferral; refresh request/ack handshake
`mc_xddr2_init_sequencer`	Step-by-step DDR2 and LPDDR2 JEDEC MR init sequences
`mc_xddr2_mode_register`	CL / CWL / BL / AL decode (DDR2 and LPDDR2 MR2 RL/WL enum)

Module	Test focus a common-ip unit test should cover
mc_xddr2_cmd_formatter	Round-trip table tests against JEDEC encoding; LPDDR2 CA-bus branch
mc_lpddr2_zq_ctrl	ZQ interval expiry, Mode-C deferral, overdue escalation, DDR2-inert
mc_lpddr2_dq_cal	One-shot DQ calibration sequence

The pumice area verifies what pumice owns: pumice_core and pumice_top (with the PeakRDL pumice_csr register block), at the default and the board geometry, plus the pure-Python oracles the top tests rely on.

33.1.2 Plane 2: Subsystem Tests

The three macro planes are common-ip too – mc_axi4_plane (wr/rd intakes + the two CAMs), mc_scheduler_plane (cmd arbiter + bank/global timers + refresh + init + mode register + page policy) and mc_dfi_2p1_plane (single CDC + cmd path + wr serializer + rd aligner). Their plane-level tests were removed from dv/tests/macro/ under the same ruling and have no common-ip replacement yet. Inside pumice they are exercised only through pumice_core and pumice_top (Plane 3). dv/tests/macro/ keeps the JEDEC command-stream oracle's own unit test (test_pumice_cmd_stream_checker.py).

33.1.3 Plane 3: End-to-End with DFI BFM Slave

The full controller drives the DFI BFM slave through our existing co-sim harness (tests/sim/dfi/test_litedram/... patterns in the DV repo).

- AXI4 stimulus generators issue representative traffic
- DFI BFM slave captures the controller's command stream
- Assertions check: command sequencing matches AXI, timing constraints are honored, data integrity round-trips through the slave's MemoryModel

33.1.4 Plane 4: External-Reference Cross-Validation

For DDR2: cross-validate against LiteDRAM's DDR2 model.

- Generate LiteDRAM controller alongside ours
- Both drive the same DFI BFM slave with the same AXI stimulus
- Compare command streams; differences flag potential design issues

This is a real, proven path. A LiteDRAM DDR2 memtest passes on the Nexys A7 board (128 MiB), which confirmed the board, PHY, pin-out, and DRAM are good. pumice itself is now board-proven on the same Nexys A7: DDR2 reads and writes are clean (0 mismatches across the characterization traffic). Both memtypes pass the full sim suite.

LPDDR2 has no LiteDRAM analog, so cross-validation for LPDDR2 is limited to the BFM master to BFM slave path. LPDDR2 is nonetheless fully functional in sim (reads and writes, bit-exact JESD209-2F CA encoding, full MR init) and passes the sim suite.

33.2 Characterization Sweep

Post-functional verification, the sweep runs each characterization parameter through its choices on the benchmark suite from §5.2:

33.2.1 Sweep Matrix

Parameter	Sweep values
`LOOKAHEAD_DEPTH`	0, 1, 2, 4
`PAGE_POLICY`	OPEN, CLOSE, HAPPY_HYBRID
`REFPB_POLICY`	ROUND_ROBIN, OLDEST_FIRST, DARP
`REFRESH_DEFER_MAX`	1, 2, 4, 8
`TXN_QUEUE_DEPTH`	8, 16, 32
`DFI_RATE`	(build-determined; usually 4)
`ADDR_MAP.bank_lsb`	ROW_MAJOR (== COL_WIDTH), interleave, hash on/off

The full cross-product would be 4 × 3 × 3 × 4 × 3 = 432 builds; in practice we sweep one parameter at a time holding others at recommended defaults, which reduces to ~17 builds per memtype. The exception is the `LOOKAHEAD_DEPTH × PAGE_POLICY` pair — these interact strongly (the fallback policy only matters when lookahead is inconclusive), so this pair is swept as a full 4 × 3 = 12-point grid.

33.2.2 Sweep Outputs

Each (parameter, value) run produces a JSON report:

```
{
  "memtype": "LPDDR2",
  "page_policy": "HAPPY_HYBRID",
  "refpb_policy": "DARP",
  "workload": "mobile-mixed",
  "metrics": {
    "read_bw_pct": 73.2,
    "write_bw_pct": 68.1,
    "read_latency_avg_ns": 142,
    "read_latency_p99_ns": 198,
    "refresh_block_pct": 4.3,
    "row_hit_rate_avg": 0.81,
    "page_predictor_accuracy": 0.91
  }
}
```

These feed into a sweep visualization tool that selects defaults from data.

33.2.3 Out-of-Order Scheduler Trade-Offs

The centralized FR-FCFS scheduler proposed in §1 (Differentiating Features) is the most architecturally aggressive choice in this controller and warrants explicit characterization beyond just the sweep matrix above. Three concrete concerns:

- **Timing-critical path scales with `TXN_QUEUE_DEPTH`.** Scanning the full queue every cycle for the best candidate is a comparator pyramid: priority resolves over (ready-flag, row-hit-flag, age) for every entry simultaneously. At `TXN_QUEUE_DEPTH = 16` this is well within an embedded-SoC clock budget; at 32 it stays comfortable; at 64 it becomes the critical path on the controller side. The characterization sweep should report scheduler combinational depth at synthesis time alongside the performance numbers — the right `TXN_QUEUE_DEPTH` is the largest value that doesn't push the scheduler off the clock budget.
- **Area cost is super-linear in queue depth.** Each queue entry needs its full metadata replicated (bank, row, col, ID, age, row-hit cache). At depth 16, ~2K gates; at depth 64, ~10K gates — comparable to the rest of the scheduler combined. The HAPPY predictor's gate cost is a separate line item.
- **Verification cost is significantly higher than FIFO.** OoO issue means assertions like "request N landed at cycle X" can't be used. The DV plan relies on protocol-level scoreboarding: the BFM master records arrival order per ID, the BFM slave records issue order, and the scoreboard verifies the AXI4 per-ID ordering invariant is preserved. Within-ID ordering violations are a higher-severity bug than performance regressions and should be caught at Plane 1 unit tests, not deferred to integration.
- **Fairness tail matters under adversarial workloads.** With `AGE_MAX` as the only anti-starvation cap, pathological patterns can drive 99th-percentile latency well above mean. The `random-narrow` and `multi-master` workloads in the benchmark suite intentionally exercise this; the `mobile-mixed` workload simulates the common case. The recommended default for `AGE_MAX` should be picked based on tail-latency targets, not throughput targets.

These concerns are unique to the OoO scheduler design choice — a simpler per-bank FIFO controller (the typical open-source pattern) would not face them, at the cost of leaving row-hit reordering performance on the table.

33.3 Verification Hooks

Internal observation outputs are exposed via CSR (see §6.3) for in-system characterization. Additionally, the controller emits internal “event” signals that can be tied to waveform-dump triggers:

- `ev_init_done`
- `ev_refresh_issued`
- `ev_page_conflict_detected`
- `ev_queue_full`
- `ev_refresh_pending_critical`

Specific events can be wired to a test-bench-controlled waveform-capture trigger during bring-up.

33.4 Coverage Targets

Functional coverage (collected during regression):

- All FSM states reached in each module that has one (the bank timers and CAMs are FSM-free; their timer/counter states are covered instead)
- All page policies exercised
- All refresh policies exercised
- All scheduler priority levels exercised
- All AXI burst types exercised (if enabled)
- All `ADDR_MAP.bank_lsb` settings (ROW_MAJOR .. interleave) exercised, with hash on and off

Code coverage targets: 100% line / branch coverage for all modules.

Assertion coverage: all design assertions (registered protocol checks) hit during regression.

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34 Open Issues and TODOs

Items where design choices were made provisionally and should be reviewed before the HAS is promoted to formal status.

34.1 Open Issues

34.1.1 1. AXI4 Narrow-Burst Handling Confirmation

AXI4 allows narrow transfers (data width less than bus width). The current plan is to handle them via byte-mask through `wrdata_mask`. **Confirm** the wstrb-to-wrdata-mask wiring honors the AXI partial-write semantics correctly, particularly for unaligned bursts.

34.1.2 2. AXI4 Read-Modify-Write Strobe Handling

Partial writes in DDR2 / LPDDR2 use `wrdata_mask` to mask byte lanes; this is straightforward. **Confirm** that per-byte masking is wired all the way through the write path (`mc_wr_intake` -> `mc_wr_data_cam` -> `mc_dfi_wr_serializer`) and that `mc_wr_intake` doesn't widen narrow writes implicitly.

34.1.3 3. CSR Address Space Allocation

The config bus is a PeakRDL passthrough cpuif on `aclk` (no APB slave, no separate `apb_pclk`). `CSR_ADDR_W = 12` provides a 4 KB register region. **Confirm** the SoC's address-map allocation has at least 4 KB available for the cpuif window, and that there's no conflict with adjacent peripherals.

34.1.4 4. Per-Bank Refresh Book-Keeping Cost

Keeping `last_ref_age` per bank costs one counter per bank. For 8 banks this is 8 small counters. **Confirm** the synthesis area impact is acceptable; we expect ~200 LUTs for the per-bank refresh tracking on a typical FPGA.

34.1.5 5. HAPPY Predictor Hash Function

RESOLVED (2026-08-25): the HAPPY address-hash predictor is retired — it was never wired into the rearchitected core. Its Ghasempour-2015 successors landed as runtime modes of `mc_page_policy` (`adapt_time` and `adapt_access`) and have since been retired themselves (2026-09-27, TASK-014): `adapt_time` measured as `fixed_open(tr_min)`, and `adapt_access` drove auto-precharge, which costs 4.9x the activations of a background precharge. The surviving mechanism is `fixed_open`'s background close, which needs no predictor and no address hash.

34.1.6 6. Self-Refresh Exit Latency

JESD79-2 requires `tXSNR` (~200 ns) before any command after SR exit. Self-refresh is not implemented (the optional `powerdown_ctrl` was removed 2026-10-11); whichever block implements SR exit must enforce this and gate the scheduler for the full duration.

34.1.7 7. Init Sequencer Retry on Error

If ZQ fails, do we retry up to 3 times then halt, or just halt? Current plan: 3 retries (parameter `INIT_ZQ_RETRIES`) then raise `init_error`. **Confirm** with the system architect — some systems

prefer fail-fast. (Note: DDR2 ZQ init differs from LPDDR2 — `init_sequencer` handles both JEDEC sequences.)

34.1.88. CSR Write Atomicity

Multi-register parameter changes (e.g., updating both `MR0` and a related timing) need a "config quiet period" to take effect atomically. Today the SoC owns the drain: software is expected to quiesce AXI traffic before reprogramming timing/mode CSRs. A dedicated `config_apply` strobe (hardware-enforced quiet period) is **not implemented** and is a possible future addition. **Document** the software drain protocol regardless.

34.1.99. Burst Splitting at Row Boundaries

AXI4 allows up to 256 beats per burst; a DRAM row contains `2^COL_WIDTH` columns. Burst splitting into fixed-BL commands happens in `mc_wr_intake` / `mc_rd_intake`. **Confirm** the split logic correctly tracks partial-burst completion for ID-based ordering.

34.1.10 10. Multi-Rank

Multi-rank scaffolding now exists: `NUM_RANKS` is a real build parameter (default 1), the DFI `dfi_cs_n_o` / `dfi_odt_o` buses are per-rank (width `NUM_RANKS * DFI_RATE`), and `addr_mapper` carries a rank field (`rank_o`) at the top of the address stack. The command path and bank timers are stamped per (rank, bank). **Remaining gap**: per-rank refresh coordination, PASR (partial-array self-refresh), and per-rank observation registers are not yet implemented. **Confirm** the multi-rank data path with a `NUM_RANKS > 1` build before relying on it.

34.1.11 11. ECC

Out of scope for this controller. ECC is expected to be handled at the SoC level or in a sideband wrapper, not the controller itself.

34.1.12 12. DFI Training Sub-Interface

Not driven in v1; the assumption is that the PHY handles its own training or training is not required for this DDR2 / LPDDR2 target. **Confirm** with the PHY vendor before silicon tape-out.

34.1.13 13. AXI Quality of Service (QoS)

Currently the `awqos` / `arqos` fields are forwarded to the scheduler as priority hints, but the scheduler does not implement explicit QoS classes. **Decide** whether to add proper QoS support (multi-class scheduler) or leave QoS as a future enhancement.

34.1.14 14. Multi-Master AXI

Currently single AXI port. **Decide** whether to add a multi-port AXI crossbar internally (simplifies SoC integration but adds area) or leave as a single-port with the SoC responsible for AXI arbitration upstream.

34.1.15 15. Observation Counter Overflow

Observation counters (row-hit, queue depth max, etc.) are 32-bit. At high traffic rates they could wrap. **Decide** on overflow handling: saturate, clear-on-read, or rely on SoC reading frequently.

34.1.16 16. Retired-Architecture Modules Still on Disk (`OLD/`)

The pre-rearchitecture modules still live under `rtl/fub/OLD/`, `rtl/macro/OLD/`, and `rtl/top/OLD/` (e.g. `scheduler.sv`, `wr_cmd_cam.sv`, `rd_cmd_cam.sv`, `xbank_timers.sv`, `axi_intake.sv`, the `*_macro.sv` blocks, `pumice_csr_slave.sv`). They are referenced only by retired sentinel tests in `dv/tests/` (`test_scheduler.py`, `test_wr_cmd_cam.py`, `test_xbank_timers.py`, `test_axi_intake.py`, and the `*_macro` tests). **Remove** the `OLD/` trees and their sentinel tests once the new architecture is fully signed off, so the live module set is the only thing that builds.

> Status (2026-07-22): done — the `OLD/` trees and their sentinel tests have been removed; only the rearchitected module set remains under `rtl/`.

34.1.17 17. Open-Page Read-Fetches-Zero Under Gapped Reads

A known scheduler/CAM interaction: under open-page policy with gapped read traffic, a read can be fetched before its data is valid (returns zero) in a narrow timing window (reproduces at `PUMICE_SEED=2`, the burst-pause / hit-miss-oscillation pattern). **Root-cause and fix** the read-fetch gating in the open-page path before promoting the HAS to formal status. The `r_fdone` fill-complete gating in the CAMs is the relevant mechanism.

34.1.18 18. Refresh-Collision Liveness Wedge at Tiny tREFI (common-IP)

The refresh-pressure soak (`pumice_master.py --soak`, Nexys A7 @75 MHz) fails the `tREFI = 0x40` corner (~850 ns, a refresh every ~4 BL8 bursts) on **both** pumice (RTL 2026-10-11) and granite (fixed entry store, 2026-10-14): writes never complete (`done=False`, `WR HANG`) in 5/5 rounds while `mism=0` everywhere — a pure liveness wedge, no data corruption. `tREFI` default and `0xFFFF` are 5/5 clean on both. The corner is far tighter than any real DDR2 tREFI, so this is robustness debt, not a shipping defect; no real DRAM runs refresh at 0x40.

Suspected area: the common scheduler-plane refresh path — the arbiter's `w_ref_col_block` all-bank column block while `refresh_req_i/drain_i` are asserted interacting with the refresh grant/drain handshake and the write drain-in-progress indication. Under 0x40 the request line is effectively permanent, so any `drain waits for idle / idle waits for grant` circularity latches.

Next crack: reproduce in `test_pumice_core_sched_order`-style sim with waves at tREFI=0x40 (the board repro is `pumice_master.py --soak`), find the circular wait, and re-soak silicon on both controllers.

34.2 TODOs Before v0.2

- Add bit-level pinout tables for the AXI, DFI, and cpuif (register) interfaces
- Add timing diagrams for the WR / RD command issue, including CWL / CL alignment
- Add sequence diagrams for `init_sequencer` (DDR2 and LPDDR2 MR init) and, once power-down exists, its entry/exit; document the FSM-free `bank_timer` countdown timers rather than an FSM diagram
- Add a draft of the Verilog package skeletons for `ddr2_init_steps_pkg.sv` and `lpddr2_init_steps_pkg.sv`
- Cross-reference each section to the corresponding pre-aspec.md bullet
- Add quantitative area / power estimates per module (synthesis pass needed)
- Add waveform examples for the canonical init sequences

34.3 Feature Roadmap — planned advanced modes (ordered)

The full catalog of planned, config-bit-selectable advanced modes lives in `docs/design-requirements.md` ("Advanced modes — selectable scheduling / paging / refresh"); the family-level split (commodity-legal here vs model-only parked for DDR3/DDR4) is in `vault/tracking/projects/components/mem-ctrl-ip/memory-controllers/ADVANCED_MODES_ROADMAP.md`. Entry gate (TASKS.md TASK-FEATURES): board reads validated at the bring-up tuple, refresh-collision fix re-soaked on silicon.

Serial pre-silicon implementation order (each step OFF-by-default with its own red-to-green model test):

1. **Foundation** — `SCHED_POLICY` / `PAGE_POLICY_CFG` / `REFRESH_MODE` mode-select CSRs + `*_STATS` telemetry + PHY capability straps (no behavior change; defaults bit-identical).
2. **Scheduling (Axis 1)** — `in_order` -> `fr_fcfs` (confirm current) -> `age_threshold` -> `most/fewest_pending` -> `ACCESS_PREF` -> write-batching -> **QoS** (AxQOS-aware pick, `QOS_EN`).
3. **Paging (Axis 2)** — `static_open/close` (confirm) -> `fixed_open`. (`adapt_time`/`adapt_access` retired 2026-09-27; `rbl_static`/`rbl_dyn` retired 2026-09-26. Modes 4..7 fall through to the build default.)
4. **Refresh (Axis 3, commodity)** — JEDEC pull-in/postpone sweep -> `refpb_rr`.

None of the mode-select CSRs above exist in the RDL yet; open issue 13 (QoS) is subsumed by step 2 of this order.